

On determinantal representations of cubic surfaces

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**Australian
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“By thirteen I gave up trying, I decided I would be a mathematician, ‘cause math is real.”

— Cady Heron

Declaration

The work in this thesis is my own except where otherwise stated.

Charlotte Sherratt

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¹the sweet treat, not the natural disaster.

Abstract

This thesis details the beautiful correspondence between the moduli space of $4 \times 3 \times 3$ tensors, the Hilbert scheme of 6 points in $\mathbb{P}^2$, and the moduli space of cubic surfaces.

Viewing a tensor φ as a 3×3 matrix of linear forms in x, y, z, w , its determinant $\det(\varphi)$ cuts out a cubic surface $S_\varphi \subset \mathbb{P}^3$. When S_φ is smooth, view φ as a 4×3 matrix of linear forms in x, y, z in two ways. It drops rank at two sets of 6 points $\Gamma_1 \subset \mathbb{P}^2$ and $\Gamma_2 \subset \mathbb{P}^2$, which are Gale-dual. The blowups of $\mathbb{P}^2$ at Γ_1 and Γ_2 are isomorphic to S_φ .

We recount this classical construction, prove it geometrically, and generalise it to normal cubic surfaces with finitely many lines (equivalently, to curvilinear length 6 subschemes of $\mathbb{P}^2$). Along the way, we prove that there are 27 lines on any smooth cubic surface, and understand when families of cubic surfaces can be lifted to families of $4 \times 3 \times 3$ tensors.

Finally, we investigate whether this correspondence extends to the GIT compactifications. We compute the semistable locus of $4 \times 3 \times 3$ tensors under the $\mathrm{SL}(4) \times \mathrm{SL}(3) \times \mathrm{SL}(3)$ -action, and show there are $\mathrm{SL}(4) \times \mathrm{SL}(3) \times \mathrm{SL}(3)$ -semistable tensors whose determinant cubic is $\mathrm{SL}(4)$ -unstable, two new results. The map sending a matrix to its determinant cubic does not induce a morphism of GIT quotients.

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Introduction: an incredible coincidence

The determinant of a $d \times d$ matrix of linear forms is a degree d polynomial. Going backwards, we ask: when can a degree d polynomial be written as the determinant of a matrix of linear forms in n variables? For very small values of both d and n , the answer is trivially *always yes* (e.g. if $d = 1$, or if both $n \leq 3$ and $d \leq 4$). For most values of d and n , a quick dimension count shows the answer is trivially *almost always no*. But when $n = 4$ and $d = 3$, that is of cubic equations in 4 variables, the answer is (non-trivially) *almost always yes*, and in finitely many ways, and turns out to be related to an incredible wealth of classical algebraic geometry. The main goal of this thesis is to understand this geometry, and exactly how far we can push it. The following facts are true:

- A smooth cubic surface is isomorphic to $\mathbb{P}^2$ blown up at 6 points $\Gamma_1 \subset \mathbb{P}^2$.
- The ideal that cuts out a general set of 6 points $\Gamma_1 \subset \mathbb{P}^2$ is generated by the maximal minors of a 4×3 matrix φ_1 of linear forms in x, y, z .
- There is an associated set of 6 points $\Gamma_2 \subset \mathbb{P}^2$ to Γ_1 , known as its *Gale-dual*. It is also cut out by the minors of another 4×3 matrix φ_2 .
- The equation cutting out a smooth cubic surface $S \subset \mathbb{P}^3$ can be written as the determinant of a 3×3 matrix φ_3 of linear forms in x, y, z, w . We call the matrix φ_3 a *determinantal representation* of S .

Remark. The first incredible coincidence is that the three matrices φ_1 and φ_2 and φ_3 can all be chosen to be adjoints of the same $4 \times 3 \times 3$ tensor $\varphi \in k^4 \otimes k^3 \otimes k^3$.

The second incredible coincidence is that this is one of the only non-trivial cases where everything lines up. The moduli spaces of cubic surfaces, $4 \times 3 \times 3$ tensors, and sets of 6 points in $\mathbb{P}^2$ are 19, 35, and 12-dimensional, respectively. But after quotienting out by the symmetry of the $\mathrm{SL}(4) \times \mathrm{SL}(3) \times \mathrm{SL}(3)$ -action,

these are all 4-dimensional. This suggests that there should be a generically finite rational map between the GIT quotients

$$\det : \mathbb{P}(\mathbb{C}^4 \otimes \mathbb{C}^3 \otimes \mathbb{C}^3) // (\mathrm{SL}(4) \times \mathrm{SL}(3) \times \mathrm{SL}(3)) \dashrightarrow |\mathcal{O}_{\mathbb{P}^3}(3)| // \mathrm{SL}(4) \quad (1)$$

of degree 72, sending a $4 \times 3 \times 3$ tensor to the cubic surface cut out by the equation $\det(\varphi_3) = 0$. This is true. But this map is *not* a morphism between the respective GIT quotients — it is merely rational.

This general correspondence was known to the classical algebraic geometers of the 19th and early 20th centuries, in the guise of the classical theorem that a general cubic surface is generated three pencils of planes [Aug62]. Salmon proved in 1849 that there are 27 lines on any smooth cubic surface [Sal49], in 1866 Clebsch proved every smooth cubic surface is the blowup of $\mathbb{P}^2$ at 6 points [Cle66], and in 1906 Segre proved the equation defining any smooth cubic surface can be written as the determinant of a 3×3 matrix of linear forms [Seg06]. A series of papers by Coble in the 1910s [Cob15; Cob16; Cob17] proved many results, including the connection between Gale-duality and cubic surfaces. In 1923, White observed the relationship between a general determinantal representation of a cubic surface, and the point configurations cut out by the minors of the 4×3 matrices [Whi23].

Giving a complete account of the history of cubic surfaces is an almost impossible task, and many results have been lost to time. For example, Dolgachev’s *Classical Algebraic Geometry* has a succinct 80 page chapter [Dol12, Chapter 9] devoted entirely to only some of the classical theory of cubic surfaces, explored in the modern language of scheme theory. His biography of Cremona [Dol04] is our source for the classical history of cubic surfaces, and for a more recent account of interesting results and questions about cubic surfaces we refer the reader to Ranestad and Sturmfels’ “Twenty-seven questions about the cubic surface” [RS20]. The study of associated sets of points in projective space has a similarly long history. We refer the reader to Eisenbud and Popescu’s “The projective geometry of the Gale transform” [EP00] for historical account of the Gale-transform.

The theory of determinantal representations of singular cubic surfaces is less well-known. Segre [Seg06] hints at how one would write down a determinantal representation for a singular cubic surface, but his construction relies on finding three mutually skew lines — there are cubic surfaces with only two lines that have determinantal representations. Extending this in 1941, Thrall partially classified $4 \times 3 \times 3$ tensors by understanding the 6 points in $\mathbb{P}^2$ they are associated to [Thr41], but ran into difficulties without the modern language of scheme theory to describe non-reduced point configurations.

In 1993, Gimigliano introduced a sheaf-theoretic approach to smooth determinantal surfaces [Gim89], viewing the various matrices as maps of vector bundles on $\mathbb{P}^2$ and $\mathbb{P}^3$. This was extended by Dolgachev and Kapranov in [DK93], detailing the specific geometry of smooth determinantal cubic surfaces. A general theory of smooth determinantal hypersurfaces was outlined by Beauville in 2000 [Bea00], and his sheaf-theoretic approach extends to singular surfaces. In 2002, Kok Onn Ng exhaustively classified all $4 \times 3 \times 3$ tensors by viewing them as presentation matrices for ideals [Ng02]. Dolgachev also follows this approach [Dol12, §9.3].

This thesis is devoted to understanding the deep geometry connecting cubic surfaces, length 6 subschemes of $\mathbb{P}^2$, and $4 \times 3 \times 3$ tensors. The rough statement of the main theorem we will prove is the following:

Theorem. *Fix a $4 \times 3 \times 3$ tensor $\varphi \in k^4 \otimes k^3 \otimes k^3$ such that the determinant cubic $\det(\varphi_3)$ cuts out a normal cubic surface $S_\varphi \subset \mathbb{P}^3$ with finitely many lines. Viewing it as a 4×3 matrix of linear forms in x, y, z in two ways φ_1 and φ_2 , the respective maximal minors generate two ideals $I_1, I_2 \subset k[x, y, z]_\bullet$ that cut out length 6 subschemes $\Gamma_1, \Gamma_2 \subset \mathbb{P}^2$. Then the following statements are true:*

- *φ_1 and φ_2 are the respective presentation matrices of a free resolution of the homogeneous ideals I_1 and I_2 .*
- *The subschemes $\Gamma_1 \subset \mathbb{P}^2$ and $\Gamma_2 \subset \mathbb{P}^2$ are Gale-dual.*
- *The linear systems of cubics through Γ_1 and Γ_2 are canonically isomorphic.*
- *These linear systems induce a pair of birational morphisms*

$$\tilde{m}_1 : \mathrm{Bl}_{\Gamma_1} \mathbb{P}^2 \longrightarrow S_\varphi, \quad \tilde{m}_2 : \mathrm{Bl}_{\Gamma_2} \mathbb{P}^2 \longrightarrow S_\varphi$$

Every normal cubic surface with finitely many lines arises as the determinant of a 3×3 matrix of linear forms in this way, except for the E_6 cubic surface.

However, not every way to write a normal singular cubic surface as a blowup of $\mathbb{P}^2$ induces a determinantal representation.

The core components of this theorem can be attributed to Ng [Ng02] and Dolgachev [Dol12, Thm. 9.3.6], who prove it starting from the Hilbert-Burch theorem. Ng provides a classification of $4 \times 3 \times 3$ tensors by enumerating all possible length 6 subschemes in $\mathbb{P}^2$ using Maple. It is also worth mentioning that in 1996, Brundu and Logar used computational techniques to provide an

explicit determinantal representation for every cubic surface, except for the E_6 cubic surface (which they prove is not determinantal) [BL98].

Our approach is similar, but aims to emphasise the deep geometry at play. Chapter 2 is devoted to fully understanding the determinantal representations of normal cubic surfaces with finitely many lines, using a mixture of classical and modern techniques. We begin with the Gale-transform, and then some purely linear-algebraic facts about determinantal cubic surfaces. We explain the connection to free resolutions of ideals, and connect the classification of determinantal representations to the resolution of singularities of a given cubic surface.

Chapter 3 uses standard techniques in deformation theory to prove that there are exactly 27 lines (counting multiplicity) on any cubic surface with finitely many lines. We then use flat families of colliding point configurations and lines in Section 3.4 to describe how every family of smooth cubic surfaces is induced by a family of determinantal representations, and provide some conditions under which this can be easily extended to singular surfaces.

Chapter 4 is devoted to proving (and understanding precisely why) the determinant map (1) is not a morphism between GIT quotients. We provide a classification of the semistable locus of $\mathbb{P}(\mathbb{C}^4 \otimes \mathbb{C}^3 \otimes \mathbb{C}^3)$ under the $\mathrm{SL}(4) \times \mathrm{SL}(3) \times \mathrm{SL}(3)$ action, and give examples of semistable tensors whose determinant cubic cuts out an $\mathrm{SL}(4)$ -unstable cubic surface.

The main original contribution of this thesis is in Chapter 4. The classification of $\mathrm{SL}(4) \times \mathrm{SL}(3) \times \mathrm{SL}(3)$ -semistability and the proof that the determinant map (1) is not a morphism of GIT quotients are both new results. Section 3.4 is original. The geometric exposition of Chapter 2 was developed mostly independently from the existing literature, except where otherwise stated, and the proofs in Chapter 3 are our own.

Chapter 1

Background

1.1 Algebro-geometric preliminaries

Some background knowledge on algebraic varieties and schemes is assumed, to the level of chapters I and II of Hartshorne's *Algebraic Geometry* [Har77], and some basic knowledge of sheaf cohomology on noetherian schemes is useful (chapter III of [Har77] is sufficient, although we don't need the full level of generality discussed there). For the theory of smooth surfaces, Beauville's *Complex Algebraic Surfaces* [Bea78] is an excellent source (we only use results from chapters II and IV), as well as chapter V of [Har77]. We will briefly recall some important cohomological tools and facts about divisors, linear systems, and blowups. We outline some of the classical geometry of cubic surfaces, and then describe the ADE resolution of anticanonical surface singularities.

We will work over a fixed algebraically closed field k of characteristic zero, unless otherwise specified. Most of the theory we develop holds over any field with minor modifications. All schemes in this thesis are noetherian and of finite type over k . A morphism of schemes will always be a morphism of k -schemes, and in Chapter 3 we will also discuss families of schemes over a different base scheme. All the terminology in this thesis will be in the sense of [Har77] unless otherwise specified.

Notation 1.1. When writing the tensor product of two k -algebras over k , we will omit the subscript. We will similarly omit the subscript for the fibre product over $\mathrm{Spec}(k)$, and when $\mathrm{Spec}(A)$ is an affine scheme we will write $X \times_A Y$ instead of $X \times_{\mathrm{Spec}(A)} Y$.

If $x \in X$ is a point, then we let $\kappa(x)$ be the residue field at x . If $\kappa(x) = k$, then we say that x is a *closed point* or k -point, and the set of k -points of a scheme X is denoted by $X(k)$. In Chapter 3, we will interchangeably switch between closed

points $x \in X(k)$ and the morphisms $x : \operatorname{Spec}(k) \rightarrow X$ that define them.

For a sheaf $\mathcal{F}$ on a scheme X , we will write $\mathcal{F}|_U = H^0(U, \mathcal{F})$ for the space of sections of $\mathcal{F}$ on some open subscheme $U \subset X$.

Definition 1.2. A S -scheme $X \rightarrow S$ is said to be *relatively projective* over S if there exists a closed immersion of S -schemes $X \hookrightarrow \mathbb{P}_S^n$ for some integer n . We will say a scheme is *projective* if it is relatively projective over k . This is projectivity in the sense of [Har77], not [EGA].

Definition 1.3. The *projectivisation* $\mathbb{P}V$ of a finite-dimensional vector space V is the algebraic variety

$$\mathbb{P}V := \operatorname{Proj}(k[V^\vee]_\bullet) \cong \mathbb{P}^{\dim V}.$$

In this way, the tautological bundle $\mathcal{O}_{\mathbb{P}V}(-1)$ of $\mathbb{P}V$ is naturally identified with a subbundle of $V \otimes \mathcal{O}_{\mathbb{P}V}$.

Some authors denote $\mathbb{P}V$ to mean $\operatorname{Proj}(k[V]_\bullet)$, the projectivisation of the dual space, which has tautological bundle $\mathcal{O}_{\mathbb{P}V^\vee}(-1) \subset V^\vee \times \mathcal{O}_{\mathbb{P}V^\vee}$. We are using the opposite convention.

This construction behaves well with respect to the Segre embedding, with the closed immersion $\mathbb{P}V \times \mathbb{P}W \rightarrow \mathbb{P}(V \otimes W)$ coming from the homogeneous ideal $I \subset k[V \otimes W]_\bullet$ that vanishes only on decomposable monomials (ones of the form $(\lambda \otimes \mu)^n$, where $(\lambda, \mu) \in V^\vee \times W^\vee$), and we have $\mathbb{P}V \times \mathbb{P}W \cong \operatorname{Proj}(k[V \otimes W]_\bullet / I)$.

Remark 1.4. Recall that any closed subscheme $Z \hookrightarrow X$ is cut out by an ideal sheaf $\mathcal{I} \subset \mathcal{O}_X$. There is a short exact sequence

$$0 \longrightarrow \mathcal{I} \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{O}_Z \longrightarrow 0 \quad (1.1)$$

of $\mathcal{O}_X$ -modules. This comes from gluing the short exact sequence of A -modules $0 \rightarrow I \rightarrow A \rightarrow A/I \rightarrow 0$ on open affine charts $\operatorname{Spec}(A) \subset X$, where $I = \mathcal{I}|_{\operatorname{Spec}(A)}$ and $Z \cap \operatorname{Spec}(A) = \operatorname{Spec}(A/I)$.

If $Z \hookrightarrow \mathbb{P}^n$ is a closed immersion to projective space, then the homogeneous ideal $I \subset k[x_0, \dots, x_n]_\bullet$ associated to Z is the largest ideal such that the associated ideal sheaf $\mathcal{I}$ is the ideal sheaf that cuts out Z .

Definition 1.5. Let X be a scheme, $x \in X$ a point and $\mathfrak{m}_x \subset \mathcal{O}_{X,x}$ the maximal ideal. The *cotangent space* to X at x is the $\kappa(x)$ -vector space $\mathfrak{m}_x / \mathfrak{m}_x^2$. This is the vector space of generators for $\mathfrak{m}_x$.

The *tangent space* to X at x is the vector space $T_x X := (\mathfrak{m}_x / \mathfrak{m}_x^2)^\vee$.

Example 1.6. (Tangent space to $\mathbb{A}^n$) If $H = \mathbb{A}^n$, then the tangent space to H at a point h is isomorphic to k^n . If S is a scheme over k smooth at a closed point $s \in S$, then $T_s S \cong k^m$ where $m = \dim \mathcal{O}_{S,s}$ is the dimension of S at s .

Definition 1.7. Let X be a variety and $x \in X$ a closed point. X is said to be *nonsingular* at x if the local ring $\mathcal{O}_{X,x}$ is regular, i.e. $\dim(\mathcal{O}_{X,x}) = \dim_k(T_x X)$. If X is nonsingular at every point $x \in X$ then we say it is a *smooth variety*. X is smooth if and only if the morphism $X \rightarrow \operatorname{Spec}(k)$ is smooth.

An equivalent condition for nonsingularity which we will use is the following:

Proposition 1.8. *If $X \hookrightarrow \mathbb{A}^n$ is a closed subvariety cut out by a principal ideal $(f) \subset k[x_1, \dots, x_n]$, then X is nonsingular at a point $x \in X$ if and only if at least one of the n partial derivatives of f is non-zero.*

Definition 1.9. Let X be a scheme, and Y be a closed subscheme cut out by a sheaf of ideals $\mathcal{I} \subset \mathcal{O}_X$, with corresponding closed immersion $i : Y \hookrightarrow X$. The *conormal sheaf* is the $\mathcal{O}_Y$ -module $\mathcal{I}/\mathcal{I}^2 \cong \mathcal{I} \otimes_{\mathcal{O}_X} \mathcal{O}_Y$. Its dual is the *normal sheaf* $\mathcal{N}_{Y/X} := \mathcal{H}om_{\mathcal{O}_Y}(\mathcal{I}/\mathcal{I}^2, \mathcal{O}_Y)$.

The normal sheaf is called the normal sheaf because it is the scheme-theoretic generalisation of the normal bundle — if Y and X are smooth varieties then $\mathcal{N}_{Y/X}$ is the normal bundle.

Definition 1.10. For a sheaf $\mathcal{F}$ on a scheme X , we denote the dimension of the cohomology ring by

$$h^i(X, \mathcal{F}) := \dim_k H^i(X, \mathcal{F}).$$

Example 1.11 (Cohomology of projective space). [Har77, III, Thm. 5.1] Let $n > 0$ and d be integers. Then

- $h^0(\mathbb{P}^n, \mathcal{O}_{\mathbb{P}^n}(d)) = \binom{n+d}{d}$ if $d \geq 0$.
- $h^0(\mathbb{P}^n, \mathcal{O}_{\mathbb{P}^n}(d)) = 0$ if $d < 0$.
- $h^i(\mathbb{P}^n, \mathcal{O}_{\mathbb{P}^n}(d)) = 0$ whenever $i \neq 0, n$.

Definition 1.12. A scheme Z is said to be *finite* if it is finite over k , in the sense that the structure morphism $Z \rightarrow \operatorname{Spec}(k)$ is finite. In this case, Z is isomorphic to an affine scheme $\operatorname{Spec}(A)$ where A is a finite-dimensional k -vector space.

The *length* of a finite scheme $Z \cong \operatorname{Spec}(A)$ is defined to be $\operatorname{len}(Z) := \dim_k A$. Equivalently, we have

$$\operatorname{len}(Z) = h^0(Z, \mathcal{O}_Z) = \sum_{z \in Z} \dim_k(\mathcal{O}_{Z,z}).$$

The length of Z at a point $z \in Z$ is also denoted $\text{len}_z(Z) = \dim_k(\mathcal{O}_{Z,z})$.

When $Z \hookrightarrow X$ is a closed subscheme of X , then the exact sequence (1.1) gives an exact sequence on stalks

$$0 \longrightarrow \mathcal{I}_z \longrightarrow \mathcal{O}_{X,z} \longrightarrow \mathcal{O}_{Z,z} \longrightarrow 0$$

for all points $z \in \text{Supp}(Z)$, and the ideal $\mathcal{I}_z$ is an ideal of length $\text{len}_z(Z)$.

Definition 1.13. Fix a coherent sheaf $\mathcal{F}$ on a projective scheme X . The *Euler characteristic* $\chi(\mathcal{F})$ of $\mathcal{F}$ is defined to be

$$\chi(\mathcal{F}) := \sum_i (-1)^i h^0(X, \mathcal{F}).$$

When $0 \rightarrow \mathcal{F}_1 \rightarrow \mathcal{F}_2 \rightarrow \mathcal{F}_3 \rightarrow 0$ is a short exact sequence of coherent sheaves, the Euler characteristic is additive, in the sense that $\chi(\mathcal{F}_2) = \chi(\mathcal{F}_1) + \chi(\mathcal{F}_3)$.

Additivity of the Euler characteristic is very useful when computing the length of a finite subscheme $Z \hookrightarrow X$. In this case, the positive cohomology of $\mathcal{O}_Z$ vanishes, and we find that the exact sequence (1.1) gives

$$\text{len}(Z) = \chi(\mathcal{O}_Z) = \chi(\mathcal{O}_X) - \chi(\mathcal{I}).$$

1.2 The basic geometry of surfaces

We very briefly recall some of the key tools for the theory of surfaces, starting with divisors. We only use divisors on smooth surfaces, so we will not worry about the general theory of Cartier divisors. See [Har77, p. II.6] for a more complete discussion of divisors.

Definition 1.14 (Divisors on a smooth variety). Let X be a smooth variety. A *prime divisor* $Y \subset X$ is an irreducible codimension one closed subvariety. The prime divisors generate a free abelian group $\text{Div}(X)$, whose elements consist of formal linear combinations $\sum_Y n_Y \cdot Y$, where each $n_Y \in \mathbb{Z}$, and only finitely many divisors have a non-zero coefficient.

Whenever $D = \sum_Y n_Y^D \cdot Y$ only has non-negative coefficients, D is said to be an *effective divisor*. More generally, we let $D_+ = \sum_{n_Y^D > 0} n_Y^D \cdot Y$ (only taking the terms in the expansion for D with positive coefficients) and $D_- = -\sum_{n_Y^D < 0} n_Y^D \cdot Y$. Both D_+ and D_- are effective divisors, and $D = D_+ - D_-$.

The local ring $\mathcal{O}_{X,\eta_Y}$ at the generic point η_Y of a prime divisor Y is a discrete valuation ring with fraction field $k(X)$. We let $v_Y : k(X) \rightarrow \mathbb{Z} \cup \{\infty\}$ be the

valuation on $\text{Frac}(\mathcal{O}_{X,\eta_Y})$. To each non-zero element $f \in k(X)$, we associate the *principal divisor* $\text{div}(f) = \sum_Y v_Y(f) \cdot Y$. Because $v_Y(f) \neq 0$ for only finitely many prime divisors, this is a finite sum and $\text{div}(f) \in \text{Div}(X)$.

Two divisors D and D' are said to be *linearly equivalent* if $D = D' + \text{div}(f)$ for some $f \in k(X)$, and we write $D \sim D'$ if they are linearly equivalent. Quotienting $\text{Div}(X)$ by $\sim$, the group of divisors up to linear equivalence is known as the *class group* $\text{Cl}(X)$. If X is affine, then all divisors are principal and $\text{Cl}(X) = 0$.

Definition 1.15 (Divisor—line-bundle correspondence). To each divisor $D \in \text{Div}(X)$, we associate the line bundle $\mathcal{O}_X(D)$. On open affines $U \subset X$, this is given by

$$\mathcal{O}_X(D)|_U = \{f \in k(X) \mid v_Y(f) + D \geq 0 \text{ for every prime divisor } Y \text{ that intersects } U\}.$$

We have $\mathcal{O}_X(\text{div}(f)) \cong \mathcal{O}_X$, and $\mathcal{O}_X(D) \sim \mathcal{O}_X(D')$ if and only if $D \sim D'$. This gives a map $\text{Cl}(X) \rightarrow \text{Pic}(X)$, the group of line bundles on X up to isomorphism. Because $\mathcal{O}_X(-D) \cong \mathcal{O}_X(D)^{-1}$ and $\mathcal{O}_X(D + D') \cong \mathcal{O}_X(D) \otimes_{\mathcal{O}_X} \mathcal{O}_X(D')$, then the map $\text{Cl}(X) \rightarrow \text{Div}(X)$ is a group homomorphism.

Conversely, to any line bundle $\mathcal{L}$ and non-zero section $\sigma \in H^0(X, \mathcal{L})$ we associate the divisor $D = \text{div}(\sigma)_+$ (viewing σ as an element of $k(X)$, and only taking the effective part of the divisor $\text{div}(\sigma)$). There is an isomorphism between $\mathcal{L} \cong \mathcal{O}_X(\text{div}(\sigma)_+)$, and $\text{div}(\sigma)_+ \sim \text{div}(\sigma')_+$ for any non-zero section $\sigma' \in H^0(X, \mathcal{L})$. So the correspondence between divisors and line bundles is a group isomorphism $\text{Cl}(X) \cong \text{Pic}(X)$ [Har77, V, Cor. 6.16].

Remark 1.16. For any prime divisor $D \hookrightarrow X$, the line bundle $\mathcal{O}_X(-D) \subset \mathcal{O}_X$ is the ideal sheaf $\mathcal{I}_D$ that cuts out $D \hookrightarrow X$.

The exact sequence (1.1) then can be rephrased in terms of divisors as

$$0 \longrightarrow \mathcal{O}_X(-D) \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{O}_D \longrightarrow 0. \quad (1.3)$$

We want to generalise the intersection number of two curves in $\mathbb{P}^2$ to divisors on an arbitrary surface. Bezout's theorem tells us that two curves f and g in $\mathbb{P}^2$ with no common components intersect at $(\deg f)(\deg g)$ points counting multiplicity, which formally can be computed as the length of their scheme-theoretic intersection in $\mathbb{P}^2$. This pairing can be extended by linearity to all divisors, giving the pairing $\text{Pic}(\mathbb{P}^2) \times \text{Pic}(\mathbb{P}^2) \rightarrow \mathbb{Z}$ sending a pair of line bundles $\mathcal{L}$ and $\mathcal{M}$ to the product of their degrees.

On a more general surface $\text{Pic}(X)$ is not necessarily isomorphic to $\mathbb{Z}$, but we can generalise this result with the following theorem:

Theorem 1.17 ([Har77, V, Thm. 1.1]). *Let X be a smooth surface. There is a unique symmetric linear pairing $\text{Div}(X) \times \text{Div}(X) \rightarrow \mathbb{Z}$ denoted by $C \cdot D$ for any divisors C and D such that:*

- (i) *if C and D are smooth curves that intersect transversally, then $C \cdot D$ is the number of points on their intersection $C \cap D$.*
- (ii) *if C and C' are linearly equivalent then $C \cdot D = C' \cdot D$.*

This gives it the structure of a symmetric bilinear pairing $\text{Pic}(X) \times \text{Pic}(X) \rightarrow \mathbb{Z}$, using the divisor-line bundle correspondence.

Proof. For a complete proof see [Har77]. We will sketch some of the important constructions involved in this definition. We first consider the case where C and D are distinct prime divisors, and define the *intersection multiplicity* of C and D at a point $p \in C \cap D$ to be the length of the local ring $\mathcal{O}_{p,X}/(f, g)$ where C and D are locally cut out by equations $f = 0$ and $g = 0$ on X . We then let

$$C \cdot D = \sum_{p \in C \cap D} (C \cdot D)_p = h^0(C \cap D, \mathcal{O}_{C \cap D})$$

which is precisely the length of the subscheme $C \cap D$, and extend it by linearity to all other divisors.

The interpretation of $C \cdot D$ as $h^0(C \cap D, \mathcal{O}_{C \cap D})$ allows for an alternative viewpoint from the perspective of cohomology. The exact sequence

$$0 \longrightarrow \mathcal{I}_{C \cap D} \longrightarrow \mathcal{O}_C \longrightarrow \mathcal{O}_{C \cap D} \longrightarrow 0$$

gives an equality $C \cdot D = \chi(\mathcal{O}_C) - \chi(\mathcal{I}_{C \cap D})$, where $\mathcal{I}_{C \cap D}$ is the ideal sheaf cutting out $C \cap D$ inside C and $\chi = h^1 - h^0$ is the *Euler characteristic*. We have $\mathcal{I}_{C \cap D} = \mathcal{I}_D \otimes_{\mathcal{O}_X} \mathcal{O}_C = \mathcal{O}_X(-D) \otimes_{\mathcal{O}_X} \mathcal{O}_C$, and so $C \cdot D$ depends only on the isomorphism class of $\mathcal{O}_X(-D)$, i.e. up to linear equivalence of D , and the same argument by symmetry applies to C .

Now, $\chi(\mathcal{O}_C) - \chi(\mathcal{O}_X(D) \otimes_{\mathcal{O}_X} \mathcal{O}_C)$ is the degree $\deg(\mathcal{O}_X(D) \otimes_{\mathcal{O}_X} \mathcal{O}_C)$ of the line bundle $\mathcal{O}_X(D) \otimes_{\mathcal{O}_X} \mathcal{O}_C$ on the curve C . Degree is additive under the tensor product, and so by defining

$$C \cdot D = \deg(\mathcal{O}_X(-D) \otimes_{\mathcal{O}_X} \mathcal{O}_C)$$

and extending by linearity to arbitrary divisors we get the intersection pairing $\text{Div}(X) \times \text{Div}(X) \rightarrow \mathbb{Z}$. $\square$

Remark 1.18 (On negative self-intersections). Intuitively, C^2 should be the number of points of intersection found by intersecting C with some infinitesimal deformation of itself. But some effective divisors have negative self-intersection, which seems to violate this intuition. This is ok:

- The self-intersection C^2 of an effective divisor C is the degree of the line bundle $\mathcal{O}_X(-C) \otimes_{\mathcal{O}_X} \mathcal{O}_C$. Because $\mathcal{O}_X(-C)$ is (isomorphic to) the ideal sheaf that cuts out C , then $\mathcal{O}_X(-C) \otimes_{\mathcal{O}_X} \mathcal{O}_C \cong \mathcal{N}_{C/X}$ and $C^2 = \deg(\mathcal{N}_{C/X})$.
- We can directly compute this when $C \cong \mathbb{P}^1$. Then $\mathcal{N}_{C/X} \cong \mathcal{O}_C(C^2)$.
- A result we prove later (Theorem 3.19) shows that C can be infinitesimally deformed inside X if and only if $h^0(C, \mathcal{N}_{C/X}) \neq 0$, which occurs if and only if $C^2 \geq 0$. So infinitesimal deformations of a curve with negative self-intersection don't exist.

Definition 1.19. The *canonical divisor* K_X on a variety X is the divisor associated to the canonical line bundle ω_X . The *anticanonical divisor* is $-K_X$. For any prime divisor $D \subset X$, there is the adjunction formula $K_D = (K_X + D)|_D$.

Recall that the anticanonical bundle of $\mathbb{P}^n$ is $\mathcal{O}_{\mathbb{P}^n}(n+1)$. The adjunction formula tells us that the anticanonical bundle of a degree d hypersurface $S \hookrightarrow \mathbb{P}^n$ is $\mathcal{O}_S(K_S) = \mathcal{O}_{\mathbb{P}^n}(n+1-d) \otimes_{\mathcal{O}_{\mathbb{P}^n}} \mathcal{O}_S$. In particular, the anticanonical divisor of a cubic surface $S \hookrightarrow \mathbb{P}^3$ is a general hyperplane section.

We will quickly recall some facts about blowing up points on smooth surfaces. Proofs and a more detailed discussion can be found in Beauville's book [Bea78].

Proposition 1.20. [Bea78, II.1-2] *Let S be a smooth surface and $p \in S$ a closed point. There is then a smooth surface $\hat{S}$ equipped with a morphism $\varepsilon : \hat{S} \rightarrow S$, such that ε is an isomorphism on $\varepsilon^{-1}(S \setminus \{p\})$, and $\varepsilon^{-1}(p) \cong \mathbb{P}^1$. We write $E := \varepsilon^{-1}(p)$, and call it the exceptional divisor of the blowup.*

*Let $C \hookrightarrow S$ be an irreducible curve. The strict transform $\hat{C}$ of C is the closure of $\varepsilon^{-1}(C \setminus \{p\})$ inside $\hat{S}$. Let $m = \text{len}_p(C)$ be the multiplicity of C at p . Then the divisor C pulls back as $\varepsilon^*C = \hat{C} + mE$.*

Proposition 1.21. [Bea78, Prop. II.3] *The following facts are true:*

- (i) *There is an isomorphism $\text{Pic}(S) \oplus \mathbb{Z} \cong \text{Pic}(\hat{S})$ sending $(D, n) \mapsto \varepsilon^*D + nE$.*
- (ii) *Let D_1 and D_2 be divisors on S . Then $(\varepsilon^*D_1) \cdot (\varepsilon^*D_2) = D_1 \cdot D_2$ and $E^2 = -1$ and $E \cdot (\varepsilon^*D_1) = 0$.*
- (iii) *The anticanonical divisor pulls back as $K_{\hat{S}} = \varepsilon^*K_S + E$.*

We will now discuss the more general theory of blowups of a scheme X along some closed subscheme $Z \hookrightarrow X$, or equivalently a coherent ideal sheaf $\mathcal{I} \subset \mathcal{O}_X$. We refer the reader to Hartshorne's book [Har77, p. II.7] for a detailed exposition with proofs. We won't need the general theory of arbitrary ideal sheaves as we are only interested in the blowup of a surface X along some finite subscheme $Z \subset X$, which can always be done in an affine chart $\text{Spec}(A) \subset X$ containing Z .

Definition 1.22 (Blowup along an ideal sheaf). Let A be a k -algebra and $I \subset A$ a finitely generated ideal. The blowup $\text{Bl}_I \text{Spec}(A)$ is defined as

$$\text{Bl}_I \text{Spec}(A) = \text{Proj} \left(A \oplus \bigoplus_{i=1}^{\infty} I^i \right).$$

This is a projective scheme over $\text{Spec}(A)$, and writing down the ideal I in terms of its generators $I = (g_1, \dots, g_n)$ gives a closed immersion $\text{Bl}_I \text{Spec}(A) \hookrightarrow \mathbb{P}_A^n$, where $\text{Bl}_I \text{Spec}(A)$ is identified with the closed subscheme cut out by the ideal of relations between the generators of I .

Let X be a scheme and $\mathcal{I} \subset \mathcal{O}_X$ a coherent ideal sheaf. Let $\text{Spec}(A_\alpha) \subset X$ be an open affine subscheme and $I_\alpha = \mathcal{I}|_{\text{Spec}(A_\alpha)}$. For an open cover $\{\text{Spec}(A_\alpha)\}$ of X , we define $\text{Bl}_{\mathcal{I}} X$ to be the scheme obtained by gluing the $\text{Bl}_{I_\alpha} \text{Spec}(A_\alpha)$ s. If $Z \hookrightarrow X$ is the closed subscheme cut out by $\mathcal{I}$, then we also call $\text{Bl}_{\mathcal{I}} X$ the blowup of X along the subscheme Z , denoted by $\text{Bl}_Z X := \text{Bl}_{\mathcal{I}} X$.

Example 1.23 (Blowing up a point on a surface). Consider the blowup of the origin on $\mathbb{A}^2 = \text{Spec}(k[x, y])$. It is cut out by the ideal (x, y) , and there is a closed immersion

$$\text{Bl}_{(x,y)} \mathbb{A}^2 = \text{Proj} (k[x, y, s, t]_{\bullet} / J) \hookrightarrow \text{Proj}(k[x, y, s, t]_{\bullet}) \cong \mathbb{A}^2 \times \mathbb{P}^1$$

where the grading is only in the s, t terms (which are each of degree one), and $J = (ys - xt)$ is the homogeneous ideal of relations between the generators s (identified with x) and t (identified with y). This is exactly the blowup of $\mathbb{A}^2$ at the origin in the sense of Proposition 1.20, and blowing up a point on a smooth surface is always locally of this form.

Example 1.24 (The blowup can be more singular). Now consider the blowup of $\mathbb{A}^2$ along the ideal $(x^n, y) \subset k[x, y]$ for some integer $n > 1$. This is the closed subscheme of $\mathbb{A}^2 \times \mathbb{P}^1$ cut out by the homogeneous ideal $(ys - x^n t)$. In the affine chart where t doesn't vanish, we have coordinates (x, y, u) where $u = s/t$, and the ideal is $(yu - x^n)$. But if $n > 1$, then all the partial derivatives of this equation vanish, so $\text{Bl}_{(x^n,y)} \mathbb{A}^2$ is singular at the point $((0, 0), [0 : 1])$.

Remark 1.25. The blowup of a scheme along an ideal sheaf is not unique. For example, $\mathrm{Bl}_{\mathcal{I}} X \cong \mathrm{Bl}_{\mathcal{I}^2} X$ as X -schemes because Proj is invariant under taking multiples of the grading. In particular, $\mathrm{Bl}_{(x,y)} \mathbb{A}^2 \cong \mathrm{Bl}_{(x^2,xy,y^2)} \mathbb{A}^2$, so the blowup of a non-reduced point can be isomorphic to the blowup of a reduced point.

We will use the following two important properties of blowups:

Proposition 1.26 (Universal property of blowing up). [Har77, II, Prop. 7.14] *Let X be a scheme, $\mathcal{I} \subset \mathcal{O}_X$ a coherent sheaf of ideals and $\pi : \mathrm{Bl}_{\mathcal{I}} X \rightarrow X$ the blowup along $\mathcal{I}$. If $f : Y \rightarrow X$ is any morphism where the inverse image of the ideal sheaf $f^{-1}\mathcal{I} \cdot \mathcal{O}_Y$ is an invertible sheaf, then f factors uniquely through π .*

Corollary 1.27. [Har77, II, Cor. 7.15] *Let $f : Y \rightarrow X$ be a morphism of schemes, and $\mathcal{I} \subset \mathcal{O}_X$ a coherent sheaf of ideals on X . There is then a unique morphism from the blowup $\mathrm{Bl}_{f^{-1}(\mathcal{I}) \cdot \mathcal{O}_Y} Y$, the blowup of Y along the inverse image of the ideal sheaf $\mathcal{I}$, such that the following diagram commutes*

$$\begin{array}{ccc} \mathrm{Bl}_{f^{-1}(\mathcal{I}) \cdot \mathcal{O}_Y} Y & \xrightarrow{\exists!} & \mathrm{Bl}_{\mathcal{I}} X \\ \downarrow & & \downarrow \\ Y & \xrightarrow{\quad} & X \end{array}$$

and if f is a closed immersion, then so is the morphism $\mathrm{Bl}_{f^{-1}(\mathcal{I}) \cdot \mathcal{O}_Y} Y \rightarrow \mathrm{Bl}_{\mathcal{I}} X$.

Definition 1.28. The strict transform of a closed subscheme $f : Y \hookrightarrow X$ is the closed subscheme $\mathrm{Bl}_{f^{-1}(\mathcal{I}) \cdot \mathcal{O}_X} Y \hookrightarrow \mathrm{Bl}_{\mathcal{I}} X$.

This is a generalisation of the strict transform in the sense of Proposition 1.20, where in the case of blowing up a maximal ideal the strict transform $\hat{C}$ is a reduced curve.

Remark 1.29. The pullback $f^*\mathcal{I}$ is not the same as $f^{-1}(\mathcal{I}) \cdot \mathcal{O}_X$ — $f^*\mathcal{I}$ may not be an ideal sheaf, but $f^{-1}(\mathcal{I}) \cdot \mathcal{O}_X$ is the ideal sheaf generated by $f^*\mathcal{I}$. We will write $f^{-1}\mathcal{I}$ for $f^{-1}(\mathcal{I}) \cdot \mathcal{O}_X$ for brevity.

Proposition 1.30 (Morphisms to projective space). *Let X be a projective scheme. A morphism $f : X \rightarrow \mathbb{P}^n$ sends points $x \mapsto [\sigma_0(x) : \cdots : \sigma_n(x)]$. This gives a bundle $\mathcal{L} = f^*\mathcal{O}_{\mathbb{P}^n}(1)$, and the $n+1$ functions σ_i are global sections $\sigma_i \in H^0(X, \mathcal{L})$, with the additional condition that for every $x \in X$, at least one σ_i doesn't vanish at x .*

Conversely, fix a scheme X , a line bundle $\mathcal{L}$, and $n+1$ global sections $\sigma_i \in H^0(X, \mathcal{L})$. Let $U \subset X$ be the open subscheme where at least one of the σ_i s doesn't vanish. There is a morphism

$$f : U \longrightarrow \mathbb{P}^n, \quad x \longmapsto [\sigma_0(x) : \cdots : \sigma_n(x)]$$

and $\mathcal{L} = f^*\mathcal{O}_{\mathbb{P}^n}(1)$ where f is defined.

Definition 1.31 (Linear system). Let X be a connected scheme and $\mathcal{L} \subset \mathcal{O}_X$ a line bundle. Let $V \subset H^0(X, \mathcal{L})$ be a non-zero vector subspace. The pair $(\mathcal{L}, V)$ is called a *linear system*, or *linear series*, on X . We identify the linear system with the closed points of the projective space $\mathbb{P}V^\vee$. The *dimension* of a linear system is $\dim V - 1$, the dimension of the projective space $\mathbb{P}V^\vee$.

When $V = H^0(X, \mathcal{L})$, we call it a *complete linear system*, and we write $(\mathcal{L}, V) = |\mathcal{L}|$. When X is a smooth variety and D is a divisor, we also write $|D| = |\mathcal{O}_X(D)|$ for the linear system of effective divisors linearly equivalent to D . When $\mathcal{F} \subset \mathcal{L}$ is a subsheaf, we will also write $|\mathcal{F}| := (\mathcal{L}, H^0(X, \mathcal{F}))$.

The subscheme of X on which all of the elements of V vanishes is called the *base-locus* of $(\mathcal{L}, V)$. When the base-locus is empty, the linear system is said to be *basepoint-free*. A basepoint-free linear system induces a morphism $f : X \rightarrow \mathbb{P}V$ as in Proposition 1.30, and more generally a linear system induces a rational map $f : X \dashrightarrow \mathbb{P}V$, defined on the complement of the base-locus (we only assumed X was connected so that this map is rational, for convenience).

If the complete linear system $|\mathcal{L}|$ is basepoint-free and the induced morphism is an immersion, then $\mathcal{L}$ is said to be *very ample*. When X is separated, a line bundle is *ample* if some positive power of it is very ample. A line bundle on a variety X is very ample if and only if it is *basepoint-free* and separates *tangent vectors*, in the sense that for every point $x \in X$ and two distinct tangent vectors $v, w \in T_x X$, there exists a global section $\sigma \in H^0(X, \mathcal{L})$ that passes through x with tangent v but not w .

If $g : Y \rightarrow X$ is a birational morphism, then a linear system $(\mathcal{L}, V)$ pulls back to the linear system $g^*(\mathcal{L}, V) := (g^*\mathcal{L}, g^*(V))$, where $g^*(V) \subset H^0(Y, g^*\mathcal{L})$ is the image of V under the natural map $g^* : H^0(X, \mathcal{L}) \rightarrow H^0(Y, g^*\mathcal{L})$ (note this map is injective). Because the pullback is just precomposition, it is immediate that the rational map $Y \dashrightarrow \mathbb{P}(g^*(V))$ induced by the linear system $g^*(\mathcal{L}, V)$ is the composition $f \circ g$, where f is defined. Blowing up the base-locus of a linear system resolves the basepoints (because the ideal sheaf that cuts out the base-locus is invertible after the blowup, this is generated by the pullback $f^*(V)$), which we outline in Example 1.34 and use throughout this thesis.

Example 1.32 (Degree d polynomials). The complete linear system $|\mathcal{O}_{\mathbb{P}^n}(d)|$ is the $\binom{n+d}{n} - 1$ -dimensional linear system of degree d hypersurfaces in $\mathbb{P}^n$ (when $d > 0$, if $d < 0$ then $h^0(\mathbb{P}^n, \mathcal{O}_{\mathbb{P}^n}(d)) = 0$ so the linear system is empty).

Definition 1.33. A length 6 subscheme $\Gamma \hookrightarrow \mathbb{P}^2$ is said to be in *general position with respect to lines* (resp. conics, resp. plane curves) if no length three subscheme

of Γ lies on a line (resp. Γ is not contained in a conic, resp. both of these conditions hold). We also say *linearly general position*.

Example 1.34 (Resolving basepoints). Let $\Gamma = \{p_1, \dots, p_r\} \subset \mathbb{P}^2$ be $r \leq 9$ points in general position with respect to plane curves. Consider the linear system $\mathfrak{d}$ of cubics through Γ . It has basepoints along Γ , so we pass to the blowup $\varepsilon : \text{Bl}_\Gamma \mathbb{P}^2 \rightarrow \mathbb{P}^2$. We write $S = \text{Bl}_\Gamma \mathbb{P}^2$, it is a smooth surface. We let $E_i = \varepsilon^{-1}(p_i)$ be the exceptional divisors, these are (-1) -curves.

Each divisor $D \in \mathfrak{d}$ is a cubic through Γ . The resulting linear system $\varepsilon^*\mathfrak{d}$ of divisors is the linear system of strict transforms $\varepsilon^{-1}(D)$ for each $D \in \mathfrak{d}$ (this is true whenever ε is a blowup, for example). The system $\varepsilon^*\mathfrak{d}$ is the complete anticanonical system $|-K_S| = |\varepsilon^*D - E_1 - \dots - E_r|$. It is easy to check that $|-K_S|$ is basepoint-free (see [Bea78, Prop. IV.9]), and hence induces a morphism to $\mathbb{P}^3$.

Example 1.35 (The twisted cubic). Let $X = \mathbb{P}^1$, and consider the complete linear system $|\mathcal{O}_{\mathbb{P}^1}(3)|$ of cubics. This induces the map $f : \mathbb{P}^1 \rightarrow \mathbb{P}^3$ sending $[s : t] \mapsto [s^3 : s^2t : st^2 : t^3]$. The image of f is a curve $C \hookrightarrow \mathbb{P}^3$, called a *twisted cubic*.

1.3 The geometry of smooth cubic surfaces

We will now briefly discuss some well-known facts about cubic surfaces.

Notation 1.36 (Projective spans). Let $x, y \in \mathbb{P}^n$ be two points. We write $\langle x, y \rangle \subset \mathbb{P}^n$ to denote the projective line between them. If X, Y are subsets of $\mathbb{P}^n$, then we write $\langle X, Y \rangle$ to be the union of all the lines $\langle x, y \rangle$, ranging over $(x, y) \in X \times Y$.

Proposition 1.37 (The blowup of 6 general points in $\mathbb{P}^2$ is a smooth cubic surface). [Bea78, Prop. IV.9] *Let $\Gamma = \{p_1, \dots, p_6\} \subset \mathbb{P}^2$ be 6 points in general position with respect to plane curves. Then the blowup $S = \text{Bl}_\Gamma \mathbb{P}^2$ is isomorphic to a smooth cubic surface, embedded by the (pullback of the) linear system of cubics through Γ .*

Proof. Recall Example 1.34. Let $\mathcal{I}$ be the ideal sheaf that cuts out Γ . The linear system of cubics through Γ is $|\mathcal{I}(3)|$. And $|\varepsilon^{-1}\mathcal{I}(3)| = \varepsilon^*|\mathcal{I}(3)|$. We just need to check that $|\varepsilon^{-1}\mathcal{I}(3)|$ separates points and tangent vectors.

This amounts to checking different combinations of lines and conics separate points and tangents, the complete proof can be found in [Bea78]. $\square$

We will postpone the proof of the following theorem to Chapter 3. Remember this theorem, we will use it a lot (and it is also quite famous!).

Theorem 1.38. [Sal49] *All smooth cubic surfaces contain 27 distinct lines. In particular, they all contain a line.*

The pairwise intersection graph of the 27 lines on any smooth cubic surface is the same.

Remark 1.39 (Eckardt points). While the intersection graph of all smooth cubics are identical, the behaviour of multiple intersections depends on the surface. A point where three lines on a cubic surface intersect is called an *Eckardt point*. Most smooth cubics have no Eckardt points, but there can be as many as 18 (which occurs on the Fermat cubic, cut out by $x^3 + y^3 + z^3 + w^3 = 0$).

Proposition 1.40. [Bea78, Thm. IV.13] *Every smooth cubic surface is isomorphic to a blowup of $\mathbb{P}^2$ at 6 distinct points.*

Proof. We provide a sketch of Beauville's proof. On any smooth cubic surface S there exist two skew lines L and L' . Any point $s \in S \setminus (L \cup L')$ is contained in the line $\langle p, p' \rangle$ for a unique pair of points $(p, p') \in L \times L'$. This extends to a birational morphism $\psi : S \rightarrow L \times L' \cong \mathbb{P}^1 \times \mathbb{P}^1$.

Then ψ is a composition of blowups and blowdowns. There are five lines that intersect both L and L' (any smooth cubic surface has the same pairwise intersection graph), and these lines are of the form $L_i = \langle p_i, p'_i \rangle$ for some set of five points $Z = \{(p_i, p'_i)\}_{i=1}^5 \in L \times L'$. And ψ is an isomorphism everywhere else, so $\psi : S \rightarrow \mathbb{P}^1 \times \mathbb{P}^1$ is the blowup of $\mathbb{P}^1 \times \mathbb{P}^1$ at Z . And the blowup of $\mathbb{P}^1 \times \mathbb{P}^1$ at one point is isomorphic to the blowup of $\mathbb{P}^2$ at two points, so the blowup of $\mathbb{P}^1 \times \mathbb{P}^1$ at five points is isomorphic to $\mathbb{P}^2$ blown up at 6 points, i.e. $S \cong \text{Bl}_\Gamma \mathbb{P}^2$ for some configuration of 6 points $\Gamma \subset \mathbb{P}^2$. $\square$

Corollary 1.41. *Given any configuration of six skew lines on smooth cubic surface S , there is a birational morphism $\pi : S \rightarrow \mathbb{P}^2$ contracting these six lines.*

Proof. Since S is smooth, it is the blowup $\varepsilon : S \rightarrow \mathbb{P}^2$ of 6 points in $\mathbb{P}^2$. After some computation, it can be seen that there is a sequence of quadratic transformations $\mathbb{P}^2 \dashrightarrow \mathbb{P}^2$ that factors through ε^{-1} (where it is defined) and extends to a morphism $\pi : S \rightarrow \mathbb{P}^2$ that blows down any set of six (-1) -curves on S . The full computation can be found in [Har77, IV, Prop. 4.10]. $\square$

Remark 1.42. A birational morphism $\pi : S \rightarrow \mathbb{P}^2$ is induced by the line bundle $\pi^* \mathcal{O}_{\mathbb{P}^2}(1)$ on S . So $\pi^* \mathcal{O}_{\mathbb{P}^2}(1) = \mathcal{O}_S(\pi^{-1}(L))$ for a general line $L \subset \mathbb{P}^2$ that doesn't pass through the point configuration Γ . The map $\pi^{-1} : \mathbb{P}^2 \setminus \Gamma \rightarrow S$ is induced by

the linear system $|\mathcal{I}_\Gamma(3)|$, on a general line $L \cong \mathbb{P}^1$ this restricts to the complete linear system $|\mathcal{O}_L(3)|$, so $\pi^{-1}(L)$ is a twisted cubic (Example 1.35). A birational morphism $S \rightarrow \mathbb{P}^2$ is uniquely determined by a choice of twisted cubic on S .

Notation 1.43. A configuration of six skew lines on a smooth cubic surface is called a *six* (by skew we mean pairwise disjoint). This is a standard name used throughout the literature, to disambiguate it from the number 6 we will only use the word “six” when we are either referring to a set of six lines on a smooth cubic surface.

There are 72 sixes on any smooth cubic surface. Fix a six $s = \{M_1, \dots, M_6\}$ on S . Let N_i be the line that intersects M_j for every $j \neq i$ — this exists and is unique. We let $\bar{s} = \{N_1, \dots, N_6\}$, this is also six (the N_i s are mutually skew). We call the pair $\{s, \bar{s}\}$ a *double-six*, and $\bar{s}$ the *dual six* to s . Note that the ordering of lines is only important for the natural pairing between a six and its dual (the bijection sending $M_i \mapsto N_i$).

The other fifteen lines on S intersect exactly two lines in s . We write $L_{ij} = L_{ji}$ to be the line that intersects M_i and M_j , it is also the line that intersects N_i and N_j . The intersection numbers of the lines on S are the following (note Beauville’s definition [Bea78, Ex. 4.12] has a few typos):

$$\begin{aligned} M_i^2 = N_i^2 = L_{ij}^2 &= -1, & M_i \cdot N_i &= 0, & M_i \cdot L_{ij} &= N_i \cdot L_{ij} = 1 \\ M_i \cdot M_j &= N_i \cdot N_j = 0, & M_i \cdot N_j &= 1 & \text{if } i \neq j \\ L_{ij} \cdot M_k &= L_{ij} \cdot N_k = 0 & \text{if } k \notin \{i, j\}, & L_{ij} \cdot L_{lk} &= 0 & \text{if } \{i, j\} \neq \{k, l\}. \end{aligned}$$

There other 35 double-sixes on S come in the pairs

$$\begin{aligned} s_{abc} &= \{M_a, M_b, M_c, L_{ef}, L_{df}, L_{de}\}, & \overline{s_{abc}} &= \{L_{bc}, L_{ac}, L_{ab}, N_d, N_e, N_f\} \\ s_{ab} &= \{M_a, N_a, L_{bc}, L_{bd}, L_{be}, L_{bf}\}, & \overline{s_{ab}} &= s_{ba}, \end{aligned}$$

where a, b, c, d, e, f are distinct elements of $\{1, \dots, 6\}$. There are $20 = \binom{6}{3}$ choices of $\{s_{abc}, \overline{s_{abc}}\}$ and $15 = \binom{6}{2}$ choices of $\{s_{ab}, \overline{s_{ab}}\} = \{\overline{s_{ba}}, s_{ba}\}$. The role of any six on a smooth cubic surface is symmetric.

Remark 1.44. Let $S \hookrightarrow \mathbb{P}^3$ and $S' \hookrightarrow \mathbb{P}^3$ be smooth cubic surfaces. Any isomorphism $f : S \rightarrow S'$ must preserve the canonical divisor, so induces a linear isomorphism $f^* : H^0(S', \mathcal{O}_{S'}(-K_{S'})) \rightarrow H^0(S, \mathcal{O}_S(-K_S))$. And because cubic surfaces are anticanonically embedded, then this extends to an isomorphism of the ambient projective space. So any isomorphism of cubic surfaces is the restriction of a projective automorphism in $\text{PGL}(4)$.

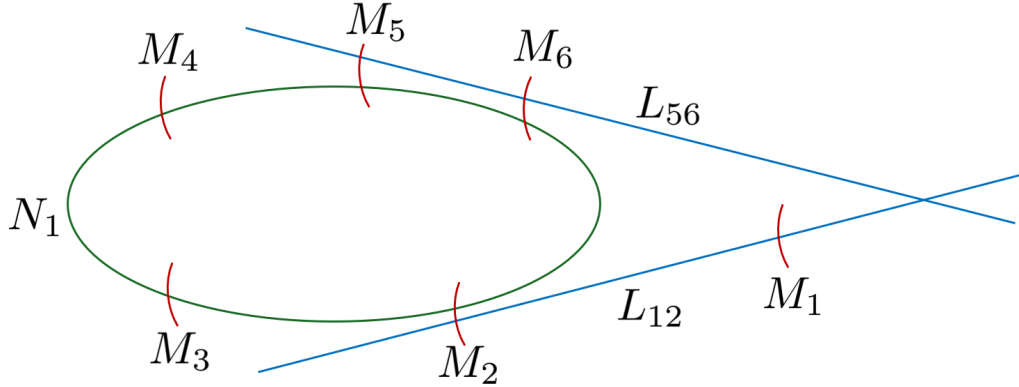


Figure 1.1: Some of the lines on smooth cubic surface S , represented as a blowup of 6 points in $\mathbb{P}^2$. We label the lines so that the morphism $\pi : S \rightarrow \mathbb{P}^2$ contracts the six $\{M_1, \dots, M_6\}$, these are the exceptional curves. Let $p_i = \pi(M_i)$. The line L_{12} intersects M_1 and M_2 , L_{12} is the strict transform $\pi^{-1}(l_{12})$ of the line l_{12} through p_1 and p_2 . The line N_1 intersects every M_i , except for M_1 . It is the strict transform of the conic Q_i through all the p_i s except p_1 . The lines L_{12} and L_{56} intersect (as pictured), but the lines L_{12} and L_{13} do not, because l_{12} and l_{13} have different tangent vectors at p_1 .

As a result, any isomorphism of smooth cubic surfaces must preserve the line configuration. And if an automorphism of a smooth cubic surface fixes the lines, then it is trivial. Note that a general smooth cubic surface has no non-trivial automorphisms.

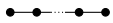
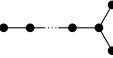
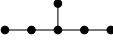
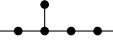
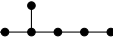
1.4 Resolving singularities on cubic surfaces

We follow Reid's exposition on ADE singularities [Rei87].

Definition 1.45. Let S be a singular surface. A *resolution of singularities* for S is a smooth surface $\tilde{S}$ equipped with a birational morphism $r : \tilde{S} \rightarrow S$.

The resolution $\tilde{S}$ is said to be *minimal* if the map r doesn't factor through any other non-isomorphic resolution $r' : \tilde{S}' \rightarrow S$.

Definition 1.46. The following table labels certain polynomials $f \in k[[x, y, z]]$ and graphs of n vertices according to types A_n and D_n and E_n .

Label	Equation	Diagram
A_n	$x^2 + y^2 + z^{n+1} \quad n \geq 1$	
D_n	$x^2 + y^2 z + z^{n-1} \quad n \geq 4$	
E_6	$x^2 + y^3 + z^4$	
E_7	$x^2 + y^3 + yz^3$	
E_8	$x^2 + y^3 + z^5$	

The graphs are called *Dynkin diagrams*, and there is a rich connection between these diagrams and the classification of semisimple Lie algebras, among other interesting related objects. We will only discuss the connection with singularities on surfaces.

Definition 1.47. Let S be a surface with a singularity at a point $s \in S$. Let $\mathfrak{m} \subset \mathcal{O}_{S,s}$ be the maximal ideal of the local ring. We let $\widehat{\mathcal{O}}_{S,s}$ be the $\mathfrak{m}$ -adic completion of the local ring. If there exists an isomorphism $\widehat{\mathcal{O}}_{S,s} \cong k[[x, y, z]]/(f)$ where f is one of the above equations, then s is a singularity of type A_n or D_n or E_n (depending on the type of f). We call singularities of this form *ADE singularities*.

Example 1.48 (A_n singularity when blowing up a curvilinear point). Recall Example 1.24, which shows the exceptional fibre of the blowup of $\text{Bl}_{(x^{n+1}, y)} \mathbb{A}^2$ is cut out by the equation $ys - x^{n+1}t$ inside $\mathbb{A}^2 \times \mathbb{P}^1$. In the chart where $s = 1$ this is smooth, and in the chart where $t = 1$ this is the equation $ys - x^{n+1}$. And $ys = \frac{1}{4}(y+s)^2 - \frac{1}{4}(y-s)^2$ so after a change of variables this is the equation of an A_n singularity at $x = y = s = 0$. Any length $n + 1$ curvilinear point Γ is locally isomorphic to (x^{n+1}, y) (the ideal cutting out Γ is of the form $(x^{n+1}f, yg)$ where f and g are polynomials that don't vanish at $(0, 0)$, and f and g are units in the local ring $\mathcal{O}_{\mathbb{A}^2, (0,0)}$).

Proposition 1.49. Let S be a surface with only ADE singularities. Then there exists a smooth surface $\widetilde{S}$ and a birational morphism $r : \widetilde{S} \rightarrow S$ such that:

- (i) r is the composition of n blowups of points.
- (ii) The exceptional divisor E_i above a singular point s_i is the union of n distinct (-2) -curves, with intersection graph of type A_n or D_n or E_n corresponding to the singularity type.
- (iii) The restriction of r to $\widetilde{S} \setminus \bigcup_{i=1}^N E_i$ is an isomorphism onto its image $S \setminus \{s_i\}_{i=1}^N$, the nonsingular locus of S .

(iv) $r : \tilde{S} \rightarrow S$ is a minimal resolution of singularities of S .

Proof. We provide a sketch. In an affine chart $\text{Spec}(R)$ for a surface S around an order n ADE singularity s , we blow up s (this is blowing up the ideal sheaf $\mathfrak{m}_s \mathcal{O}_{\text{Spec}(R)}$, where $\mathfrak{m}_s$ is the maximal ideal corresponding to this point). We base change to the $\mathfrak{m}_s$ -adic completion of the local ring $R_{\mathfrak{m}_s}$, by the map of rings $\phi : R \rightarrow \hat{R}_{\mathfrak{m}_s} \cong k[[x, y, z]]/(f)$ (the composition of localising at $\mathfrak{m}_s$ and then the completion map), where f is the ADE equation describing s .

The base change of $\text{Bl}_s \text{Spec}(R)$ by $\text{Spec}(k[[x, y, z]]/(f)) \rightarrow \text{Spec}(R)$ is isomorphic to $\text{Bl}_{(x,y,z)} \text{Spec}(k[[x, y, z]]/(f))$. And this is isomorphic to the subscheme of $\text{Bl}_{(x,y,z)} \text{Spec}(k[x, y, z])$ cut out by the strict transform of f . The strict transform of f is less singular (it is an ADE equation of order $n - 1$), which decreases the order of the singularity by one. For example, if $f = x^2 + y^2 + z^{n+1}$ cuts out an A_n singularity then $\hat{f} \equiv z(x^2 + y^2 + z^n)$ (locally, after some change of coordinates), and so cuts out an A_{n-1} singularity on $\text{Bl}_s \text{Spec}(R)$. We repeat this process n times, and resolve the singularity by blowups. Properties (i), (iii), (iv) are then immediate, and property (ii) is easy to manually check. $\square$

Example 1.50 (Non-curvilinear blowup is not normal). A non-curvilinear point on a smooth surface is locally cut out by the ideal $(xf, yg) \in k[x, y]$, where x and y are local coordinates and f and g both vanish at $(0, 0)$. The blowup $\text{Bl}_{(xf, yg)} \mathbb{A}^2$ is cut out by the equation $xf(x, y) + tyg(x, y) = 0$, the entire exceptional divisor is singular (and this surface is not normal).

Remark 1.51. An ADE singularity on a surface S is also known as a *canonical singularity*. They satisfy the property that under the resolution $r : \tilde{S} \rightarrow S$ of singularities for S given in Proposition 1.49, the anticanonical divisor pulls back as $K_{\tilde{S}} = f^* K_S$.

Recall that any smooth cubic surface is anticanonically embedded. Normal cubic surfaces with finitely many lines satisfy the same property, and the minimal resolution of singularities of a normal cubic surface with finitely many lines is isomorphic to $\mathbb{P}^2$ blown up 6 times. These can be viewed as degenerations of blowing up 6 points in $\mathbb{P}^2$ for a smooth cubic surface, which we explore in detail throughout this thesis.

There is a very neat classification of normal cubic surfaces with finitely many lines, by understanding intersections of conics and cubics in $\mathbb{P}^2$. This is the *normal form*, first discussed by Bruce and Wall [BW79], and Sakamaki has a more recent exposition that explicitly computes the resolution of the ADE singularities using this normal form [Sak10]. We sketch it here as an example, but with an important

warning: the blowup of a length six subscheme contained in a conic *never* comes from a determinantal representation of a cubic surface. We will prove this later (Proposition 2.58). The failure of the normal form to be determinantal is central to the latter part of this thesis, where we try to understand the moduli space of determinantal cubic surfaces.

Theorem 1.52 (Normal form). [BW79; Sak10] *The equation defining any normal singular cubic surface $S \hookrightarrow \mathbb{P}^3 = \text{Proj}(k[x, y, z, w]_\bullet)$ with finitely many lines can be written as $F = wf_2 - f_3 = 0$, where f_2 and f_3 are a (possibly non-reduced or reducible) conic C_2 and cubic C_3 on $\mathbb{P}^2 = \text{Proj}(k[x, y, z]_\bullet)$, respectively, with no common components.*

The linear system of cubics through $C_2 \cap C_3$ induces a birational morphism $\text{Bl}_{C_2 \cap C_3} \mathbb{P}^2 \rightarrow S \hookrightarrow \mathbb{P}^3$, and blowing up $C_2 \cap C_3$ 6 times gives a minimal resolution of singularities of S .

Proof. We sketch the classification (see [Sak10] for a detailed explanation). We choose coordinates on $\mathbb{P}^3$ so that the singularity is at $[0 : 0 : 0 : 1]$. Then $F = wf_2 - f_3$ for some forms f_2 and f_3 . The normal cubic surfaces are classified by the various possibilities for f_2 and f_3 .

The ideal I that cuts out $C_2 \cap C_3$ is generated by (f_2, f_3) . Its degree 3 terms are xf_2, yf_2, zf_2, f_3 . Because f_2 and f_3 are homogeneous, then it is easy to verify (by direct substitution) that

$$F(xf_2(x, y, z), yf_2(x, y, z), zf_2(x, y, z), f_3((x, y, z))) = (f_3f_2^3)(x, y, z) - (f_3f_2^3)(x, y, z) = 0$$

so the image of the rational map $[x : y : z] \mapsto [xf_2 : yf_2 : zf_2 : f_3]$ is contained in the surface S . And the basepoints of this map can be resolved by a sequence of 6 blowups, where we read off the (-2) -curves to understand the singularity types, and the image of the (-1) -curves are the lines on S . $\square$

Corollary 1.53. *Any normal singular cubic surface S with finitely many lines has at most four singularities. These are ADE singularities, of types $A_1, \dots, A_5, D_4, D_5, E_6$. The total multiplicity of the singularities on S is at most 6.*

Proof. This directly follows from the classification. $\square$

Example 1.54 (A_1 normal form). An A_1 cubic surface (a cubic surface with one A_1 singularity and no other singularities) has normal form $wf_2 - f_3$, where f_2 is an irreducible conic and f_3 is a cubic such that the intersection $C_2 \cap C_3$ is reduced. Then the blowup $\text{Bl}_{C_2 \cap C_3} \mathbb{P}^2$ is a smooth surface. It has a (-2) -curve — the strict transform of C_2 . And the (-1) -curves are the 6 exceptional divisors, and the 15 lines through two points in $C_2 \cap C_3$.

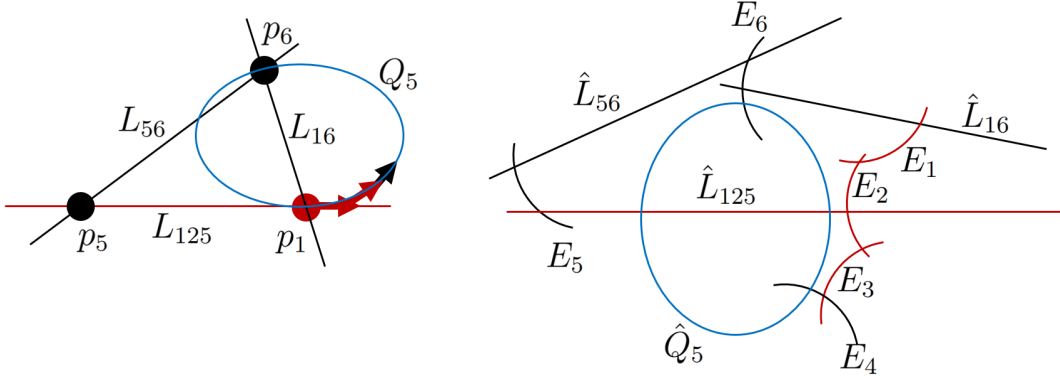


Figure 1.2: $(-n)$ -curves on $\tilde{S}$, from Example 1.55. Note that $\hat{L}_{125}$ and $\hat{Q}_5$ don't intersect, they just look like they do because it is impossible to draw this as a planar graph. We have drawn (-2) -curves in red, (-1) -curves in black, and the conic Q_5 is drawn in blue for clarity, it is also a (-1) -curve.

Here is an example of a resolution of singularities of a D_4 cubic surface. Note this is not one of Sakamaki's resolutions — the subscheme Γ is not contained in a conic. We choose this resolution because it induces a determinantal representation, see Example 2.62.

Example 1.55 (A resolution of singularities for a D_4 cubic surface). Consider a conic Q_6 and a line L_{125} that have a double intersection at a point p_1 . Let p_5 be another point on the line L_{125} , p_6 another point on the conic Q_5 , and consider the length six scheme $\Gamma \subset \mathbb{P}^2$, where Γ has a length 4 point at p_1 contained in the conic Q_5 , and length 1 points at p_5, p_6 . The blowup $\text{Bl}_\Gamma \mathbb{P}^2$ has an A_3 singularity on the exceptional fibre above p_1 , and resolving it by a sequence of 6 blowups of $\mathbb{P}^2$ gives a smooth surface $\tilde{S}$.

The (-1) -curves on $\tilde{S}$ are $E_4, E_5, E_6, \hat{L}_{56}, \hat{L}_{16}, \hat{Q}_5$. The (-2) -curves are $\hat{L}_{125}$, and E_1, E_2, E_3 . The intersections of these curves can be seen in Figure 1.2. The linear system of cubics $|\mathcal{I}_\Gamma(3)|$ induces a morphism $\tilde{S} \rightarrow \mathbb{P}^3$, its image is a cubic surface with a D_4 singularity — the intersection graph of the (-2) -curves on $\tilde{S}$ is a D_4 Dynkin diagram.

Remark 1.56. There are exactly two non-normal cubic surfaces (up to projective equivalence) with finitely many lines. In coordinates, these are also cut out by the equations $wf_2 - f_3 = 0$, where $f_3 = x^2z + y^3$ and f_2 is either y^2 or xy [LPS11, Thm. 3.1]. The intersection $f_2 \cap f_3$ is not curvilinear, so the blowup $\text{Bl}_{(f_2, f_3)} \mathbb{P}^2$ has infinitely many singularities, as expected.

Chapter 2

Determinantal cubic surfaces and Gale-duality

Writing a given cubic polynomial $f \in |\mathcal{O}_{\mathbb{P}^3}(3)|$ as the determinant of a matrix seems like an initially intractable problem — manually solving for the coefficients involves solving 20 simultaneous cubic equations in 36 variables (very hard!). Like many problems about cubic surfaces, there is a very elegant solution that emerges from the geometry of the surface. The exposition in this chapter has its roots in linear algebra and the classical theory of varieties, which we rephrase in the modern language of sheaves and schemes for its clarity and generality.

The key to fully understanding the determinantal representations of a given cubic surface S lies in the observation that a determinantal representation of a smooth cubic surface is uniquely determined by a set of 6 points in $\mathbb{P}^2$. To properly describe this we need to look at several associated objects to a given determinantal equation, we quickly introduce the main characters:

Notation 2.1. Let U and V and W be three fixed vector spaces of dimensions $\dim U = 4, \dim V = 3, \dim W = 3$, respectively. The roles of V and W are always interchangeable. We call an element $\varphi \in U^\vee \otimes V^\vee \otimes W^\vee$ a $4 \times 3 \times 3$ tensor.

A $4 \times 3 \times 3$ tensor φ can be viewed as a matrix in three canonical ways: as a 3×3 matrix of linear forms in $k[U^\vee]_\bullet$, or a 4×3 matrix of linear forms in either $k[V^\vee]_\bullet$ or $k[W^\vee]_\bullet$, which are all partial adjoints of φ . We denote these by

$$\varphi_U \in U^\vee \otimes \operatorname{Hom}(V, W^\vee), \quad \varphi_V \in V^\vee \otimes \operatorname{Hom}(W, U^\vee), \quad \varphi_W \in W^\vee \otimes \operatorname{Hom}(V, U^\vee).$$

We also view each φ_X as a map of $k[X^\vee]_\bullet$ -modules, because $X^\vee$ is the degree one component of the graded ring $k[X^\vee]_\bullet$. Fixing a basis for $V \cong k^3$ and $W^\vee \cong k^3$, the determinant cubic $\det(\varphi_U)$ is a homogeneous cubic equation in $k[U^\vee]_\bullet$. The

determinant cubic cuts out a cubic surface, which we denote by $S_\varphi \hookrightarrow \mathbb{P}U$.

Similarly, in a basis for W and $U^\vee$, the four maximal 3×3 minors of φ_V generate an ideal $I_V^\varphi \subset k[V^\vee]_\bullet$. We let $\mathcal{I}_V^\varphi \subset \mathcal{O}_{\mathbb{P}V}$ be the corresponding ideal sheaf, and $\Gamma_V^\varphi \hookrightarrow \mathbb{P}V$ the subscheme cut out by $\mathcal{I}_V^\varphi$. We define Γ_W^φ and I_W^φ and $\mathcal{I}_W^\varphi$ identically.

For each cubic surface $S \hookrightarrow \mathbb{P}U$, a tensor φ is said to be a *determinantal representation* of S if $\det(\varphi_U)$ is the cubic polynomial that cuts out S . Two tensors φ and φ' are said to be *equivalent* if they are isomorphic as trilinear maps (related by an element of $\mathrm{GL}(U) \times \mathrm{GL}(V) \times \mathrm{GL}(W)$).

We provide a glossary of important notation in Appendix D.

The main theorem we will prove is the following:

Theorem. *Let $\varphi \in \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ be a $4 \times 3 \times 3$ tensor, such that the associated cubic surface S_φ is normal with finitely many lines.*

The subschemes Γ_V^φ and Γ_W^φ are length 6, curvilinear, and Gale-dual. The matrix φ_V is the presentation matrix of a free resolution of the ideal $\mathcal{I}_V^\varphi$. After blowing up along $\mathcal{I}_V^\varphi$, the linear system of cubics $|\mathcal{I}_V^\varphi(3)|$ through Γ_V^φ induces a birational morphism

$$\tilde{m}_V : \mathrm{Bl}_{\mathcal{I}_V^\varphi} \mathbb{P}V \rightarrow S_\varphi \subset \mathbb{P}U.$$

Conversely, every determinantal representation of a normal cubic surface S with finitely many lines is uniquely determined by the point configuration Γ_V^φ , or equivalently a possibly degenerate six on S . There are 72 inequivalent determinantal representations for a general smooth cubic surface S , and at most 72 inequivalent representations for any normal cubic surface with finitely many lines.

This is a combination of Theorem 2.42, Corollary 2.43, Corollary 2.54, and Corollary 2.61.

This chapter tells this story, beginning with the simple observation that the data of a general pair of Gale-dual point configurations is given by a $4 \times 3 \times 3$ tensor. We introduce the Gale-transform and this linear-algebraic description in Section 2.1, and set up more important notation. We then detail how the classical geometry of cubic surfaces naturally emerges from linear algebra in Section 2.2, and find the lines on a determinantal cubic surface. We connect this to free resolutions of ideals in Section 2.3, and provide our own geometric proof that $\det(\varphi_U)$ gives a cubic relation between the minors of φ_V .

As a corollary, we prove various classical results in Section 2.4. Finally, we complete our proof of the main theorem and geometrically classify determinantal representations of normal cubic surfaces with finitely many lines in Section 2.5,

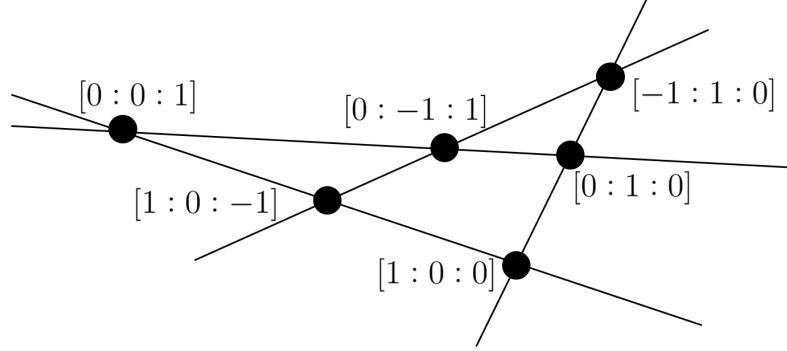


Figure 2.1: The point configuration $\Gamma_V^{4A_1}$, where φ^{4A_1} is a determinantal representation of a $4A_1$ cubic surface.

and describe when a representation of a singular cubic surface as a blowup of a length 6 subscheme is not induced by a $4 \times 3 \times 3$ tensor (when the subscheme is contained in a conic).

There are a few key examples that should be kept in mind while reading this chapter. The first is the representation of any smooth cubic surface by blowing up 6 points. The second is an explicit determinantal representation of the $4A_1$ cubic surface, which we use because of its very simple defining equations. We will also mention some examples where Γ_V^φ is non-reduced, but the equations are harder to work with. The following example will reappear 11 times throughout the thesis, we list all of the examples together in Appendix A for the reader's benefit.

Example 2.2 (A point configuration representing the $4A_1$ cubic surface). To start with, consider the following pair of ordered point sets

$$\begin{aligned}\Gamma_V^{4A_1} &:= \{[1 : 0 : 0], [0 : 1 : 0], [0 : 0 : 1], [0 : -1 : 1], [1 : 0 : -1], [-1 : 1 : 0]\} \subset \mathbb{P}^V \\ \Gamma_W^{4A_1} &:= \{[0 : -1 : 1], [1 : 0 : -1], [-1 : 1 : 0], [1 : 0 : 0], [0 : 1 : 0], [0 : 0 : 1]\} \subset \mathbb{P}^W.\end{aligned}$$

They are Gale-dual. See Figure 2.1 for a picture of the point arrangement.

2.1 The Gale-transform

The classical Gale-transform gives a correspondence between sets of points in one projective space and another set of points in another projective space. Specifically, given a set of $n = r + s + 2$ ordered points $\Gamma_1 \subset \mathbb{P}^r$, the Gale-transform produces another ordered set $\Gamma_2 \subset \mathbb{P}^s$. This correspondence is purely

linear-algebraic, but ends up giving a very rich geometric structure. The Gale-transform is also known under the name of *associated points*. Eisenbud and Popescu describe the Gale-transform in great generality in their two papers [EP99; EP00], which are our primary reference.

Definition 2.3. Let r and s be positive integers. To an ordered collection of $n = r + s + 2$ distinct points $\Gamma = \{v_1, \dots, v_n\} \subset \mathbb{P}^r$ we associate a matrix $M : k^{\oplus n} \rightarrow k^{r+1}$, whose i^{th} column is a non-zero lift of v_i . Conversely, any matrix $M : k^{\oplus n} \rightarrow k^{r+1}$ with no zero columns gives an ordered collection of n points $\{v_1, \dots, v_n\} \subset \mathbb{P}^r$, where v_i is the projective equivalence class of the i^{th} column of M . If M is of this form, then M is said to *represent* a point configuration $\Gamma \subset \mathbb{P}^r$. A representative matrix is unique up to rescaling its columns, or equivalently up to right multiplication by a non-degenerate $n \times n$ diagonal matrix D .

Two ordered collections of points $\Gamma_1 \subset \mathbb{P}^r$ and $\Gamma_2 \subset \mathbb{P}^s$ are said to be *Gale-dual* if there exist representative matrices $M_1 : k^{\oplus n} \rightarrow k^{r+1}$ of Γ_1 and $M_2 : k^{\oplus n} \rightarrow k^{s+1}$ of Γ_2 such that $M_2^T \cdot M_1 = 0$. Equivalently, fixing two representative matrices M_1 of Γ_1 and M_2 of Γ_2 , the configurations Γ_1 and Γ_2 are *Gale-dual* if and only if there exists a non-degenerate $n \times n$ diagonal matrix D where $M_2^T \cdot D \cdot M_1 = 0$.

Note that this explicitly depends on an identified basis of $k^{\oplus n}$ — we write $k^{\oplus n}$ instead of k^n to emphasise the importance of this distinguished basis. There is a canonical pairing $k^{\oplus n} \cong (k^{\oplus n})^\vee = (k^\vee)^{\oplus n}$, unique up to multiplication by a non-degenerate diagonal matrix D (rescaling each copy of k). The total ordering of the point configuration is unimportant, the important part is that Γ_1 and Γ_2 have an ordering relative to each other — the Gale transform sends the i^{th} point in Γ_1 to the i^{th} point in Γ_2 .

When $r = s$, a point configuration Γ is said to be *self-associated* if Γ is Gale-dual to itself.

Proposition 2.4. *Every ordered collection of $n = r + s + 2$ distinct points $\Gamma_1 = \{v_1, \dots, v_n\} \subset \mathbb{P}^r$ that spans $\mathbb{P}^r$ has a unique Gale-dual $\Gamma_2 = \{w_1, \dots, w_n\} \subset \mathbb{P}^s$, up to projective equivalence.*

Proof. When Γ_1 spans $\mathbb{P}^r$, its representative matrix $M_1 : k^{r+1} \rightarrow k^n$ is injective. Then $\dim(\text{coker}(M_1)) = n - r - 1 = s + 1$. Picking a basis for $\text{coker}(M_1) \cong k^{s+1}$, the matrix $M_2 : k^{s+1} \rightarrow (k^{\oplus n})^\vee \cong k^{\oplus n}$ represents a point configuration Γ_2 , unique up to $\text{PGL}(s + 1)$. Conversely, if $\Gamma \subset \mathbb{P}^{s+1}$ is Gale-dual to Γ_1 , then its representative matrix M is equivalent to M_2 up to an element $g \in \text{GL}(s + 1)$, and rescaling the column vectors (picking a different projective lift). The linear isomorphism $g : k^{s+1} \rightarrow k^{s+1}$ induces an isomorphism $g : \mathbb{P}^2 \rightarrow \mathbb{P}^2$ that sends

$\Gamma \rightarrow \Gamma_2$ and preserves the order, so Γ and Γ_2 are projectively equivalent ordered point configurations. $\square$

Remark 2.5. The Gale-transform can be written very easily in a specific basis. Choosing a basis for $\mathbb{P}^r$ where $v_1, \dots, v_{r+1}$ are the basis vectors, we have a block form $M_1 = \begin{pmatrix} I & A \end{pmatrix}^T$ for some $(r+1) \times (s+1)$ matrix A . Then matrix representing the Gale-dual is equivalent to $M_2 = \begin{pmatrix} -A^T & I \end{pmatrix}^T$. In particular, this shows that a set of $r+1$ points in Γ_1 are linearly independent if and only the complementary set of $s+1$ points in Γ_2 are linearly independent.

Example 2.6 ($4A_1$ Gale-duality). We continue Example 2.2. In this case, the matrix is

$$A = \begin{pmatrix} 0 & -1 & 1 \\ 1 & 0 & -1 \\ -1 & 1 & 0 \end{pmatrix}.$$

Notice that $A = -A^T$ is skew-symmetric, hence $\Gamma_W^{4A_1}$ is the same as $\Gamma_V^{4A_1}$ as sets. But they are not the same ordered point configuration — the first three points of $\Gamma_V^{4A_1}$ are linearly independent, whereas the first three points of $\Gamma_W^{4A_1}$ are linearly dependent.

We will be working with point configurations up to projective equivalence, and so whenever Γ_1 and Γ_2 are Gale-dual we will say that Γ_2 is *the Gale-transform* of Γ_1 . Since the condition $M_2^T \cdot M_1 = 0$ is equivalent to $M_1^T \cdot M_2 = 0$, the Gale-transform gives an involution between sets of n points in $\mathbb{P}^r$ and n points in $\mathbb{P}^{n-r-2}$. We are most interested in the connection between Gale-duality and the Segre variety, which gives an excellent coordinate-free description of Gale-duality.

Definition 2.7. To two ordered sets of $n = r + s + 2$ distinct points $\Gamma_1 = \{v_1, \dots, v_n\} \subset \mathbb{P}^r$ and $\Gamma_2 = \{w_1, \dots, w_n\} \subset \mathbb{P}^s$ we associate their *tensor product*

$$\Gamma_1 \otimes \Gamma_2 := \{v_1 \otimes w_1, v_2 \otimes w_2, \dots, v_n \otimes w_n\} \subset \mathbb{P}(k^{r+1} \otimes k^{s+1}).$$

The point configuration $\Gamma_1 \otimes \Gamma_2$ is then contained in the image of the Segre embedding $\mathbb{P}^r \times \mathbb{P}^s \rightarrow \mathbb{P}(k^{r+1} \otimes k^{s+1})$. We recover Γ_1 and Γ_2 from $\Gamma_1 \otimes \Gamma_2$ by computing the image of $\Gamma_1 \otimes \Gamma_2$ under the canonical projections from $\mathbb{P}^r \times \mathbb{P}^s$ to $\mathbb{P}^r$ and $\mathbb{P}^s$.

Proposition 2.8. *Two ordered configurations of $n = r + s + 2$ points $\Gamma_1 \subset \mathbb{P}^{r+1}$ and $\Gamma_2 \subset \mathbb{P}^{s+1}$ are Gale-dual if and only $\Gamma_1 \otimes \Gamma_2$ is contained in a $r+s$ -dimensional linear subspace of $\mathbb{P}(k^{r+1} \otimes k^{s+1})$.*

Proof. The configuration $\Gamma_1 \otimes \Gamma_2$ is contained in a $r + s$ -dimensional linear subspace if and only if there is a non-trivial linear relation between the $v_i \otimes w_i$ s (note the linear subspace is $r + s$ -dimensional as a projective space, but is a $r + s + 1$ -dimensional vector space). Up to rescaling the v_i s by a different projective representative, this can be written as $\sum v_i \otimes w_i = 0$. And $\sum v_i \otimes w_i = M_2^T \cdot M_1$. $\square$

Proposition 2.9. [Cob22] *A set of 6 points $\Gamma \subset \mathbb{P}^2$ is self-associated if and only if Γ is contained in a conic.*

Proof. This is a well-known classical result. The configuration $\Gamma \otimes \Gamma$ is the image of Γ under the Veronese embedding $\mathbb{P}^2 \rightarrow \mathbb{P}(\text{Sym}^2(k^3)) \subset \mathbb{P}(k^3 \otimes k^3)$, sending a point $[x : y : z] \mapsto [x^2 : y^2 : z^2 : xy : xz : yz]$. A hyperplane in $\mathbb{P}(\text{Sym}^2(k^3))$ is cut out by the equation $Ax^2 + By^2 + Cz^2 + Dxy + Exz + Fyz = 0$ (viewing the monomials as coordinates), this cuts out a conic in $\mathbb{P}^2$. $\square$

We are also interested in a generalisation of Gale-duality to certain finite subschemes of $\mathbb{P}^r$. The key property we want to preserve is the canonical identification of $k^{\oplus n} \cong (k^{\oplus n})^\vee$. Because $\text{Spec}(k^{\oplus n})$ is the coordinate ring of n reduced points, the appropriate generalisation is to a class of finite schemes whose coordinate ring exhibits a similar duality property:

Definition 2.10. Let B be a finite-dimensional local k -algebra. B is *Gorenstein* if there is a unique k -linear map (up to a constant multiple) $e : B \rightarrow k$ such that the pairing $\tau : B \times B \rightarrow k$ sending $(x, y) \mapsto e(xy)$ is non-degenerate. The induced identification $B \cong B^\vee$ is canonical.

An finite-dimensional k -algebra A uniquely splits into the direct sum $A = A_1 \oplus \cdots \oplus A_n$ of local k -algebras A_i . The algebra A is said to be *locally Gorenstein* if each A_i is Gorenstein.

A finite scheme is said to be *locally Gorenstein* if its coordinate ring $H^0(\Gamma, \mathcal{O}_\Gamma)$ is. The pairing $\tau : H^0(\Gamma, \mathcal{O}_\Gamma) \times H^0(\Gamma, \mathcal{O}_\Gamma) \rightarrow k$ is the pairing given by Serre duality, canonically identifying $H^0(\Gamma, \mathcal{O}_\Gamma) \cong H^0(\Gamma, \mathcal{O}_\Gamma)^\vee$.

Definition 2.11. Let $\Gamma = \text{Spec}(A)$ be a finite locally Gorenstein scheme, and $X \subset A = H^0(\Gamma, \mathcal{O}_\Gamma)$ a vector subspace (note that we use the opposite convention for $\mathbb{P}X$ to [EP00]). We let $X^\perp = Y$ be the annihilator of X under the pairing $\tau : A \times A \rightarrow k$. The pair of linear series $(\mathcal{O}_\Gamma, X)$ and $(\mathcal{O}_\Gamma, X^\perp)$ are said to be *Gale-dual*.

The two linear series $(\mathcal{O}_\Gamma, X)$ and $(\mathcal{O}_\Gamma, Y)$ induce a pair of morphisms $\iota_X : \Gamma \rightarrow \mathbb{P}X$ and $\iota_Y : \Gamma \rightarrow \mathbb{P}Y$. We write Γ_X and Γ_Y for the respective images of ι_X and ι_Y . Whenever Z is a vector space and $\Gamma_Z \hookrightarrow \mathbb{P}Z$ is a closed

subscheme, if there is an isomorphism $f : \mathbb{P}Y \rightarrow \mathbb{P}Z$ that restricts to an isomorphism $\Gamma_Y \rightarrow \Gamma_Z$ then the subschemes Γ_X and Γ_Z are said to be *Gale-dual*. Equivalently, there is an isomorphism of linear series $f^* : (\mathcal{O}_{\Gamma_Z}, Z) \rightarrow (\mathcal{O}_{\Gamma_Y}, Y)$.

When $(\mathcal{O}_\Gamma, X)$ and $(\mathcal{O}_\Gamma, Y)$ are any two linear series, there is the product morphism $(\iota_X, \iota_Y) : \Gamma \rightarrow \mathbb{P}X \times \mathbb{P}Y$. We let $\Gamma_X \otimes \Gamma_Y$ be the image of Γ inside $\mathbb{P}X \times \mathbb{P}Y$. The standard projections of $\Gamma_X \otimes \Gamma_Y$ onto $\mathbb{P}X$ and $\mathbb{P}Y$ recover Γ_X and Γ_Y .

Proposition 2.12. *The scheme-theoretic Gale duality is a generalisation of point-wise Gale duality.*

Proof. If $\Gamma \hookrightarrow \mathbb{P}^r$ is a list of $n = r + s + 2$ points, then its coordinate ring is $A = k^{\oplus n}$. The line bundle $\mathcal{O}_\Gamma(1)$ on Γ is the pullback of $\mathcal{O}_{\mathbb{P}^r}(1)$ under the closed immersion $\iota : \Gamma \hookrightarrow \mathbb{P}^r$, but $\mathcal{O}_\Gamma(1) \cong \mathcal{O}_{\Gamma_1}$ because Γ is a finite scheme. So the immersion ι identifies $\mathbb{P}^r$ with an injective linear map $M_1 : (k^{r+1})^\vee \rightarrow k^{\oplus n}$. The columns of the dual map $k^{\oplus n} \rightarrow k^{r+1}$ are n vectors in k^{r+1} , which are the coordinates of the n points (because $k^{\oplus n}$ is split into the direct sum of n copies of k , and so has a naturally given basis as a vector space). The orthogonal complement of $(k^{r+1})^\vee$ is given by some other linear map $M_2 : (k^{s+1})^\vee \rightarrow k^{\oplus n}$, and the image of $\Gamma \hookrightarrow \mathbb{P}^s$ is the Gale dual in the point set sense, because $M_2^T \cdot M_1 = 0$. Picking a different set of pairings $k^{\oplus n} \cong (k^{\oplus n})^\vee$ involves picking n scalars — this is the diagonal matrix D .

Other results like $(\mathcal{O}_{\Gamma_X}, X)$ and $(\mathcal{O}_{\Gamma_Y}, Y)$ being Gale-dual if and only if $\Gamma_X \otimes \Gamma_Y$ is contained in a $r + s$ -dimensional linear subspace, and $(\mathcal{O}_\Gamma, X)$ being self-associated if and only if Γ_X lies on a conic (when $\text{len}(\Gamma) = 2 \dim X = 6$) are true by almost the same argument as the reduced case. $\square$

The case $r = s = 2$ is quite special — it is the only non-trivial case where the span of the length 6 subscheme $\Gamma_V \otimes \Gamma_W$ has codimension $r + s$, the dimension of the Segre variety. This is also the only non-trivial case where the Segre variety has degree $\binom{r+s}{r} = n$, and so the intersection of a general 4-dimensional linear subspace with the Segre embedding gives a general pair of Gale-dual point configurations. Recall that V and W are three-dimensional vector spaces. As a result, we have:

Proposition 2.13. *The intersection of the Segre variety $\mathbb{P}V \times \mathbb{P}W$ with a general four-dimensional linear subspace $\Lambda \hookrightarrow \mathbb{P}(V \otimes W)$ gives a pair of Gale-dual point configurations Γ_V and Γ_W , where $\Lambda \cap (\mathbb{P}V \times \mathbb{P}W) = \Gamma_V \otimes \Gamma_W$.*

Now, a 4-dimensional subspace of $\mathbb{P}(V \otimes W)$ is cut out by 4 independent linear equations. So Λ is the projectivised kernel of some surjective linear map

$\varphi^\vee : V \otimes W \rightarrow U^\vee$. This is a $4 \times 3 \times 3$ tensor. We will now take this $4 \times 3 \times 3$ tensor as the starting object, and see how much of the structure of a point configuration can be recovered.

Example 2.14. We continue Example 2.6. The tensor product $\Gamma_V^{4A_1} \otimes \Gamma_W^{4A_1}$ of the point configurations consists of the 6 points

$$\begin{pmatrix} 0 & 0 & 0 \\ -1 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & -1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & -1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 0 \\ -1 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ -1 & 1 & 0 \end{pmatrix}.$$

Consider the map $\varphi_U^{4A_1} : U \rightarrow V^\vee \otimes W^\vee$, defined by:

$$\varphi_U = \begin{pmatrix} u_0 & u_1 & u_1 \\ u_1 & u_2 & u_1 \\ u_1 & u_1 & u_1 + u_3 \end{pmatrix}. \quad (2.1)$$

The kernel of the dual map $(\varphi_U^{4A_1})^\vee : V \otimes W \rightarrow U^\vee$ is the span of $\Gamma_V^{4A_1} \otimes \Gamma_W^{4A_1}$.

To start with, we set up some important notation. For the rest of this chapter, the surface S_φ is *assumed to have finitely many lines* unless otherwise specified. We will sometimes impose the additional condition of S_φ being normal, especially in Section 2.5. The theory of determinantal representations of non-normal cubic surfaces and cubic surfaces with infinitely many lines is also possible to develop, but doesn't arise as directly as a generalisation of the theory for smooth surfaces.

Notation 2.15. Recall Notation 2.1. The vector spaces U and V and W will always be fixed 4, 3, and 3-dimensional vector spaces, respectively. We will be working over various projective spaces and will very frequently perform calculations by lifting closed points $x \in \mathbb{P}X$ to a non-zero element of their equivalence class in X , which we will again denote by $\bar{x}$, or sometimes x by abuse of notation (we use an overline instead of a tilde to avoid confusion with maps of sheaves of modules). Everything will only depend on homogeneous coordinates, so the specific choice of lift will never matter.

The tensor φ will always be an element of $\mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$. When the map $\varphi_U : U \rightarrow V^\vee \otimes W^\vee$ is injective, we will also write $\varphi : \mathbb{P}U \rightarrow \mathbb{P}(V^\vee \otimes W^\vee)$ to be the induced linear embedding of projective spaces. We will write $\varphi^\vee = \varphi_U^\vee$ when viewing φ as an element of $\text{Hom}(V \otimes W, U^\vee)$.

Because the roles of V and W are completely symmetric, we will interchangeably refer to elements of $V^\vee \otimes W^\vee$ as either pairings $V \otimes W \rightarrow k$ or linear maps

$V \rightarrow W^\vee$, this allows us to easily refer to the kernel (a subspace of V) and cokernel (a subspace of W) but this ordering is for convenience of notation and is completely arbitrary. All results that apply for V equivalently apply for W , we will only prove results for either V or W with the understanding that they apply to both, instead of continuously using the cluttered phrasing ... V (resp. ... W).

Notation 2.16. A point $u \in S_\varphi$ is said to be *rank one* (resp. *rank two*) if $\varphi_U(\bar{u})$ is a rank one (resp. rank two) linear map when viewed as an element of $\text{Hom}(V, W^\vee)$. The scheme of rank one (resp. rank two) points in S_φ will be called the *rank one locus* (resp. *rank two locus*), denoted by $S_{\varphi, \text{rk } 1}$ (resp. $S_{\varphi, \text{rk } 2}$). As a scheme, $S_{\varphi, \text{rk } 1}$ is cut out by the vanishing of the nine 2×2 minors, and its complement $S_{\varphi, \text{rk } 2}$ inside S_φ is an open subscheme. If every point in S_φ is rank two then S_φ is said to be a *rank two cubic surface*.

The rank two locus of $\mathbb{P}V$ is Γ_V^φ (it is cut out by the vanishing of the maximal minors of φ_V). Similarly, the *full rank locus* of $\mathbb{P}V$ will be denoted by $\mathbb{P}V_{\varphi, \text{rk } 3} = \mathbb{P}V \setminus \Gamma_V^\varphi$. We define $\mathbb{P}W_{\varphi, \text{rk } 3}$ identically.

When φ_U is injective, we will identify U with its image $\varphi_U(U) \subset V^\vee \otimes W^\vee$, and write $U \subset V^\vee \otimes W^\vee$. The dual map $\varphi^\vee : V \otimes W \rightarrow U^\vee$ is then surjective, and its projectivised kernel $\mathbb{P}(\ker(\varphi^\vee)) \subset \mathbb{P}(V \otimes W)$ is a four-dimensional linear subspace. We let $\Gamma_{V \otimes W}^\varphi = \mathbb{P}(\ker(\varphi^\vee)) \cap (\mathbb{P}V \times \mathbb{P}W)$ be the scheme-theoretic intersection, where $\mathbb{P}V \times \mathbb{P}W \subset \mathbb{P}(V \otimes W)$ is the Segre variety.

Definition 2.17. A cubic surface $S \hookrightarrow \mathbb{P}^3$ is said to be *determinantal* if it has a determinantal representation. Recall that two determinantal representations φ and φ' are said to be equivalent if they are isomorphic as trilinear maps.

If φ and φ' are equivalent then the cubic surfaces S_φ and $S_{\varphi'}$ are isomorphic, but the converse is not true (we prove this later). We will see that there are finitely many determinantal representations of a normal cubic surface with finitely many lines, up to equivalence. Note the group $\text{GL}(U) \times \text{GL}(V) \times \text{GL}(W)$ acts as $\text{PGL}(U) \times \text{PGL}(V) \times \text{PGL}(W)$ on $\mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$.

Lemma 2.18. *Any rank one point on a determinantal cubic surface is singular. A normal cubic surface has finitely many rank one points.*

Proof. The derivative of the determinant cubic equation (in coordinates) is $\det(\varphi_U)' = \text{tr}(\text{Adj}(\varphi_U)(\varphi_U'))$ where $\text{Adj}(\varphi_U)$ is the adjugate matrix of φ_U whose ij^{th} entry is the ij^{th} 2×2 minor of φ_U . When $\varphi_U(\bar{u})$ is rank one, $\text{Adj}(\varphi_U(\bar{u})) = 0$ so all derivatives of $\det(\varphi_U)$ vanish and $\varphi_U(\bar{u})$ is singular. Normal cubic surfaces have finitely many singularities. $\square$

Example 2.19. We continue Example 2.14. The determinant cubic is

$$\det(\varphi_U^{4A_1}) = u_3(u_0u_2 - u_1^2) - u_1(u_0 - u_1)(u_1 - u_2). \quad (2.2)$$

This is exactly the normal form of the $4A_1$ cubic surface (see [BW79]). We call it *the* $4A_1$ cubic surface, because every cubic surface with $4A_1$ singularities is isomorphic to $S_{4A_1} = S_{\varphi^{4A_1}}$.

There are four rank one points on S_{4A_1} , having coordinates

$$S_{4A_1, \text{rk } 1} = \{[1 : 0 : 0 : 0], [1 : 1 : 1 : 0], [0 : 0 : 1 : 0], [0 : 0 : 0 : 1]\} \quad (2.3)$$

The rank two loci $\Gamma_V^{\varphi^{4A_1}}$ and $\Gamma_W^{\varphi^{4A_1}}$ are the point configurations $\Gamma_V^{4A_1}$ and $\Gamma_W^{4A_1}$ introduced in Example 2.2.

We want to construct maps from S_φ to $\mathbb{P}V$ and $\mathbb{P}W$. In the smooth case, this will recover the description of a smooth cubic surface as the blowup of $\mathbb{P}^2$ at 6 points given by a birational morphism $\pi : S \rightarrow \mathbb{P}^2$, and our description directly generalises to the singular case. Each rank two point $u \in S_\varphi$ lifts to a rank two linear map $\varphi_U(\bar{u}) \in \text{Hom}(V, W^\vee)$. Its kernel is a line in V , i.e. a point in $\mathbb{P}V$, and its cokernel is a line in W , i.e. a point in $\mathbb{P}W$. We will formalise this statement by first passing to the land of sheaves.

Notation 2.20. The map φ_U (resp. φ_V , resp. φ_W) also induces a map of free $k[V^\vee]_\bullet$ - (resp. $k[W^\vee]_\bullet$ -) modules after tensoring with the respective graded ring, which we will again use the same notation for. After taking the projective spectrum, these three maps of free modules give three maps of coherent sheaves on $\mathbb{P}U$ (resp. $\mathbb{P}V$, resp. $\mathbb{P}W$) which will be denoted by

$$\begin{aligned} \tilde{\varphi}'_U &: V \otimes \mathcal{O}_{\mathbb{P}U}(-1) \longrightarrow W^\vee \otimes \mathcal{O}_{\mathbb{P}U}, \\ \tilde{\varphi}_V &: W \otimes \mathcal{O}_{\mathbb{P}V}(-1) \longrightarrow U^\vee \otimes \mathcal{O}_{\mathbb{P}V}, \\ \tilde{\varphi}_W &: V \otimes \mathcal{O}_{\mathbb{P}W}(-1) \longrightarrow U^\vee \otimes \mathcal{O}_{\mathbb{P}W}. \end{aligned}$$

These are not maps of constant rank — $\tilde{\varphi}'_U$ (resp. $\tilde{\varphi}_V$, resp. $\tilde{\varphi}_W$) drops rank on a closed subscheme, but we can restrict to various open subvarieties where the maps are indeed maps of constant rank. Base-changing $\tilde{\varphi}'_U$ by the map $S_\varphi \hookrightarrow \mathbb{P}U$ gives a (generically) rank two map of coherent sheaves

$$\tilde{\varphi}_U : V \otimes \mathcal{O}_{S_\varphi}(-1) \longrightarrow W^\vee \otimes \mathcal{O}_{S_\varphi}$$

whose rank only drops on the rank one locus of S_φ . After restriction to the open subscheme $S_{\varphi, \text{rk } 2}$ (resp. $\mathbb{P}V_{\varphi, \text{rk } 3}$, resp. $\mathbb{P}W_{\varphi, \text{rk } 3}$), the map $\tilde{\varphi}_U|_{S_{\varphi, \text{rk } 2}} = \tilde{\varphi}_U|_{\text{rk } 2}$

(resp. $\tilde{\varphi}_V|_{\mathbb{P}V_{\varphi, \text{rk } 3}} = \tilde{\varphi}_V|_{\text{rk } 3}$, resp. $\tilde{\varphi}_W|_{\mathbb{P}W_{\varphi, \text{rk } 3}} = \tilde{\varphi}_W|_{\text{rk } 3}$) is of constant rank and is a map of vector bundles. In particular, $\tilde{\varphi}_V|_{\text{rk } 3}$ and $\tilde{\varphi}_W|_{\text{rk } 3}$ are injective.

Definition 2.21. To the map $\tilde{\varphi}'_U$ we associate the sheaf $\text{coker}(\tilde{\varphi}'_U)$ on $\mathbb{P}U$. There is an exact sequence

$$0 \longrightarrow V \otimes \mathcal{O}_{\mathbb{P}U}(-1) \xrightarrow{\tilde{\varphi}'_U} W^\vee \otimes \mathcal{O}_{\mathbb{P}U} \longrightarrow \text{coker}(\tilde{\varphi}'_U) \longrightarrow 0. \quad (2.4)$$

Because $\mathcal{O}_{\mathbb{P}U}(-1)$ has no 0th or 1st cohomology, then taking cohomology gives an isomorphism $W^\vee \cong H^0(\mathbb{P}U, \text{coker}(\tilde{\varphi}'_U))$. We are interested in the restriction $\text{coker}(\tilde{\varphi}'_U)|_{S_\varphi}$ of $\text{coker}(\tilde{\varphi}'_U)$ to S_φ . Because $\tilde{\varphi}'_U$ only drops rank on S_φ , then $\text{coker}(\tilde{\varphi}'_U)$ is only supported on S_φ and so $H^0(S_\varphi, \text{coker}(\tilde{\varphi}'_U)) \cong H^0(\mathbb{P}U, \text{coker}(\tilde{\varphi}'_U)) \cong W^\vee$. Because $\tilde{\varphi}_U|_{\text{rk } 2}$ is constant rank, then $\text{coker}(\tilde{\varphi}_U)$ is a line bundle on $S_{\varphi, \text{rk } 2}$. The linear system $|\text{coker}(\tilde{\varphi}_U)|$ induces a morphism $\pi_W : S_{\varphi, \text{rk } 2} \rightarrow \mathbb{P}W$, which sends a point $u \in S_{\varphi, \text{rk } 2}$ to its projectivised cokernel $\mathbb{P}(\text{coker}(\varphi_U(\bar{u}))) \subset \mathbb{P}W$, which is a point because $\varphi_U(\bar{u})$ is rank two.

Similarly, there is a morphism $\pi_V : S_{\varphi, \text{rk } 2} \rightarrow \mathbb{P}V$ sending $u \mapsto \mathbb{P}(\ker(\varphi_U(\bar{u})))$ (take the cokernel of the dual map $\tilde{\varphi}_U^\vee : W \otimes \mathcal{O}_{\mathbb{P}U}(-1) \rightarrow V^\vee \otimes \mathcal{O}_{\mathbb{P}U}$, this is a line bundle on $S_{\varphi, \text{rk } 2}$, and $\text{coker}(\varphi_U^\vee(\bar{u})) = \ker(\varphi_U(\bar{u}))$).

Lemma 2.22. π_V and π_W are birational morphisms from $S_{\varphi, \text{rk } 2}$. The fibre of π_V (resp. π_W) above a closed point $v \in \mathbb{P}V$ (resp. $w \in \mathbb{P}W$) is an open subset of the linear subspace $\mathbb{P}(\text{coker}(\varphi_V(\bar{v})))$ (resp. $\mathbb{P}(\text{coker}(\varphi_W(\bar{w})))$). This is either a point, a line (possibly minus some finite set of points), or empty.

Proof. A point $u \in S_\varphi$ is in the fibre above a point $v \in \mathbb{P}V$ if and only if both $v \in \ker(\varphi_U(\bar{u}))$ and $u \in S_{\varphi, \text{rk } 2}$. This is equivalent to $\varphi(\bar{u}, \bar{v}, -) = 0 \in W^\vee$ and $u \in S_{\varphi, \text{rk } 2}$ (viewing φ as a trilinear map $\varphi : U \otimes V \otimes W \rightarrow k$), which is equivalent to $\bar{u} \in \text{coker}(\varphi_V(\bar{v}))$ and $u \in S_{\varphi, \text{rk } 2}$. So $\pi_V^{-1}(v) = \mathbb{P}(\text{coker}(\varphi_V(\bar{v}))) \cap S_{\varphi, \text{rk } 2}$, an open subset of $\mathbb{P}(\text{coker}(\varphi_V(\bar{v})))$.

Since S_φ has finitely many lines then the fibre above a $v \in \mathbb{P}V$ is only not a singleton finitely many times. So π_V restricts to a injective morphism of two-dimensional quasi-projective varieties on an open set $(S_{\varphi, \text{rk } 2} \setminus \{\text{lines}\})$, so π_V is birational. If S_φ is a rank two surface, then $\pi_V : S_\varphi \rightarrow \mathbb{P}V$ and $\pi_W : S_\varphi \rightarrow \mathbb{P}W$ are birational morphisms of projective varieties, and are also surjective. $\square$

Example 2.23. We continue Example 2.19. The fibre of π_V above $[1 : 0 : 0]$ is the line $[0 : 0 : s : t]$. The fibre above $[1 : 1 : 0]$ is empty. The fibre above $[1 : 2 : 0]$ is the point $[4 : 2 : 1 : 0]$.

Corollary 2.24. If S_φ is normal with finitely many lines, then the rank two loci Γ_V^φ and Γ_W^φ are finite.

Proof. A point v is in Γ_V^φ if and only if $\text{coker}(\varphi_V(\bar{v}))$ is at least two-dimensional, so $\mathbb{P}(\text{coker}(\varphi_V(\bar{v}))) \subset S_\varphi$ is a line. If two distinct points v and v' had the same line $\mathbb{P}U'$ in their cokernel, then $\text{Span}\{v, v'\} \subset \ker(\varphi_U(\bar{u}))$ for any $u \in \mathbb{P}U'$ and every point on the line $\mathbb{P}U'$ is rank one (so singular), a contradiction on S_φ being normal. So the number of points in Γ_V^φ is less than or equal to the finite number of lines on S_φ , and is finite. $\square$

Remark 2.25. The reason we restrict to cubic surfaces S_φ with finitely many lines is to ensure:

- That S_φ is irreducible and contains no hyperplanes.
- That there are finitely many rank one points (Lemma 2.18).
- That the fibres of π_V and π_W are finite except at finitely many points (Lemma 2.22).

Cubic surfaces with infinitely many lines are ruled, and so there exist rational maps $\pi : S_\varphi \dashrightarrow \mathbb{P}^2$ whose image is the union of several curves (π_V might not even be birational!).

This is enough for some results, but the additional hypothesis that S_φ is normal ensures that the rank two loci Γ_V^φ and Γ_W^φ are finite (Corollary 2.24), and curvilinear (we will prove this later).

2.2 Lines on determinantal cubic surfaces

One of the keys to understanding determinantal cubic surfaces lies in the configuration of lines. Because of the immersion $S_\varphi \hookrightarrow \mathbb{P}(V^\vee \otimes W^\vee)$, the lines on S_φ are also lines inside $\mathbb{P}(V^\vee \otimes W^\vee)$. We introduce some important terminology to label these lines, which is also found in Appendix D for convenience.

Definition 2.26. A rank at most two linear subspace $\mathbb{P}U' \subset \mathbb{P}(V^\vee \otimes W^\vee)$ is:

- Of *type K* (resp. *type C*) if there is a non-zero vector $\bar{v} \in V$ (resp. $\bar{w} \in W$) annihilated by every element $\bar{u} \in U'$.
- Of *type L* if there is a pair of two-dimensional subspaces $V' \subset V$ and $W' \subset W$ where $\bar{u}(V' \otimes W') = 0$ for every $\bar{u} \in U'$.
- *Rank one* if every $u \in \mathbb{P}U'$ is rank one. Rank one with *shared kernel* (resp. *shared cokernel*) if the kernel (resp. cokernel) of every $u \in \mathbb{P}U'$ is the same.

One should be careful — not every rank one line is a line with shared kernel or cokernel!

The reason we choose this naming is as follows: under the identification $V^\vee \otimes W^\vee = \text{Hom}(V, W^\vee)$, a subspace is type K (resp. C) if all the kernels (resp. cokernels) intersect nontrivially. This is not the same as having a shared kernel (resp. cokernel), since the rank isn't necessarily constant, but all rank two matrices in the subspace have the same kernel (resp. cokernel). And it is of type L if there is a basis where all the entries of each matrix are non-zero in an L shape (see the proof of Lemma C.1 in the appendix if you are curious!), L does not stand for “line”. Every line of rank at most two in $\mathbb{P}(V^\vee \otimes W^\vee)$ is of type C, K or L (possibly of multiple such types). More generally, we have the following result:

Lemma 2.27. [AL81, Lem. 7] *Rank at most two linear subspaces $U' \subset V^\vee \otimes W^\vee$ are of exactly one of the following forms:*

- *Type C or K or L (possibly multiple of these types at the same time).*
- *U' is the three-dimensional subspace of skew-symmetric matrices under a basis for $V \cong W \cong k^3$.*

Proof. We provide our own proof for the case for two-dimensional linear subspaces in the appendix (Lemma C.1). Atkinson and Lloyd prove this at the end of their paper [AL81], and prove a range of results related to matrices of bounded rank (and generalise the notion of types C or K or L) in their earlier paper [AL80]. $\square$

Corollary 2.28. *The lines on S_φ are of least one of the following forms:*

- (i) *Of type K — the closure of the fibre of π_V above a point $v_i \in \Gamma_V^\varphi$.*
- (ii) *Of type C — the closure of the fibre of π_W above a point $w_i \in \Gamma_W^\varphi$.*
- (iii) *Of type L.*

Proof. $\pi_V^{-1}(v)$ is a line if and only if $\text{coker}(\varphi_V(\bar{v}))$ is two-dimensional, or equivalently if and only if v is in the rank two locus Γ_V^φ , so every line of type K is the fibre above a point $v_i \in \Gamma_V^\varphi$. The result for lines of type C holds identically. $\square$

Proposition 2.29. *Let M be a line of type L. If M is also of type C or K, then M contains a rank one point, and $V' = \ker(\varphi_U(\bar{u}))$ and $W' = \text{coker}(\varphi_U(\bar{u}))$. Otherwise, π_V and π_W are both defined and injective when restricted to M and the image of M under π_V (resp. π_W) is then $\mathbb{P}V'$ (resp. $\mathbb{P}W'$).*

Proof. This follows by directly computing. $\square$

Example 2.30. We continue Example 2.23. The lines of type C and K (they are of both types at the same time for this specific example, because $\varphi_U^{4A_1}$ is symmetric, and are also of type L) are

$$s = \{[0 : 0 : s : t], [s : 0 : 0 : t], [s : 0 : t : 0], [s : s : s : t], [s : s : t : 0], [t : s : s : 0]\}.$$

These are ordered to be the fibres above the points in $\Gamma_V^{4A_1}$ in the same order as given in Example 2.2. We use the notation s , because this is a six on S_{4A_1} . Each line $M_i \in s$ passes through exactly two singularities on S_{4A_1} . The lines of type L only are

$$\{[s : t : 0 : t - s], [0 : t : s : t - s], [s + 2t : t : -s : -t]\}.$$

These do not pass through any of the singularities. There are 9 lines on the $4A_1$ cubic surface.

Corollary 2.31. *If S_φ is a normal rank two cubic surface with finitely many lines, then π_V is the blowup of $\mathbb{P}V$ along some finite subscheme $Z \hookrightarrow \mathbb{P}V$. The reduced subschemes $\Gamma_{V,\text{red}}^\varphi$ and Z_{red} are equal.*

Proof. Because S_φ is rank two, π_V is a birational morphism of projective varieties and hence (isomorphic to) a blowup along some closed subscheme $Z \hookrightarrow \mathbb{P}V$ [Har77, II, Thm. 7.17]. Z is supported only where the fibres of π_V are not singletons, i.e. on Γ_V^φ . $\square$

We will later see that $Z = \Gamma_V^\varphi$, and that Γ_V^φ and Γ_W^φ are length 6 and that the subschemes Γ_V^φ and Γ_W^φ are Gale-dual in the generalised sense, but for now we only have the tools to prove this in the case that S_φ is smooth. The following theorem is a result from Coble [Cob15] rephrased in the language of determinantal representations.

Theorem 2.32. *If S_φ is a smooth determinantal cubic surface represented by a $\varphi \in U^\vee \otimes V^\vee \otimes W^\vee$ then $\Gamma_V^\varphi = \{v_1, \dots, v_6\}$ and $\Gamma_W^\varphi = \{w_1, \dots, w_6\}$ are reduced and consist of exactly 6 distinct ordered points. The ordered point configurations Γ_V^φ and Γ_W^φ are Gale-dual, and the configuration $\Gamma_V^\varphi \otimes \Gamma_W^\varphi$ is the intersection $\Gamma_{V \otimes W}^\varphi = \mathbb{P}(\ker(\varphi^\vee)) \cap \mathbb{P}V \times \mathbb{P}W$.*

The lines on S_φ are then exactly one of the following types:

- (i) *The six exceptional fibres E_{v_i} of π_V above a point $v_i \in \Gamma_V^\varphi$.*
- (ii) *The six exceptional fibres E_{w_i} of π_W above a point $w_i \in \Gamma_W^\varphi$.*

(iii) The 15 lines L_{ij} of type L (where $i \neq j$ and $L_{ij} = L_{ji}$).

The twelve lines $\{E_{v_i}\}_{i=1}^6$ and $\{E_{w_j}\}_{j=1}^6$ form two halves of a double-six. The line E_{w_i} is the strict transform under π_V of the conic in $\mathbb{P}V$ that passes through the five points v_j for $j \neq i$, but doesn't intersect v_i . The line L_{ij} is the strict transform under π_V of the line in $\mathbb{P}V$ through v_i and v_j .

Proof. The morphism $\pi_V : S_\varphi \rightarrow \mathbb{P}V$ is a birational morphism of smooth surfaces. The fibres of π_V are either a point or a line, then π_V is the composition of some number of contractions of distinct lines. These lines are all (-1) -curves, so π_V is a blowup at finitely many points (the subscheme Γ_V^φ). And because S_φ is a cubic surface, this number is 6 (this can be seen from, for example, the rank of the Picard group being $1 + 6 = 7$). So $\Gamma_V^\varphi = \{v_1, \dots, v_6\}$ is a length 6 reduced subscheme. No three points of Γ_V^φ lie on a line, and no 6 points lie on a conic, because then the strict transforms of these curves would be $(-n)$ -curves for $n > 1$, but the only curves of negative self-intersection on a smooth cubic surface are the 27 (-1) -curves. Similarly, $\Gamma_W^\varphi = \{w'_1, \dots, w'_6\}$ is a reduced set of 6 points in $\mathbb{P}W$.

Combined with the analogous result for W , there are two sets of six lines $\{E_{v_1}, \dots, E_{v_6}\}$ of type K and $\{E_{w'_1}, \dots, E_{w'_6}\}$ of type C, and every remaining line is of type L. Every line on S_φ is a (-1) -curve, and so is either one of the exceptional curves E_{v_i} , or the strict transform L_{ij} of the line through v_i and v_j , or the strict transform Q_i of the conic through $\Gamma_V^\varphi \setminus \{v_i\}$. There are then at least fifteen lines in S_φ of type L, whose images under π_V give fifteen distinct lines because S_φ has no rank one points. But the image of a line of type L under π_V must be a line in $\mathbb{P}V$, of which there are only fifteen possibilities, so there are exactly fifteen lines of type L on S_φ , being the lines L_{ij} .

Then each $E_{w'_j}$ is the strict transform under π_V of the conic in $\mathbb{P}V$ that passes through five points in Γ_V^φ , but doesn't pass through some v_i . We relabel the w'_j s to w_i s, such that E_{w_i} is the strict transform of the conic through the five points $\Gamma_V^\varphi \setminus \{v_i\}$. This gives a bijection

$$\Gamma_V^\varphi = \{v_1, \dots, v_6\} \longrightarrow \{w_1, \dots, w_6\} = \Gamma_W^\varphi$$

sending $v_i \mapsto w_i$.

To show Γ_V^φ and Γ_W^φ are Gale-dual, it suffices to show that $\Gamma_V^\varphi \otimes \Gamma_W^\varphi$ is contained in the four-dimensional linear subspace $\mathbb{P}(\ker(\varphi^\vee))$. For each i , the lines E_{v_i} and E_{w_i} are disjoint, so they span $\mathbb{P}U$. Then any $\bar{u} \in U$ can be decomposed (non-uniquely) into $\bar{u} = \alpha \bar{u}_{v_i} + \beta \bar{u}_{w_i}$ for some $u_{v_i} \in E_{v_i}$ and $u_{w_i} \in E_{w_i}$. The first term is annihilated by $\bar{v}_i$, the second term is annihilated by $\bar{w}_i$, so any $u \in U = \text{im}(\varphi)$

is annihilated by $\bar{v}_i \otimes \bar{w}_i$. The annihilator of $\text{im}(\varphi)$ is $\ker(\varphi^\vee)$, so $v_i \otimes w_i \in \Gamma_{V \otimes W}^\varphi$. Hence $\Gamma_V^\varphi \otimes \Gamma_W^\varphi \subset \mathbb{P}(\ker(\varphi^\vee))$.

Conversely, whenever $v \otimes w \in \mathbb{P}(\ker(\varphi^\vee))$ then $\bar{w} \in \ker(\varphi_V(\bar{v}))$, so $\varphi_V(\bar{v})$ is rank at most two and $v \in \Gamma_V^\varphi$, and similarly $w \in \Gamma_W^\varphi$. Then $v = v_i$ for some i , and so $\text{Span}\{w, w_i\} \subset \ker(\varphi_V(\bar{v}))$, and either $w = w_i$ or $\varphi_V(\bar{v})$ is rank one. In the second case, S_φ then contains the plane $\mathbb{P}(\text{coker}(\varphi_V(\bar{v})))$ and is not irreducible, so $w = w_i$ and hence

$$\Gamma_V^\varphi \otimes \Gamma_W^\varphi = \Gamma_{V \otimes W}^\varphi.$$

□

Corollary 2.33. *The Gale-transform of 6 general points in $\mathbb{P}^2$ extends to a birational map $\mathbb{P}^2 \dashrightarrow \mathbb{P}^2$ given by blowing up the 6 points, and then blowing down the strict transforms of the conics through five of these points.*

Proof. The birational map is $\pi_W \circ \pi_V^{-1}$. □

Remark 2.34. Corollary 2.33 *does not* generalise to the case where Γ_V^φ has three collinear points, because there is no isomorphism $f : \text{Bl}_{\Gamma_V^\varphi} \mathbb{P}V \rightarrow \text{Bl}_{\Gamma_W^\varphi} \mathbb{P}W$ that sends $E_{v_i} \mapsto E_{w_i}$.

Proof. Fix an ordering on Γ_V^φ so that v_1, v_2, v_3 are linearly independent and v_4, v_5, v_6 are collinear. The construction in Corollary 2.33 cannot work, because the conic that passes through v_1, v_2, v_3, v_4 and v_5 is the product of two lines. It isn't immediate that there isn't some other generalisation, however.

We prove no such isomorphism exists by contradiction. Suppose such an f exists. Let the line through v_4, v_5, v_6 be L_V , with strict transform $\hat{L}_V$ under the blowup $\pi_V : \text{Bl}_{\Gamma_V^\varphi} \mathbb{P}V \rightarrow \mathbb{P}V$. Then the image of $\hat{L}_V$ under f is a (-2) -curve that passes through the three exceptional divisors $E_{w_4}, E_{w_5}, E_{w_6}$, and is the strict transform of a curve $L_W := (\pi_W \circ f)(\hat{L}_V) \subset \mathbb{P}W$ that passes through w_4, w_5, w_6 . Then L_W must be an irreducible conic passing through all 6 points in Γ_W^φ , since w_4, w_5, w_6 are linearly independent (Remark 2.5). But no such conic exists, since w_1, w_2, w_3 are all contained in a line, a contradiction.

We will later see that Γ_V^φ and Γ_W^φ are always isomorphic as schemes, but the isomorphism $\Gamma_V^\varphi \rightarrow \Gamma_W^\varphi$ is never induced from the Gale-transform. □

Remark 2.35. Corollary 2.33 *does* generalise to the case of 6 points on an irreducible conic. 6 points on a conic are self-associated, so the blowups of Γ_V and Γ_W are trivially isomorphic. However, we will soon see that Γ_V^φ and Γ_W^φ can never be contained in a conic.

2.3 The minors of the matrix

Recall that representing a smooth cubic surface S_φ by blowing up 6 points $\Gamma_V^\varphi \hookrightarrow \mathbb{P}^2$ induces the anticanonical embedding $S_\varphi \hookrightarrow \mathbb{P}(H^0(\mathbb{P}^2, \mathcal{I}_V^\varphi(3))^\vee)$. The blowup $\pi_V : S \cong \text{Bl}_{\Gamma_V^\varphi} \mathbb{P}V \rightarrow \mathbb{P}V$ then induces an isomorphism $U^\vee \cong H^0(\mathbb{P}V, \mathcal{I}_V^\varphi(3))$ for any determinantal representation of a smooth cubic surface.

The mere existence of this isomorphism is uninteresting because they are just vector spaces of the same dimension. It turns out that this isomorphism arises canonically from the structure of φ , with the linear system of cubics through Γ_V^φ being generated by the minors of the matrix φ_V even when S_φ is a singular cubic surface with finitely many lines. As a result, φ_V is the presentation matrix of the ideal I_V^φ whenever S_φ has finitely many lines.

To see this, we will first briefly revisit our discussion of the various kernel and cokernel sheaves associated to φ , where S_φ is a normal cubic surface with finitely many lines.

Definition 2.36. To the cokernel sheaves $\text{coker}(\tilde{\varphi}_V|_{\text{rk } 3})$ on $\mathbb{P}V$ and $\text{coker}(\tilde{\varphi}_W|_{\text{rk } 3})$ on $\mathbb{P}W$ we associate the morphisms

$$m_V : \mathbb{P}V_{\varphi, \text{rk } 3} \longrightarrow \mathbb{P}U, \quad m_W : \mathbb{P}W_{\varphi, \text{rk } 3} \longrightarrow \mathbb{P}U.$$

These are defined similarly to π_V and π_W on closed points, where $m_V(v)$ (resp. $m_W(w)$) is the $u \in \mathbb{P}U$ that is in the kernel of $\varphi_V(\bar{v})$ (resp. $\varphi_W(\bar{w})$).

The maps m_V and m_W are indeed morphisms, in the exact same way as π_V and π_W (Definition 2.21, from the identification $U^\vee \cong H^0(\mathbb{P}V, U^\vee \otimes \mathcal{O}_{\mathbb{P}V}) \cong H^0(\mathbb{P}V, \text{coker}(\tilde{\varphi}_V))$ and the identical version for W).

Lemma 2.37. *The maps m_V and m_W are birational maps from $\mathbb{P}V$ to S_φ and $\mathbb{P}W$ to S_φ , respectively. m_V is the birational inverse of π_V and m_W is the birational inverse of π_W .*

Proof. A closed point u is in the image of m_V if and only if $\varphi_U(\bar{u})$ is rank two, so the image of m_V is indeed contained in S_φ .

We now directly evaluate $(m_V \circ \pi_V)(u)$ to show they are inverses. For any rank two point $u \in S_\varphi$ that isn't contained in a line of type K, $\pi_V(u)$ is the unique $v \in \mathbb{P}V_{\varphi, \text{rk } 3}$ such that $\bar{v} \in \ker(\varphi_U(\bar{u}))$. Then m_V is defined, and $\bar{u} \in \text{coker}(\varphi_V(\bar{v}))$, so $m_V(v) = m_V(\pi_V(u)) = u$, and so $m_V \circ \pi_V = \text{id}$ on an open subscheme of S_φ , hence m_V is birational onto S_φ and its inverse is π_V . $\square$

Recall that I_V^φ and I_W^φ are the ideals generated by the minors of φ_V and φ_W , respectively.

Lemma 2.38. *Fix an identification $U \cong U^\vee$. For any tensor $\varphi \in U^\vee \otimes V^\vee \otimes W^\vee$, there is a canonical isomorphism of linear systems*

$$\mathbb{P}U^\vee \cong |\mathcal{I}_V^\varphi(3)| \cong |\mathcal{I}_W^\varphi(3)|$$

and the isomorphism $|\mathcal{I}_V^\varphi(3)| \cong |\mathcal{I}_W^\varphi(3)|$ is independent of the chosen identification of $U \cong U^\vee$.

Proof. We will first formalise what we mean by “the ideal generated by the minors of φ_V ”. Choose a basis $\{\bar{u}_0, \bar{u}_1, \bar{u}_2, \bar{u}_3\}$ for U , and a basis for V and W . The i^{th} minor m_i of φ_V is found by deleting the i^{th} row of φ_V , taking the determinant, and multiplying by $(-1)^i$. This gives a map $U^\vee \rightarrow (I_V^\varphi)_3$, sending the basis vector $u_i^\vee \mapsto m_i$ (since I_V^φ is generated by the m_i s by construction).

The i^{th} minor vanishes at a point $v \in \mathbb{P}V$ if and only if $\bar{u}_i \in \text{coker}(\varphi_V(\bar{v}))$. More generally, a linear combination $c_0 m_0 + \cdots + c_3 m_3$ of minors vanishes when $c_0 \bar{u}_0 + \cdots + c_3 \bar{u}_3 \in \text{coker}(\varphi_V(\bar{v}))$. But when changing basis, the minors are identified with elements of $U^\vee$. So this relies on choosing an isomorphism $U \cong U^\vee$, there is no coordinate-free map $U^\vee \cong (I_V^\varphi)_3$ (but the specific basis chosen for U was unimportant).

Doing the same process for I_W^φ , we get an isomorphism $(I_V^\varphi)_3 \cong (I_W^\varphi)_3$, and changing the basis for V and W only scales these minors by a constant. So there is a coordinate-free isomorphism of projective spaces. $\square$

Remark 2.39. The point of phrasing this as depending on an identification $U \cong U^\vee$ instead of depending on a basis for U is that this is really a statement about the maximal minors of an element of $(U^\vee \otimes V^\vee \otimes W^\vee) \otimes A$ for any ring A (by an identical proof). Given any morphism $\text{Spec}(A) \rightarrow \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$, this identification of linear systems can be made globally and does not depend on the tensor φ .

Lemma 2.40. *Fix an identification $U \cong U^\vee$ and a tensor $\varphi \in U^\vee \otimes V^\vee \otimes W^\vee$. The determinant cubic $\det(\varphi_U)$ gives a cubic relation between the coordinate minors of φ_V , and also between the coordinate minors of φ_W . When the associated cubic surface S_φ is irreducible, this relation is minimal.*

Proof. We first show this is true when S_φ is smooth. Given a fixed isomorphism $U \cong U^\vee$, the linear system of cubics through Γ_V^φ (i.e. the linear system of minors of φ_V) gives a rational map $m'_V : \mathbb{P}V \dashrightarrow \mathbb{P}U$, sending a point $v \in \mathbb{P}V$ to the point $[m_0(v) : m_1(v) : m_2(v) : m_3(v)] \in \mathbb{P}U$. The image of this map is contained in S_φ , and so $\det(\varphi_U)(m_0(v), m_1(v), m_2(v), m_3(v)) = 0$ on the open subscheme

$\mathbb{P}V_{\varphi, \text{rk } 3} \subset \mathbb{P}V$, so $\det(\varphi_U)(m_0, m_1, m_2, m_3) = 0$ (as it is a degree nine polynomial in $k[V^\vee]_\bullet$ that vanishes on an open set).

A general tensor cuts out a smooth cubic surface, and so on a dense open subset of $U^\vee \otimes V^\vee \otimes W^\vee$ the determinant cubic gives a relation between the minors. But $\det(\varphi_U)(m_0, m_1, m_2, m_3) = 0$ is a polynomial condition in coordinates that holds generally, therefore it holds everywhere. $\square$

Example 2.41. We continue Example 2.30. The 4×3 matrix is

$$\varphi_V^{4A_1} = \begin{pmatrix} v_0 & 0 & 0 \\ v_1 + v_2 & v_0 + v_2 & v_0 + v_1 + v_2 \\ 0 & v_1 & 0 \\ 0 & 0 & v_2 \end{pmatrix}. \quad (2.5)$$

The minors of $\varphi_V^{4A_1}$ are

$$m_0 = v_1 v_2 (v_0 + v_2), \quad m_1 = -v_0 v_1 v_2, \quad m_2 = v_0 v_2 (v_0 + v_2), \quad m_3 = -v_0 v_1 (v_0 + v_1 + v_2)$$

and one can verify that $\det(\varphi_U(m_0, m_1, m_2, m_3)) = 0$ by substituting into (2.2).

Theorem 2.42. Fix a tensor $\varphi \in U^\vee \otimes V^\vee \otimes W^\vee$. The matrix φ_V is the matrix of linear relations between the four generators of $I_V^\varphi(3)$. If the cubic surface S_φ has finitely many lines, then φ_V is the presentation matrix of a free resolution of the ideal I_V^φ and there is an isomorphism $\mathcal{I}_V^\varphi(3) \cong \text{coker}(\tilde{\varphi}_V)$ and

$$0 \longrightarrow W \otimes \mathcal{O}_{\mathbb{P}V}(-1) \xrightarrow{\tilde{\varphi}_V} U^\vee \otimes \mathcal{O}_{\mathbb{P}V} \longrightarrow \mathcal{I}_V^\varphi(3) \longrightarrow 0. \quad (2.6)$$

Proof. The columns of φ_V give linear relations between its minors because of the cofactor expansion: the linear combination $\sum_{i=0}^3 (\varphi_V)_{ij} m_i = 0$ because this is the determinant of the 4×4 matrix obtained by duplicating a column of φ_V . Then the generators of $I_V^\varphi(3)$ are all contained in $\text{coker}(\varphi_V)$, so $I_V^\varphi(3) \subset \text{coker}(\varphi_V)$ is a submodule and $\mathcal{I}_V^\varphi(3) \subset \text{coker}(\tilde{\varphi}_V)$ is a subsheaf (this is true for any tensor φ).

When S_φ has finitely many lines, $\text{coker}(\tilde{\varphi}_V)$ has rank at most one and so must be isomorphic to a twisted ideal sheaf $\mathcal{I}(d) \subset \mathcal{O}_{\mathbb{P}V}(d)$ for some ideal sheaf $\mathcal{I}$ and integer d . The sheaf $\text{coker}(\tilde{\varphi}_V)$ drops rank only on Γ_V^φ (when all the minors vanish!), and so after blowing up $\varepsilon : \text{Bl}_{\Gamma_V^\varphi} \mathbb{P}V \rightarrow \mathbb{P}V$ the inverse image ideal sheaf $\varepsilon^{-1}\mathcal{I}$ is invertible. But then $\mathcal{I} \subset \mathcal{I}_V^\varphi$. We find that $\mathcal{I} = \mathcal{I}_V^\varphi$ and $d = 3$, so $\text{coker}(\tilde{\varphi}_V) = \mathcal{I}_V^\varphi(3)$.

Then the sequence $0 \rightarrow W \otimes k[V^\vee]_\bullet(-4) \xrightarrow{\varphi_V} U^\vee \otimes k[V^\vee]_\bullet \rightarrow I_V^\varphi \rightarrow 0$ is exact, so φ_V is the presentation matrix for a free resolution of I_V^φ . $\square$

Corollary 2.43. *When S_φ has finitely many lines, the linear system of cubics $|\mathcal{I}_V^\varphi(3)|$ through Γ_V^φ induces a birational morphism $\tilde{m}_V : \text{Bl}_{\Gamma_V^\varphi} \mathbb{P}V \rightarrow S_\varphi$ after blowing up $\varepsilon : \text{Bl}_{\Gamma_V^\varphi} \mathbb{P}V \rightarrow \mathbb{P}V$. The map $\tilde{m}_V$ factors as $m_V \circ \varepsilon$ where $m_V \circ \varepsilon$ is defined, and $\tilde{m}_V$ is the blowup of S_φ along $S_{\varphi, \text{rk } 1}$.*

Proof. Blowing up $\mathbb{P}V$ along Γ_V^φ resolves the basepoints of the linear system $|\text{coker}(\tilde{\varphi}_V)| \cong |\mathcal{I}_V^\varphi(3)|$. So $\tilde{m}_V$ is a morphism, and $\tilde{m}_V = m_V \circ \varepsilon$ where defined. Then $\tilde{m}_V$ is birational. To see it is the blowup of S_φ along $S_{\varphi, \text{rk } 1}$, we consider the ideal sheaf that cuts out $S_{\varphi, \text{rk } 1}$ (generated by the 9 maximal minors of φ_U), and apply a similar argument. $\square$

Example 2.44. We continue Example 2.41. The map $\tilde{m}_V : \text{Bl}_{\Gamma_V^{4A_1}} \mathbb{P}V \rightarrow S_{4A_1}$ contracts the strict transforms of 4 lines: the lines cut out by the equations $v_0 = 0$, $v_0 + v_1 + v_2 = 0$, $v_1 = 0$, $v_2 = 0$. The respective images of these lines are the rank one singularities

$$[1 : 0 : 0 : 0], [1 : 1 : 1 : 0], [0 : 0 : 1 : 0], [0 : 0 : 0 : 1].$$

These singularities are all A_1 singularities, so the surface $\text{Bl}_{\Gamma_V^{4A_1}} \mathbb{P}V$ is smooth — each A_1 singularity is resolved after only one blowup. This is expected: $\text{Bl}_{\Gamma_V^{4A_1}} \mathbb{P}V$ is the blowup of the 6 reduced points $\Gamma_V^{4A_1}$, so had to be smooth.

Remark 2.45. The *Hilbert-Burch theorem* (see [Eis95, Thm. 20.15]) is normally invoked as the starting point when classifying determinantal representations of a cubic surface (cf. [Ng02], [Dol12, §9.3.2]). When R is a polynomial ring and $I \subset R$ is a homogeneous ideal with n generators where $\text{depth}(I) = \text{codim}(I) = 2$ (I cuts out a codimension 2 Cohen-Macaulay subscheme of $\text{Spec}(R)$), there is a free resolution

$$0 \longrightarrow R^{n-1} \xrightarrow{\phi} R^n \longrightarrow R \longrightarrow R/I \longrightarrow 0$$

for some n such that I is generated by the minors of ϕ under the map $R^n \rightarrow I \subset R$ (up to taking radicals).

When I is generated by four cubics and cuts out a finite subscheme $\Gamma \hookrightarrow \mathbb{P}^2$, this is the converse to Theorem 2.42 (take an affine chart $\mathbb{A}^2 \subset \mathbb{P}^2$ containing Γ , and dehomogenise I to be an ideal in the polynomial ring $k[x, y]$). This is essentially our approach, except we use the geometry of determinantal representations to prove this special case of the Hilbert-Burch theorem.

Care must be taken when generalising the case for smooth surfaces. Dolgachev claims that the morphism $\tilde{m}_V$ is an isomorphism $\text{Bl}_{\Gamma_V^\varphi} \mathbb{P}V \rightarrow S_\varphi$. By Corollary 2.43, this is only true when S_φ is a rank two surface. And we can also just directly

compute that whenever $u \in S_\varphi$ is a rank one singularity, $\tilde{m}_V$ contracts the strict transform of the line $\mathbb{P}(\ker(\varphi_U(\bar{u})) \subset \mathbb{P}V$ to u .

We now have the tools to fully understand the determinantal representations of a smooth cubic surface, so we will briefly pause our general discussion to complete our discussion of the classical theory of determinantal representations of smooth cubic surfaces.

2.4 Smooth determinantal representations

Let S be smooth cubic surface. Recall that there are exactly 72 sixes on S , and that given any six s there is a morphism $\pi_s : S \rightarrow \mathbb{P}^2$ that contracts the six s (Corollary 1.41), blowing up a set of 6 points $\Gamma_s \hookrightarrow \mathbb{P}^2$. The point configuration Γ_s is unique up to projective equivalence.

Corollary 2.46. *Every six s on a smooth cubic surface S induces a determinantal representation φ of $S = S_\varphi$, where the six s is the six of lines of type K on S_φ .*

A smooth cubic surface with no non-trivial automorphisms has exactly 72 inequivalent determinantal representations.

Proof. For any six s , the presentation matrix of the ideal of cubics through Γ_s induces a determinantal representation of S .

Any non-trivial automorphism σ of S acts non-trivially on the set of lines. Suppose it sends a six s to a six s' (we could have $s = s'$, as long as σ non-trivially permutes the order of the lines in s). Then the composition $\pi_{s'} \circ \sigma \circ \pi_s^{-1}$ is an automorphism of $\mathbb{P}^2$ interchanging the two point configurations, so they induce equivalent determinantal representations. Conversely, if two point configurations Γ_s and $\Gamma_{s'}$ are isomorphic then the projective isomorphism $\nu : \mathbb{P}V \rightarrow \mathbb{P}V$ sending Γ_s to $\Gamma_{s'}$ extends to a non-trivial automorphism $\pi_{s'}^{-1} \circ \nu \circ \pi_s$ of S .

A general smooth cubic surface has no non-trivial automorphisms, these occur only when there is an Eckardt point (a point where three lines intersect), a codimension one condition. $\square$

Remark 2.47. Fix a determinantal representation φ of a smooth cubic surface S , and let $v_1, v_2, v_3 \in \Gamma_V^\varphi$ and $w_1, w_2, w_3 \in \Gamma_W^\varphi$ be the first half of the two Gale-dual point configurations. Letting these three vectors be the basis of V and W , the tensor φ vanishes at $v_1 \otimes w_1, v_2 \otimes w_2, v_3 \otimes w_3$, and so has a matrix representation of the form

$$\varphi_U = \begin{pmatrix} 0 & \varphi_{12} & \varphi_{13} \\ \varphi_{21} & 0 & \varphi_{23} \\ \varphi_{13} & \varphi_{23} & 0 \end{pmatrix}$$

for some linear forms $\varphi_{ij} \in k[U^\vee]_1$. Every determinantal representation of a smooth cubic surface can be written in this off-diagonal way.

But this isn't true for a general determinantal representation of a cubic surface — we would need to be able to find three distinct lines, and the A_1A_5 cubic surface only has two lines (but is still determinantal).

The 6 planes φ_{ij} are called *tritangent planes* to S_φ , since the intersection $S_\varphi \cap \{\varphi_{ij} = 0\}$ is the union of the three lines E_{v_i} and E_{w_j} and L_{ij} .

Remark 2.48. There is a classical construction of smooth determinantal cubic surfaces, dating back to the 19th century. In coordinates $av_1 + bv_2 + cv_3$ for V , we get a system of equations from solving the

$$\varphi_U(u) \begin{pmatrix} a \\ b \\ c \end{pmatrix} = 0 \iff \begin{aligned} b\varphi_{12}(u) + c\varphi_{13}(u) &= 0 \\ a\varphi_{21}(u) + c\varphi_{23}(u) &= 0 \\ a\varphi_{31}(u) + b\varphi_{32}(u) &= 0 \end{aligned}$$

which has a non-trivial solution if and only if $u \in S$. Dividing out a, b, c when they are not all zero (which for a general u is the case, because π_V is birational), we rewrite this system of equations as

$$\begin{aligned} \varphi_{12}(u) + \frac{\lambda'}{\lambda}\varphi_{13}(u) &= 0 \\ \frac{\mu'}{\mu}\varphi_{21}(u) + \varphi_{23}(u) &= 0 \\ \varphi_{31}(u) + \frac{\nu'}{\nu}\varphi_{32}(u) &= 0 \end{aligned}$$

subject to the relation $\frac{\lambda\mu\nu}{\lambda'\mu'\nu'} = \frac{abc}{abc} = 1$, and so the three plane pencils $\pi_1 = \lambda\varphi_{12} + \lambda'\varphi_{13}$, $\pi_2 = \mu'\varphi_{21} + \mu\varphi_{23}$, $\pi_3 = \nu\varphi_{31} + \nu'\varphi_{32}$ have an intersection point whenever $\lambda\mu\nu = \lambda'\mu'\nu'$, and the set of points satisfying this relation is S_φ .

Conversely, let S be a cubic surface with three skew lines l_1, l_2, l_3 . The three pencils of planes $\mathcal{P}_i$ through the lines l_i are families $\pi_i : \mathcal{P}_i \rightarrow \mathbb{P}^1$. A general plane $\pi_1(\lambda) \in \mathcal{P}_1$ and $\pi_2(\mu) \in \mathcal{P}_2$ intersect at three points on S , two contained in the lines l_1 and l_2 , and another point p somewhere else on S . There is a unique $\pi_3(\nu) \in \mathcal{P}_3$ containing p , and this incidence relation gives a two-dimensional subvariety $G \subset \mathbb{P}^1 \times \mathbb{P}^1 \times \mathbb{P}^1$. The image of G under the morphism

$$\Psi : \mathbb{P}^1 \times \mathbb{P}^1 \times \mathbb{P}^1 \longrightarrow \mathbb{P}^3, \quad (\lambda, \mu, \nu) \longmapsto \pi_1(\lambda) \cap \pi_2(\mu) \cap \pi_3(\nu)$$

is an open subvariety of S . There is a change of coordinates on the three copies

of $\mathbb{P}^1$ so that the plane pencils are given by the equations

$$\pi_1(\lambda : \lambda') = \lambda A_2 + \lambda' A_3, \quad \pi_2(\mu : \mu') = \mu' B_1 + \mu B_3, \quad \pi_3(\nu : \nu') = \nu C_1 + \nu' C_2$$

such that the subvariety $G \subset \mathbb{P}^1 \times \mathbb{P}^1 \times \mathbb{P}^1$ is cut out by the relation $\lambda\mu\nu = \lambda'\mu'\nu'$. Then the equation defining the cubic surface is the determinant of the matrix

$$\varphi_U = \begin{pmatrix} 0 & A_2 & A_3 \\ B_1 & 0 & B_3 \\ C_1 & C_2 & 0 \end{pmatrix}.$$

This argument was first discovered by Segre [Seg06], some work is needed to show that the relation defining G is of this form up to a coordinate change (details can be found in [Dol12, Thm. 9.3.2], or in [Aug62]¹), and a longer discussion of this classical correspondence and other related constructions can be found in Dolgachev's survey [Dol04].

Remark 2.49 (What if $k \neq \bar{k}$?). [Bea00, Cor. 6.5] Everything almost works over a non-closed field. If $S \hookrightarrow \mathbb{P}_k^2$ is a smooth cubic surface defined over some field k , then the following four statements are equivalent

- S is determinantal.
- S contains a twisted cubic.
- There is a birational morphism $\pi : S \rightarrow \mathbb{P}_k^2$.
- S contains a k -point and six skew lines defined over k .

When S is not smooth, then S is determinantal if and only if there is a birational morphism $\pi : S \rightarrow \mathbb{P}_k^2$ that is the blowup of a length 6 subscheme not contained in a conic. We will properly describe this in the following two sections.

As an example, giving a determinantal representation for a real cubic surface is the same as giving a self-adjoint determinantal representation over $\mathbb{C}$. Buckley and Kosir show that there are either 0, 2, 4, 6 or 24 inequivalent self-adjoint determinantal representations of a given smooth real cubic surface [BK07]. Their paper is also where I first learnt about the off-diagonal representation.

¹You may need a Latin dictionary.

2.5 Singular determinantal representations

We return to the discussion of determinantal representations of normal cubic surfaces with finitely many lines. We approach this by noting that finding a resolution of singularities for S_φ is the same as resolving the singularities on $\mathrm{Bl}_{\Gamma_V^\varphi} \mathbb{P}V$, which we will use to fully understand determinantal representations of normal cubic surfaces with finitely many lines.

Recall Sakamaki's description of the minimal resolution of singularities $r : \tilde{S} \rightarrow S$ for any normal cubic surface S with finitely many lines [Sak10], coming from blowing up $\mathbb{P}^2$ 6 times. The important observations are that $\mathrm{Pic}(\tilde{S}) \cong \mathbb{Z}^7$, and because r is a minimal resolution of ADE singularities it is minimal. So any other resolution $r' : \tilde{S}' \rightarrow S$ factors uniquely through r . In particular, given any other resolution $\tilde{S}' \rightarrow S$ where $\mathrm{Pic}(\tilde{S}') \cong \mathbb{Z}^7$, then $\tilde{S} \cong \tilde{S}'$.

Recall Examples 1.48 and 1.50, which show that the exceptional fibres of the blowup of a curvilinear subscheme $\Gamma \hookrightarrow \mathbb{P}^2$ has only A_m singularities if and only if Γ is curvilinear (and otherwise $\mathrm{Bl}_\Gamma \mathbb{P}^2$ is not even normal).

Definition 2.50. Let $\Gamma \hookrightarrow \mathbb{P}^2$ be a finite, curvilinear subscheme. We define the *smooth blowup* $\mathrm{SmBl}_\Gamma \mathbb{P}^2$ of $\mathbb{P}^2$ at Γ to be the minimal resolution of the A_m singularities of $\mathrm{Bl}_\Gamma \mathbb{P}^2$.

If $\tilde{S}$ is the smooth blowup of $\mathbb{P}^2$ at Γ , then the morphism $\tilde{\varepsilon} : \tilde{S} \rightarrow \mathbb{P}^2$ is the composition of $\mathrm{len}(\Gamma)$ repeated blowups, and the exceptional fibre above a point $p \in \Gamma$ consists of $(\mathrm{len}_p(\Gamma) - 1)$ (-2) -curves, and one (-1) -curve. Note that if Γ is reduced, then $\mathrm{Bl}_\Gamma \mathbb{P}^2 = \mathrm{SmBl}_\Gamma \mathbb{P}^2$. And we have $\mathrm{Pic}(\mathrm{SmBl}_\Gamma \mathbb{P}^2) \cong \mathbb{Z}^{\mathrm{len}(\Gamma)+1}$.

Proposition 2.51. *Whenever S_φ is normal with finitely many lines, the smooth blowup $\tilde{\varepsilon}_V : \mathrm{SmBl}_{\Gamma_V^\varphi} \mathbb{P}V \rightarrow \mathbb{P}V$ gives a minimal resolution of singularities for S_φ .*

Proof. We first need to prove that Γ_V^φ is curvilinear. The scheme $\mathrm{Bl}_{\Gamma_V^\varphi} \mathbb{P}V$ has finitely many singularities if and only if Γ_V^φ is curvilinear. And the morphism $\tilde{m}_V : \mathrm{Bl}_{\Gamma_V^\varphi} \mathbb{P}V \rightarrow S_\varphi$ only contracts the strict transforms of lines in $\mathbb{P}V$ (and is an isomorphism away from these points). The exceptional curves have finitely many singularities, and so does S_φ , so $\mathrm{Bl}_{\Gamma_V^\varphi} \mathbb{P}V$ is curvilinear. Then Γ_V^φ is cut out by four independent cubic equations, so $\mathrm{len}(\Gamma_V^\varphi) \leq 6$ and $\dim \mathrm{Pic}(\mathrm{SmBl}_{\Gamma_V^\varphi} \mathbb{P}V) \leq 7$. And the morphism $\tilde{m}_V \circ \tilde{\varepsilon}_V : \mathrm{SmBl}_{\Gamma_V^\varphi} \mathbb{P}V \rightarrow S_\varphi$ is a resolution of singularities of S_φ , so $\mathrm{SmBl}_{\Gamma_V^\varphi} \mathbb{P}V$ is a minimal resolution of singularities. $\square$

Remark 2.52. While the blowup of a finite curvilinear subscheme $\Gamma \hookrightarrow \mathbb{P}^2$ only has A_n singularities, the resolution $\mathrm{SmBl}_\Gamma \mathbb{P}^2$ can have (-2) -curves that intersect as the D_n and E_n diagrams (recall Example 1.55). It can also have (-3) or worse

curves, but on a length 6 subscheme, these only occur if there is a line L where $\text{len}(L \cap \Gamma) \geq 4$. We will see that Γ_V^φ can never be of this form (Proposition 2.57).

Corollary 2.53. *Whenever S_φ is normal with finitely many lines, Γ_V^φ and Γ_W^φ are length 6, curvilinear, and isomorphic as schemes.*

Proof. The schemes $\text{Bl}_{\Gamma_V^\varphi} \mathbb{P}V$ and $\text{Bl}_{\Gamma_W^\varphi} \mathbb{P}W$ are isomorphic (Corollary 2.43), so have the same A_n singularities. So Γ_V^φ and Γ_W^φ are curvilinear, have the same length, and have the same multiplicities at non-reduced points. $\square$

Now that we know that Γ_V^φ and Γ_W^φ are isomorphic and curvilinear, we can show that they are Gale-dual (note that we actually needed them to be isomorphic to make this claim, otherwise we don't have a canonical isomorphism between the coordinate rings $H^0(\Gamma_V, \mathcal{O}_{\Gamma_V})$ and $H^0(\Gamma_W, \mathcal{O}_{\Gamma_W})$).

Corollary 2.54. *Whenever S_φ is normal with finitely many lines, the linear series $(\mathcal{O}_{\Gamma_V^\varphi}, V)$ and $(\mathcal{O}_{\Gamma_W^\varphi}, W)$ are Gale-dual.*

Proof. This can be computed using a cohomological argument from Dolgachev's exposition [Dol12] (note that his U is our W and his V is our $U^\vee$). Recall that Γ_V^φ is cut out by the maximal minors of φ_V , whenever Γ_V^φ is finite and length 6 then φ_V is the presentation matrix of $\mathcal{I}_V^\varphi$ and we have an exact sequence

$$0 \longrightarrow W \otimes \mathcal{O}_{\mathbb{P}V}(-1) \xrightarrow{\tilde{\varphi}_V} U^\vee \otimes \mathcal{O}_{\mathbb{P}V} \longrightarrow \mathcal{I}_V^\varphi(3) \longrightarrow 0. \quad (2.7)$$

Twisting the exact sequence (2.7) by $\mathcal{O}_{\mathbb{P}V}(-2)$ then gives a long exact sequence on cohomology

$$\begin{aligned} \cdots \longrightarrow U^\vee \otimes H^1(\mathbb{P}V, \mathcal{O}_{\mathbb{P}V}(-2)) &\longrightarrow H^1(\mathbb{P}V, \mathcal{I}_V^\varphi(1)) \\ &\longrightarrow W \otimes H^2(\mathbb{P}V, \mathcal{O}_{\mathbb{P}V}(-3)) \longrightarrow U^\vee \otimes H^2(\mathbb{P}V, \mathcal{O}_{\mathbb{P}V}(-2)) \longrightarrow \cdots \end{aligned} \quad (2.8)$$

and the first and last terms are both zero because all cohomology of $\mathcal{O}_{\mathbb{P}^2}(-2)$ vanishes. The trace isomorphism $H^2(\mathbb{P}V, \mathcal{O}_{\mathbb{P}V}(-3)) \cong k$ then gives a canonical identification $H^1(\mathbb{P}V, \mathcal{I}_V^\varphi(1)) = W$. We also have the standard exact sequence

$$0 \longrightarrow \mathcal{I}_V^\varphi(1) \longrightarrow \mathcal{O}_{\mathbb{P}V}(1) \longrightarrow \mathcal{O}_{\Gamma_V^\varphi}(1) \longrightarrow 0 \quad (2.9)$$

found by twisting the sequence $0 \rightarrow \mathcal{I}_V^\varphi \rightarrow \mathcal{O}_{\mathbb{P}V} \rightarrow \mathcal{O}_{\Gamma_V^\varphi} \rightarrow 0$ by $\mathcal{O}_{\mathbb{P}V}(1)$. The subscheme Γ_V^φ is finite, so $\mathcal{O}_{\Gamma_V^\varphi}(1) \cong \mathcal{O}_{\Gamma_V^\varphi}$ and we take cohomology of this sequence, noting that $H^0(\mathbb{P}V, \mathcal{I}_V^\varphi(1)) = 0$ vanishes because $\mathcal{I}_V^\varphi$ is generated in degree three and up, and $H^1(\mathbb{P}V, \mathcal{O}_{\mathbb{P}V}(1)) = 0$. This gives an exact sequence

$$0 \longrightarrow H^0(\mathbb{P}V, \mathcal{O}_{\mathbb{P}V}(1)) \longrightarrow H^0(\Gamma_V^\varphi, \mathcal{O}_{\Gamma_V^\varphi}) \longrightarrow H^1(\mathbb{P}V, \mathcal{I}_V^\varphi(1)) \longrightarrow 0.$$

Or more cleanly using the previous isomorphism and $H^0(\mathbb{P}V, \mathcal{O}_{\mathbb{P}V}(1)) = V^\vee$,

$$0 \longrightarrow V^\vee \longrightarrow H^0(\Gamma_V^\varphi, \mathcal{O}_{\Gamma_V^\varphi}) \longrightarrow W \longrightarrow 0 \quad (2.10)$$

and so the linear series $(\mathcal{O}_{\Gamma_V^\varphi}, V)$ and $(\mathcal{O}_{\Gamma_W^\varphi}, W)$ are Gale-dual. $\square$

Remark 2.55. The linear series $(\mathcal{O}_{\Gamma_V^\varphi}, V)$ and $(\mathcal{O}_{\Gamma_W^\varphi}, W)$ are also Gale-dual whenever Γ_V^φ is finite and S_φ has finitely many lines. This additional hypothesis is needed, as there are determinantal representations of non-normal cubic surfaces with finitely many lines.

We want to use Gale-duality to find all determinantal representations of a given normal cubic surface with finitely many lines. To do this, we need to first rule out when a length 6 subscheme does not come from a $4 \times 3 \times 3$ tensor.

Definition 2.56. A length 6 subscheme $\Gamma \hookrightarrow \mathbb{P}V$ is said to be *determinantal* if there exists a $4 \times 3 \times 3$ tensor φ where $\Gamma = \Gamma_V^\varphi$. The blowup $\text{Bl}_\Gamma \mathbb{P}^2$ is said to be determinantal if Γ is.

Proposition 2.57. *Let φ be any $4 \times 3 \times 3$ tensor, and $L \hookrightarrow \mathbb{P}V$ a line. If $\text{len}(L \cap \Gamma_V^\varphi) \geq 4$, then $L \subset \Gamma_V^\varphi$.*

Proof. The restriction of the minors m_i of φ_V to L are four homogeneous cubics $m_i|_L$ on $L \cong \mathbb{P}^1$. But each $m_i|_L$ has at least four roots on the subscheme $(\Gamma_V^\varphi \cap \mathbb{P}^1) \hookrightarrow \mathbb{P}^1$, so $m_i|_L = 0$ and $L \subset \Gamma_V^\varphi$. $\square$

Proposition 2.58. *Any curvilinear length 6 subscheme contained in a conic is not determinantal. In particular, Sakamaki's resolution of singularities [Sak10] is non-determinantal.*

Proof. Suppose φ is a $4 \times 3 \times 3$ tensor where Γ_V^φ is curvilinear, length 6, and contained in a conic C . Then the strict transform $\hat{C}$ of C under the resolution $\tilde{\varepsilon}: \tilde{S} \rightarrow \mathbb{P}V$ is a (-2) -curve. So r contracts $\hat{C}$ to a point $u \in S_\varphi$, and $m_V^{-1}(u) = C \setminus \Gamma_V^\varphi$. But this is impossible — the fibres of m_V can only be points, lines, or empty. A similar argument holds when Γ_V^φ is contained in a reducible conic C , it reduces to the case that Γ_V^φ has a length 4 collinear subscheme, but then Γ_V^φ is not finite.

This can also be proved algebraically, by contradiction on the possible linear relations between the four cubic generators of the ideal that cuts out Γ_V . $\square$

Theorem 2.59. *Let $\Gamma_V \hookrightarrow \mathbb{P}V$ be a length 6 curvilinear scheme. The scheme Γ_V is determinantal if and only if it is not contained in a conic and has no collinear length 4 subscheme.*

Proof. We take the linear series $(\mathcal{O}_{\Gamma_V}, V)$ and its Gale-dual $(\mathcal{O}_{\Gamma_W}, W)$, and take the tensor product $\Gamma_V \otimes \Gamma_W \hookrightarrow \mathbb{P}(V \otimes W)$. This is cut out by four linear relations, which we write as a matrix $\varphi^\vee : V \otimes W \rightarrow U^\vee$. This is a $4 \times 3 \times 3$ tensor. Then Γ_V^φ and Γ_W^φ are either finite, or contain a line or a conic. Note that Γ_V^φ and Γ_W^φ contain Γ_V and Γ_W , respectively. The latter two options can only happen when one of Γ_V or Γ_W have a length 4 collinear subscheme, or are contained in a conic, which is not true by assumption. So Γ_V^φ and Γ_W^φ are finite, and hence equal to Γ_V and Γ_W (since they are at most length 6 and contain length 6 subschemes). $\square$

Remark 2.60. This proves that whenever $\mathcal{I} \subset \mathcal{O}_{\mathbb{P}^2}$ is an ideal sheaf that cuts out a curvilinear length 6 subscheme, not contained in a conic and not having a length 4 collinear subscheme, then $\mathcal{I}$ has a free resolution presented by a 4×3 matrix of linear forms, and is cut out by the minors of this matrix. This is the special case of the Hilbert-Burch theorem that is needed to classify determinantal representations of normal cubic surfaces with finitely many lines.

The connection between free resolutions is one of the motivating reasons to study Gale-duality, and Eisenbud and Popescu prove various related results to this example. The study of determinantal representations of cubic surfaces gives a strong motivation for why Gale-duality is such a rich geometric construction, however the geometry is usually more difficult to, the case of $r = s = 2$ has a lot of extra symmetry. Theorem 2.59 is a special case of [Eis95, Thm. 6.1].

Corollary 2.61. *Let S be a normal cubic surface with finitely many lines, and $r : \tilde{S} \rightarrow S$ a minimal resolution of singularities. Any morphism $\tilde{\varepsilon} : \tilde{S} \rightarrow \mathbb{P}^2$ identifies $\tilde{S}$ as the smooth blowup of a length 6 subscheme Γ .*

The surface S is determinantly represented by some tensor φ , where $\Gamma = \Gamma_V^\varphi$, if and only if Γ does not lie on a conic.

Proof. The subscheme Γ cannot have a collinear length 4 subscheme because the strict transform of this subscheme is a (-3) -curve. $\square$

Example 2.62. Recall the resolution of singularities for a D_4 cubic surface (Example 1.55). Because this does not involve blowing 6 points on a conic, then this resolution is determinantal. It is the only determinantal representation of a D_4 cubic surface, up to equivalence.

Corollary 2.63. *There are no rank two representations of a cubic surface with A_4 or A_5 singularities.*

Proof. If φ is a rank two representation of an A_5 cubic surface, then Γ_V^φ is a length 6 curvilinear point. This is contained in a conic, a contradiction. Similarly, if

Bl_{Γ_V} has an A_4 singularity and Γ_V has no collinear length 3 subscheme, then Γ_V is contained in a conic.

These surfaces are of course still determinantal, but any determinantal representation will always have a rank one singularity. $\square$

Corollary 2.64. *The E_6 cubic surface is not determinantal. Every other cubic surface is.*

Proof. The E_6 cubic surface S only has one line, so a resolution of singularities $\tilde{S} \rightarrow S$ can only be the smooth blowup of a length 6 subscheme supported at a single point, which does not have a length 4 intersection with a line. But then this subscheme is contained in a conic. $\square$

Remark 2.65. The non-determinantality of the E_6 cubic surface was first noted in the 19th century by Segre [Seg06], and Dolgachev's exposition [Dol12] proves the E_6 cubic is the only non-determinantal normal cubic surface because every length 6 subscheme associated to an E_6 cubic surface is contained in a conic, i.e. is not determinantal. Brundu and Logar explicitly compute a determinantal representation of every other cubic surface (up to projective equivalence) [BL98], and also prove the E_6 cubic is the only non-determinantal cubic surface.

Remark 2.66. We have arrived at a simple way to find determinantal representations of any normal cubic surface S with finitely many lines. All we need to do is:

- Find a composition of 6 blowups of $\mathbb{P}^2$ that has the same $(-n)$ -curves as the ADE resolution of singularities for S , and is not contained in a conic.
- Find the associated length 6 subscheme Γ_V , by contracting the exceptional (-2) -curves of the blowup.
- Option 1: find the four cubic equations that cut out Γ_V , find the matrix of linear relations between them, this is φ_V .
- Option 2: find the Gale-dual Γ_W , find $\text{Span}(\Gamma_V \otimes \Gamma_W)$, find the four linear equations that cut this vector space out, this is $\varphi^\vee$.

This is exactly how we started the repeated $4A_1$ example, using option 2. But option 1 is easier when Γ_V is non-reduced, or when using a computer algebra package to directly compute free resolutions.

Chapter 3

Families of lines on cubic surfaces

This chapter is essential to understand precisely under what conditions we can use our intuition from classical geometry as to how the lines in a cubic surface ‘should’ vary as we vary the surface. We want to be able to make claims like “as we continuously vary a smooth cubic surface, the lines inside it vary continuously”, and understand the line configuration on a singular cubic surface as a limit of the line configuration on a family of smooth cubic surfaces that limit to it.

In the Euclidean topology, when we have a family of geometric objects parametrised by a coordinate t we can often just compute the limit as $t \rightarrow t_0$ and the limiting object is often reasonably similar to what we started with. The algebraic analogue is taking a family (e.g. over $\mathbb{A}^1$), and computing the fibre above some closed point $t = t_0$. But on closed sets, the fibre dimension can jump up, and things can spontaneously appear and disappear.

Example 3.1 (Algebraic limits can be bad). Consider the family

$$X_T = \operatorname{Spec}(k[x, t]/(tx)) \rightarrow \operatorname{Spec}(k[t]).$$

The fibre at any $t \neq 0$ is the point $x = 0 \in \operatorname{Spec}(k[x])$. But the fibre at $t = 0$ is the affine line $\mathbb{A}^1$. So the “limit” at $t = 0$ of a family of points is a line.

The right condition to put on a family of schemes to stop pathological behaviour like this is called *flatness*, which among other things implies a constant fibre dimension (over a connected base). In this chapter, we introduce flatness and the Hilbert scheme (Section 3.1), and prove that the line configuration on a smooth cubic surface varies étalely if the cubic surface is varied flatly (the relative Hilbert scheme $\operatorname{Hilb}_{m+1}(X_T/T)$ of lines on a flat family $X_T \rightarrow T$ of smooth cubic surfaces is étale over T) in Section 3.3. This tells us that when a smooth cubic surface is varied flatly, lines cannot spontaneously appear or disappear (flat) and

do not collide (unramified), so the number of lines is always 27 and they vary nicely. We also show that the intersection number of a pair of lines in a flat family of smooth cubic surfaces is constant, meaning that the intersection graph of lines on any smooth cubic surface is the same. A consequence of this theorem is the well-known statement that every smooth cubic surface is the blowup of $\mathbb{P}^2$ in 72 different ways (Proposition 1.40), which we used in proving that there are exactly 72 inequivalent determinantal representations of a smooth cubic surface with no non-trivial automorphisms (Corollary 2.46).

We then show that any the Hilbert scheme of lines on any family of cubic surfaces $X_T \rightarrow T$ with finitely many lines is flat over T , which means that there are 27 lines on any cubic surface with finitely many lines, counting multiplicity. This lets us make sense of what it means for lines to collide in a limiting family of cubic surfaces, and normal cubic surfaces with finitely many lines are in fact characterised entirely by their line configuration. The key conclusions of this chapter are, roughly, that:

- Any cubic surface with finitely many lines has exactly 27 lines, counted with multiplicity. A smooth cubic surface has 27 distinct lines.
- The intersection graph of lines on any smooth cubic surface is the same.
- There is a scheme that parametrises the lines on a family of cubic surfaces $X_T \rightarrow T$, called the *Hilbert scheme* $\text{Hilb}_{m+1}(X_T/T)$. When $X_T \rightarrow T$ is a family of cubic surfaces with finitely many lines, the Hilbert scheme is flat over T .

Most of this chapter can be treated as a black box if you are not sufficiently familiar with scheme theory to follow the discussion of the Hilbert scheme.

In Section 3.4 we apply some of these properties to understand the relationship between families of point configurations, cubic surfaces, and determinantal representations. Using this, we outline a geometric method for finding determinantal representations of a given normal singular cubic surface with finitely many lines. And we provide some example applications: we explain why there are 50 inequivalent determinantal representations of a general A_1 cubic surface; why there are only 2 for a $4A_1$ cubic surface; and provide a proof that the E_6 cubic surface is not determinantal. These statements are all consequences of Theorem 2.59, but these three proofs do not rely heavily on the interpretation of φ_V as a presentation matrix, they only need an understanding of the Hilbert scheme of lines on a family of cubic surfaces.

3.1 Flatness and the Hilbert scheme

Definition 3.2 (Flat module). [Bou61, I.2, Defn. 1, p. 23] Let A be a ring and M an A -module. M is *flat* over A (or *A -flat*) if the following condition is satisfied: given any pair of A -modules N, N' and an injective R -module homomorphism $\nu : N \rightarrow N'$, the map

$$\mathrm{id}_M \otimes_R \nu : M \otimes_R N \longrightarrow M \otimes_R N'$$

is injective.

A map of rings $A \rightarrow B$ is *flat* if B is flat as an A -module.

Flatness satisfies many nice properties. The main one we will need is that flatness is *preserved under base-change*: If M is a flat A -module and B is any A -algebra, then the module $M \otimes_A B$ is flat over B .

Proposition 3.3. [Bou61, I.2, Prop. 1, p. 26] *M is A -flat if and only if the canonical map $J \otimes_A M \rightarrow M$ sending $j \otimes m \mapsto jm$ is injective. This is equivalent to the sequence $0 \rightarrow J \rightarrow A$ remaining exact after tensoring with M — it is enough to check flatness only on ideals $J \subset A$, instead of every single A -module N .*

Definition 3.4 (Flat morphism). [Har77, p. III.9] Let $f : X \rightarrow T$ be a morphism of schemes. f is flat at a point $x \in X$ if the stalk $\mathcal{O}_{X,x}$ is a flat $\mathcal{O}_{T,f(x)}$ -module, where the $\mathcal{O}_{Y,f(x)}$ -module structure is given by the map $f^\# : \mathcal{O}_{Y,f(x)} \rightarrow \mathcal{O}_{X,x}$. The map f is a *flat morphism* if f is flat at every $x \in X$. A family $f : X \rightarrow T$ is *flat over T* if f is a flat morphism.

If $f : X \rightarrow T$ and $g : T' \rightarrow T$ are flat morphisms of schemes, then the base change $X \times_T T'$ is flat over T' .

We will often parametrise the points of T by t , and for any point $t \in T$ we write $X_t = X \times_T \mathrm{Spec}(\kappa(t))$ (under the morphism $t : \mathrm{Spec}(\kappa(t)) \rightarrow T$). These are the *fibres* of the family $X \rightarrow T$. We are only really interested in the fibres above closed points $t \in T(k)$, but the general language is useful.

Definition 3.5 (Smooth, étale, unramified morphism). [Har77, p. III.10] A morphism $f : X \rightarrow Y$ is *smooth* if f is flat, locally of finite presentation, and all the fibres are smooth (i.e. a nonsingular scheme, in the sense of Definition 1.7). Concretely, if f is locally of finite presentation then f locally factors as $X' \hookrightarrow \mathbb{A}_{Y'}^n \rightarrow Y'$ for sufficiently small open affines $X' \subset X$ and $Y' \subset Y$. And so X_y is smooth if and only if each X'_y is a smooth subvariety of $\mathbb{A}_{\kappa(y)}^n$.

A morphism $f : X \rightarrow Y$ is *étale* if f is smooth and finite.

There is another condition, known as *unramified*, that we can place on morphisms. We won't worry about the precise definition, but it is known that

$$f \text{ is étale} \iff f \text{ is flat and unramified} \iff f \text{ is smooth and unramified.}$$

Definition 3.6 (Hilbert polynomial). [Har77, V, Thm. 9.9] Let κ be a field and $X \hookrightarrow \mathbb{P}_{\kappa}^n$ be a projective scheme over κ . Let $\mathcal{F}$ be a coherent sheaf on X .

Then $\chi(\mathcal{F}(m))$ is a polynomial in n . We call this polynomial $P \in \mathbb{Q}[m]$ the *Hilbert polynomial* of $\mathcal{F}$. The polynomial $\chi(\mathcal{F}(m))$ may have rational coefficients, but is integer-valued whenever $m \in \mathbb{Z}$. The higher cohomology $h^i(X, \mathcal{F})$ for $i > 0$ vanishes when m is sufficiently large, so $\chi(\mathcal{F}(m))$ can also be computed by $P(m) = h^0(X, \mathcal{F}(m))$ when $m \gg 0$.

If $Z \hookrightarrow X$ is a closed subscheme, then the *Hilbert polynomial* of Z is the Hilbert polynomial of the sheaf $\mathcal{O}_Z$ on X . The Hilbert polynomial of a subscheme $Z \hookrightarrow X$ depends on the ambient projective space X is immersed in, or equivalently on a choice of ample line bundle on X .

Example 3.7 (Hilbert polynomial of a finite scheme). Let Γ be a finite scheme over k of length m . Then for any sheaf $\mathcal{F}$ on Γ , its positive cohomology vanishes and $\chi(\mathcal{F}) = h^0(\Gamma, \mathcal{F}) = m$. So the Hilbert polynomial of a finite scheme of length m is the constant m , independent of any projective morphism $\Gamma \hookrightarrow \mathbb{P}^n$.

Example 3.8 (Hilbert polynomial of a line). Let $i : \mathbb{P}^1 \hookrightarrow \mathbb{P}^n$ be a linear embedding of a line into $\mathbb{P}^n$. Then $\mathcal{O}_{\mathbb{P}^1}(1) = i^* \mathcal{O}_{\mathbb{P}^n}(1)$. We have $h^0(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(m)) = m + 1$, because $H^0(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(m)) = k[s, t]_m$ is the $m + 1$ -dimensional space of degree m homogeneous polynomials in s and t . For $i > 0$ the higher cohomology $h^i(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(m)) = 0$ vanishes, so

$$\chi(\mathcal{O}_{\mathbb{P}^1}(m)) = P_{\text{line} \subset \mathbb{P}^n}(m) = m + 1.$$

Example 3.9 (Hilbert polynomial of a hypersurface). Let $S \hookrightarrow \mathbb{P}^n$ be a hypersurface of degree d . We have the exact sequence (1.3)

$$0 \longrightarrow \mathcal{O}_{\mathbb{P}^n}(-d) \longrightarrow \mathcal{O}_{\mathbb{P}^n} \longrightarrow \mathcal{O}_S \longrightarrow 0 \quad (3.1)$$

and after twisting (3.1) by $\mathcal{O}_{\mathbb{P}^n}(m)$, we get the sequence

$$0 \longrightarrow \mathcal{O}_{\mathbb{P}^n}(-d + m) \longrightarrow \mathcal{O}_{\mathbb{P}^n}(m) \longrightarrow \mathcal{O}_S(m) \longrightarrow 0$$

and so using the additivity of Euler characteristic in short exact sequences,

$$\chi(\mathcal{O}_S(m)) = \chi(\mathcal{O}_{\mathbb{P}^n}(m-d)) - \chi(\mathcal{O}_{\mathbb{P}^n}(m)) = \binom{n+m}{n} - \binom{n-d+m}{n}.$$

In particular, the Hilbert polynomial of any cubic surface is $\frac{3}{2}m^2 + \frac{3}{2}m + 1$.

The reason we care about the Hilbert polynomial is because it captures every invariant preserved in a flat family over an integral base.

Theorem 3.10. [Har77, III, Thm. 9.9] *Let T be an integral scheme and $X_T \hookrightarrow \mathbb{P}_T^n$ a closed subscheme. For each point $t \in T$, let $P_t \in \mathbb{Q}[m]$ be the Hilbert polynomial of the fibre $X_t \hookrightarrow \mathbb{P}_{\kappa(t)}^n$. Then X_T is flat over T if and only if P_t is a constant function of t .*

Corollary 3.11 (Length is constant in a flat family). *Let $X_T \rightarrow T$ be a finite projective morphism. Let X_t be the fibre above a point $t \in T$. The length $\text{len}(X_t)$ is independent of t if and only if $X_T \rightarrow T$ is flat.*

Proof. $\text{len}(X_t) = h^0(X_t, \mathcal{O}_{X_t}(m)) = \chi(\mathcal{O}_{X_t}(m))$ is the Hilbert polynomial, which is independent of t if and only if $X_T \rightarrow T$ is flat. $\square$

Corollary 3.12 (Intersection number is constant in a flat family). *Fix an integral scheme T . Let $S_T \rightarrow T$ be a smooth, projective morphism, and $C_T \hookrightarrow S_T$ and $D_T \hookrightarrow S_T$ be two flat families of divisors. Then the intersection number $C_t \cdot D_t = \chi(\mathcal{O}_{C_t \cap D_t})$ is independent of t .*

Proof. This follows identically. $\square$

We are now ready to introduce the Hilbert scheme. Its existence and properties are all extremely non-trivial, and a complete discussion is too complex to devote even an entire thesis to, so we will treat it as a definition, without any proof. The first chapter of Kollar's book [Kol96, I.1] has an excellent discussion of the Hilbert scheme, and proves many results in the deformation theory of the Hilbert scheme. The specific results we are interested in this thesis are covered in more detail in Hartshorne's book on deformation theory [Har09].

Definition 3.13 (Hilbert scheme). [Kol96, I.1] Let T be an integral noetherian scheme. Let $X_T \hookrightarrow \mathbb{P}_T^n \rightarrow T$ be a projective scheme over T . Let $P \in \mathbb{Q}[m]$ be a polynomial. Then there is another scheme $\text{Hilb}_P(X_T/T)$ projective over T called the *Hilbert scheme*, that parametrises flat families of subschemes of X_T in the

sense that there is a natural isomorphism

$$\mathrm{Hom}_T(S, \mathrm{Hilb}_P(X_T/T)) \cong \left\{ \begin{array}{l} \text{closed subschemes } Y_S \hookrightarrow X_T \times_T S \\ \text{that are flat and proper over } S \text{ with} \\ \text{Hilbert polynomial } P \end{array} \right\}.$$

If T is not connected then a similar decomposition occurs, but the Hilbert polynomial P can vary between connected components of T .

A key property of the Hilbert scheme is that it has a *universal family* $\mathcal{U} \hookrightarrow \mathrm{Hilb}_P(X_T/T) \times \mathbb{P}_T^n \rightarrow \mathrm{Hilb}_P(X_T/T)$, which is the flat family induced by the identity morphism $\mathrm{Hilb}_P(X_T/T) \rightarrow \mathrm{Hilb}_P(X_T/T)$. Every flat family of closed subschemes of $X_T \times_T S$ with Hilbert polynomial P is a pullback of the universal family by some morphism $S \rightarrow \mathrm{Hilb}_P(X_T/T)$.

The Hilbert scheme behaves nicely with respect to base change by a morphism $T' \rightarrow T$, in the sense that

$$\mathrm{Hilb}_P(X_T \times_T T'/T') \cong \mathrm{Hilb}_P(X_T/T) \times_T T'$$

which comes from base changing the Hom functor. And if $X_T \subset \mathbb{P}_T^n$ is a closed subscheme, then there is a closed immersion

$$\mathrm{Hilb}_P(X_T/T) \hookrightarrow \mathrm{Hilb}_P(\mathbb{P}_T^n/T) \cong \mathrm{Hilb}_P(\mathbb{P}^n/k) \times_k T.$$

When $T = \mathrm{Spec}(k)$, we omit k and write $\mathrm{Hilb}_P(X/k) = \mathrm{Hilb}_P(X)$.

The two main Hilbert schemes we are interested in are the Hilbert schemes of cubic surfaces, and of lines inside $\mathbb{P}^3$. Proving the following two examples rigorously is non-trivial — one needs to show the functorial property of the Hilbert scheme, we only motivate these spaces by explaining why they have the same points as the Hilbert scheme.

Example 3.14. The Hilbert scheme of degree d hypersurfaces inside $\mathbb{P}^n$ is the projective space $|\mathcal{O}_{\mathbb{P}^n}(d)|$. This is true on points, because a closed subscheme $S \hookrightarrow \mathbb{P}^n$ has Hilbert polynomial $P(m) = \binom{n+m}{n} - \binom{n-d+m}{n}$ if and only if S is cut out by a homogeneous degree d polynomial.

Example 3.15. The Hilbert scheme of lines inside $\mathbb{P}^n$ is the Grassmannian of lines $\mathrm{Gr}(2, n)$. Any line in $\mathbb{P}^n$ is parametrised by coordinates $[r : s] \in \mathbb{P}^1$, and its coordinates $u_i(r, s) = r_i r + s_i s$ are linear forms in r, s . At least two of these are linearly independent (say u_a, u_b), and so we apply a change of coordinates so that $u_a(r, s) = r$ and $u_b(r, s) = s$. Then the remaining $n - 2$ coordinates are described

by the $2 \times (n - 2)$ coefficients r_i, s_i . Affine charts of this form give a covering of $\mathrm{Gr}(2, n)$ by the $\binom{n}{2}$ open sets $G_{ab} \cong \mathbb{A}^{2(n-2)}$. In particular, $\mathrm{Gr}(2, n)$ is a smooth projective variety of dimension $2(n - 2)$.

Example 3.16. For any scheme T , the Hilbert scheme $\mathrm{Hilb}_{m+1}(\mathbb{P}_T^n/T)$ is isomorphic to the product $\mathrm{Gr}(2, n) \times T$. For any flat family of projective schemes $X_T \hookrightarrow \mathbb{P}_T^n \rightarrow T$, then $\mathrm{Hilb}_{m+1}(X_T/T) \hookrightarrow \mathrm{Gr}(2, n) \times T$ is a closed subscheme of the trivial product, which is a $2(n - 2)$ -dimensional smooth and projective T -scheme.

3.2 A smooth cubic surface has finitely many lines

The lines on a cubic surface S are parametrised by the Hilbert scheme $\mathrm{Hilb}_{m+1}(S) \hookrightarrow \mathrm{Hilb}_{m+1}(\mathbb{P}^3)$. To show that there are finitely many lines when S is smooth, we show that $\mathrm{Hilb}_{m+1}(S)$ is a zero-dimensional projective subscheme (and so finite as a set), by showing that the tangent space to the Hilbert scheme is zero-dimensional. This is a slightly stronger result, and any singular cubic surface with finitely many lines has a finite Hilbert scheme but this Hilbert scheme does not have a zero-dimensional tangent space.

Definition 3.17 (The dual numbers, tangent spaces, first order deformations). The *dual numbers* are the ring $k[\varepsilon]/(\varepsilon^2)$. This is a k -algebra with a unique maximal ideal (ε) . For any scheme X and point $x \in X$, the *tangent space* $T_x X$ is the set of morphisms $v : \mathrm{Spec}(k[\varepsilon]/(\varepsilon^2)) \rightarrow X$ that reduce to the morphism $x : \mathrm{Spec}(k) \rightarrow X$ on closed points.

There is a unique morphism $\mathrm{Spec}(k) \rightarrow \mathrm{Spec}(k[\varepsilon]/(\varepsilon^2))$ coming from the quotient $k[\varepsilon]/(\varepsilon^2) \rightarrow k$ by the maximal ideal (ε) . Given any family of schemes $X_\varepsilon \rightarrow \mathrm{Spec}(k[\varepsilon]/(\varepsilon^2))$, the *special fibre* X_k is the fibre $X_\varepsilon \times_{k[\varepsilon]/(\varepsilon^2)} \mathrm{Spec}(k)$. The special fibre is homeomorphic to X_ε as a topological space, but has a different structure sheaf (unless $X_\varepsilon \rightarrow \mathrm{Spec}(k[\varepsilon]/(\varepsilon^2))$ factors through the morphism $\mathrm{Spec}(k) \rightarrow \mathrm{Spec}(k[\varepsilon]/(\varepsilon^2))$, in which case $X_\varepsilon \cong X_\varepsilon \times_{k[\varepsilon]/(\varepsilon^2)} \mathrm{Spec}(k)$ and the family is not flat).

A flat family $Y_\varepsilon \rightarrow \mathrm{Spec}(k[\varepsilon]/(\varepsilon^2))$ is called a *first order deformation* of $Y_\varepsilon \times_{k[\varepsilon]/(\varepsilon^2)} \mathrm{Spec}(k)$. If $Y_\varepsilon \hookrightarrow X \times \mathrm{Spec}(k[\varepsilon]/(\varepsilon^2))$ is a closed immersion, then Y_ε is said to be a first order deformation of *subschemes* of X .

Example 3.18 (Tangent space to the Hilbert scheme). The tangent space to a point $[Y] \in \mathrm{Hilb}_P(X)$ representing a closed subscheme $Y \hookrightarrow X$ is the set of first order deformations of Y as a subscheme of X , by the functoriality of the Hilbert scheme.

Recall the normal sheaf $\mathcal{N}_{Y/X} := \mathcal{H}om_{\mathcal{O}_Y}(\mathcal{I}/\mathcal{I}^2, \mathcal{O}_Y)$ (Definition 1.9).

Theorem 3.19 (Tangent space to Hilbert scheme). [Kol96, I, Thm. 2.8] *Fix a scheme X and a Hilbert polynomial P . The tangent space $T_{[Y]}H$ to the Hilbert scheme $\text{Hilb}_P(X/k)$ at a given closed subscheme $Y \subset X$ is naturally isomorphic to $H^0(Y, \mathcal{N}_{Y/X})$.*

Proof. This is a well-known result, we provide a sketch of the proof for brevity. We pass to an affine chart $X' = \text{Spec}(R)$ of X , where $Y' = \text{Spec}(R/I)$ for some ideal I . We want to find an ideal I_ε that reduces to I after tensoring with k (this is the requirement $I_\varepsilon \subset I \oplus \varepsilon R$), with the additional condition that the quotient $Y'_\varepsilon = \text{Spec}(R[\varepsilon]/(I_\varepsilon, \varepsilon^2))$ is flat over $\text{Spec}(k[\varepsilon]/(\varepsilon^2))$ (this is just finding a first order deformation of Y').

The ideal I_ε is determined by a lift of each element $f \in I$, and also maybe some other terms. Writing a list of generators $I = (f_1, \dots, f_n)$, we thus have a presentation of $I_\varepsilon = (f_1 + \varepsilon g_1, \dots, f_n + \varepsilon g_n, \varepsilon h_1, \dots, \varepsilon h_m)$ for some g_i s and h_i s in R . It turns out that flatness is equivalent to each $\varepsilon f \in I_\varepsilon$ if and only if $f \in I$ (we are about to prove Lemma 3.23, of which this is a special case). Then each $h_i \in I$ and so I_ε is only determined by the assignment of an element of $\varepsilon R \cong R$ to each element of I (a lifting function $\psi : I \rightarrow I \oplus \varepsilon R$, that reduces to the identity on I). This has to be R -linear, and this assignment is unique up to a coset of I inside R (i.e. an element of R/I).

So Y'_ε is uniquely determined by some $\psi \in \text{Hom}_R(I, R) \cong \text{Hom}_{R/I}(I/I^2, R/I)$. A global first order deformation is uniquely determined by local deformations that agree on intersections, so a first order deformation of Y is uniquely determined by a global section of $\mathcal{H}om_{\mathcal{O}_Y}(\mathcal{I}/\mathcal{I}^2, \mathcal{O}_Y)$. This is the normal sheaf, so we have

$$T_{[Y]}H = H^0(Y, \mathcal{N}_{Y/X}).$$

□

Corollary 3.20. *There are finitely many lines on any smooth cubic surface.*

Proof. Fix a smooth cubic surface $S \hookrightarrow \mathbb{P}^3$. The Hilbert scheme $\text{Hilb}_{m+1}(S)$ is a subscheme of $\text{Hilb}_{m+1}(\mathbb{P}^3)$, a projective variety. The dimension of the tangent space to any point $L \in \text{Hilb}_{m+1}(S)$ is $h^0(\mathcal{N}_{L/S}, S)$, and $\mathcal{N}_{L/S} = \mathcal{O}_L(-1)$ because L is a divisor of degree (-1) inside S .

Because $L \cong \mathbb{P}^1$, this reduces to computing $h^0(\mathcal{O}_{\mathbb{P}^1}(-1), \mathbb{P}^1) = 0$, which is zero because $\mathcal{O}_{\mathbb{P}^1}(-1)$ has no global sections. So $T_{[L]}\text{Hilb}_{m+1}(S) = 0$ and S has finitely many lines. □

3.3 The 27 lines on a smooth cubic

We want to understand how the number of lines on one cubic surface is related to another, and for this we can prove a much stronger theorem, which is that the Hilbert scheme of lines on a family of smooth cubic surfaces is étale, which we will prove via deformation theory. This proof is roughly the same as the proof for the tangent space of the Hilbert scheme, but some more care needs to be taken because we will be finding extensions of ideals over an arbitrary local Artinian k -algebra, where there may not be a splitting like there was with $k[\varepsilon]/(\varepsilon^2)$. Moreover, local extensions might not be able to be glued to form a global extension, and local extensions may not even exist.

Definition 3.21. A local Artinian k -algebra A is a local k -algebra (it has one maximal ideal $\mathfrak{m}_A$) that is Artinian, i.e. A satisfies the descending chain condition on ideals.

This implies that A is finite-dimensional as a k -vector space and that A splits as $A = k \oplus \mathfrak{m}_A$ (as an A -module).

Local Artinian k -algebras are isomorphic to some finite-dimensional quotient of the ring $k[t_1, \dots, t_n]_{(t_1, \dots, t_n)}$ localised at 0, which captures all the possible higher order infinitesimal variations of the origin inside $\mathbb{A}^n$. These are finite length quotients of the ring of formal power series $k[[t_1, \dots, t_n]]$, and reading off the coefficients of a k -valued function evaluated in this ring correspond to its higher order derivatives. We are now ready to state a valuative criterion for étaleness, without proof.

Lemma 3.22 (A Valuative criterion for étaleness). [EGA, IV₄ 17.14.2] *Suppose $f : H \rightarrow T$ is a morphism of locally finite type. Then f is smooth (resp. unramified, resp. étale) at a point $h \in H$ if and only if: given any local Artinian k -algebra A and square zero ideal $J \subset A$ satisfying $\mathfrak{m}_A J = 0$ where $\mathfrak{m}_A \subset A$ is the maximal ideal, the following diagram of morphisms of k -schemes has at least one (resp. at most one, resp. exactly one) extension:*

$$\begin{array}{ccccc} \mathrm{Spec}(k) & \longrightarrow & \mathrm{Spec}(A/J) & \longrightarrow & H \\ & & \downarrow & \nearrow \exists! & \downarrow \\ & & \mathrm{Spec}(A) & \longrightarrow & T \end{array}$$

where $\mathrm{Spec}(k) \rightarrow f(h) \subset T$ is the induced morphism on points.

Let $X_T \rightarrow T$ be some family of schemes and P a (Hilbert) polynomial. A morphism $\text{Spec}(A/J) \rightarrow \text{Hilb}_P(X_T/T)$ is then the combined data of a morphism $\text{Spec}(A/J) \rightarrow T$, giving a flat family $X_{A/J} := X_T \times_T \text{Spec}(A/J) \rightarrow \text{Spec}(A/J)$, and a closed subscheme $Y_{A/J} \hookrightarrow X_{A/J}$ flat over $\text{Spec}(A/J)$. Given the further data of a compatible morphism $\text{Spec}(A) \rightarrow T$, i.e. a flat family $X_A \rightarrow \text{Spec}(A)$ where $X_A \times \text{Spec}(A/J) = X_{A/J}$, we want to uniquely extend $Y_{A/J}$ to a subscheme $Y_A \hookrightarrow X_A$ flat over $\text{Spec}(A)$.

We will first need to understand the properties of flatness in relation to these extensions, which we note in the following lemma

Lemma 3.23. *Fix a ring A , a ring R_A flat over A and an ideal $I_A \subset R_A$. Then R_A/I_A is flat over A if and only if $J I_A = I_A \cap J R_A$ for all ideals $J \subset A$.*

The case that I_ε gives a flat family if and only if $(\varepsilon f \in I \iff f \in I_\varepsilon)$ is a special case of this theorem. We can't leverage the splitting $k[\varepsilon]/(\varepsilon^2) = k \oplus (\varepsilon)$ for an arbitrary ring, so this is slightly harder to prove.

Proof. Fix an ideal $J \subset A$ and let $\psi : J \otimes_A R_A \rightarrow A \otimes_A R_A \cong R_A$ be the standard map sending a pair $j \otimes r \mapsto jr \in R_A$, and extended by linearity. ψ is injective because the sequence $0 \rightarrow J \rightarrow A$ remains exact after tensoring with R_A , since $A \rightarrow R_A$ is flat. The map ψ induces a map

$$\phi : J \otimes_A R_A/I_A \longrightarrow R_A/I_A$$

by tensoring with R_A/I_A . The map ϕ is injective if and only if $\phi^{-1}(\{0\}) = \{0\}$, or equivalently if and only if $\psi^{-1}(I_A) = J \otimes_A I_A$.

Suppose R_A/I_A is flat over A . Then ϕ is injective, so $\psi^{-1}(I_A) = J \otimes_A I_A$. The image of ψ is contained in $J R_A$, so $\psi^{-1}(I_A \cap J R_A) = \psi^{-1}(I_A) = J \otimes_A I_A$. Applying ψ to both sides, we find $I_A \cap J R_A = \psi(J \otimes_A I_A) = J I_A$. This is true for any ideal $J \subset A$, so we have proved the backwards direction.

Conversely, suppose $J I_A = I_A \cap J R_A$. We always have $\psi(J \otimes_A I_A) = J I_A$, and because ψ is injective we can invert this to get $\psi^{-1}(J I_A) = J \otimes_A I_A$. So $\psi^{-1}(I_A \cap J R_A) = \psi^{-1}(I_A) = J \otimes_A I_A$, and then ϕ is injective. This is true for an arbitrary ideal $J \subset A$, and so the sequence $0 \rightarrow J \rightarrow A$ remains exact after tensoring with R_A/I_A for every ideal $J \subset A$. So R_A/I_A is flat over A . $\square$

It suffices to only check the maximal ideal if A is a local ring (A -flatness of R_A/I_A is equivalent to $\mathfrak{m}_A I_A = I_A \cap \mathfrak{m}_A R_A$) [Bou61, II.2, Cor. 2, p. 107].

Definition 3.24. Let A be a ring, $J \subset A$ an ideal, $A \rightarrow R_A$ an A -algebra and $I_{A/J} \subset R_A \otimes_A A/J$ an ideal.

A flat extension I_A of $I_{A/J}$ is an ideal $I_A \subset R_A$ such that $I_{A/J} = I_A \otimes_A A/J$ and the quotient R_A/I_A is flat over A .

Lemma 3.25. *Fix a local Artinian k -algebra A with maximal ideal $\mathfrak{m}_A$ and an ideal $J \subset A$ such that $\mathfrak{m}_A J = J^2 = 0$. Let $A \rightarrow R_A$ be a ring flat over A , and $R_{A/J} = R_A \otimes_A A/J$. Suppose $I_{A/J} \subset R_{A/J}$ is an ideal such that $R_{A/J}/I_{A/J}$ is flat over A/J . Let $R = R_A \otimes_A A/\mathfrak{m}_A$ and $I = I_{A/J} \otimes_{A/J} A/\mathfrak{m}_A \subset R$.*

If a flat extension I_A of $I_{A/J}$ exists, then the set of flat extensions of $I_{A/J}$ is in bijection with the vector space $J \otimes \text{Hom}_{R/I}(I/I^2, R/I)$.

Proof. This is again well-known in the literature, we provide a detailed sketch but omit some unimportant technicalities.

The decomposition $A = k \oplus \mathfrak{m}_A$ means that multiplication by an element $j \in J$ is the same as quotienting by $\mathfrak{m}_A$, by chaining through the following isomorphisms:

$$JR_A \cong J \otimes_A (A \otimes_A R_A) \cong J \otimes_A ((k \oplus \mathfrak{m}_A) \otimes_A R_A) \cong J \otimes (k \otimes_A R_A) \cong J \otimes R.$$

Similarly, multiplication by an element $m \in \mathfrak{m}_A$ is the same as quotienting by J , so

$$\mathfrak{m}_A R_A \cong J \otimes R_{A/J}.$$

Suppose a flat extension I_A of $I_{A/J}$ exists. Then $J \otimes I \cong JI_{A/J} \cong JI_A = I_A \cap JR_A$, because I_A is a flat extension. Every $i \in I_{A/J}$ has a lift $\tilde{i} \in I_A$. Another lift $\tilde{i}'$ has difference $\tilde{i} - \tilde{i}' \in I_A \cap JR_A \cong JI_{A/J}$, so each element $i \in I_{A/J}$ has a unique lift $\tilde{i} \in I_A$ up to a coset of $JI_{A/J}$.

The ideal I_A is then determined by a lifting function $\psi_A : I_{A/J} \rightarrow R_A$ that sends an element $i \in I_{A/J}$ to one of its lifts $\psi_A(i) \in I_A$. A priori, ψ_A is only a map of sets, but by only defining it on the generators of $I_{A/J}$ as an R -module and extending by R -linearity to the rest of the ideal we can equip ψ_A with the structure of a R -module homomorphism. The image of the lifts generate I_A because R_A/I_A is flat over A , so I_A is entirely determined (not necessarily uniquely!) by the R -module homomorphism $\psi_A \in \text{Hom}_R(I_{A/J}, R_A)$.

Consider any other lifting function $\psi'_A \in \text{Hom}_R(I_{A/J}, R_A)$. It gives a flat extension $I'_A := \psi'_A(I_{A/J})R_A$ of $I_{A/J}$, and conversely any flat extension arises in this way. The difference $\psi_A - \psi'_A$ is zero modulo JR_A , so its image is contained in JR_A , and hence $\psi_A - \psi'_A \in \text{Hom}_R(I_{A/J}, JR_A)$.

Because $\text{Hom}_R(I_{A/J}, JR_A) \cong \text{Hom}_R(I_{A/J}, J \otimes R) \cong J \otimes \text{Hom}(I_{A/J}, R)$, and a R -module homomorphism $I_{A/J} \rightarrow R$ must factor through the quotient $I_{A/J} \rightarrow I$, then the difference between two lifts lives in $\text{Hom}_R(I_{A/J}, JR_A) \cong J \otimes \text{Hom}_R(I, R)$. The only thing left to do is to find the appropriate notion of equivalence between lifting maps, where we want $\psi_A \sim \psi'_A$ if and only if $I_A = I'_A$.

If $I'_A = I_A$, then the image of $\psi_A - \psi'_A$ is contained in $I_A \cap JR_A$, because the images of ψ_A, ψ'_A are contained in I_A . And because I_A is a flat extension, $I_A \cap JR_A = JI_A \cong J \otimes I$. Conversely, if the image of $\psi_A - \psi'_A$ is contained in $I_A \cap JR_A$ then $\psi'_A(I_{A/J}) \subset I_A$, so $I'_A \subset I_A$. But any choice of lifts of $I_{A/J}$ generates I_A , so $I'_A = I_A$.

So $\psi_A \sim \psi'_A$ if and only if $\psi_A - \psi'_A \in J \otimes \text{Hom}_R(I, I)$. Quotienting out by this, we arrive at the space $J \otimes \text{Hom}_R(I, R/I) \cong J \otimes \text{Hom}_{R/I}(I/I^2, R/I)$, whose elements uniquely determine a flat extension I'_A of $I_{A/J}$, relative to the fixed extension I_A . $\square$

We now need to understand how these local extensions glue together to give a global extension of a deformation.

Lemma 3.26. [Kol96, I, Prop. 2.5.2] *Fix a local Artinian k -algebra A and a square zero ideal $J \subset A$ satisfying $\mathfrak{m}_A J = 0$, and a commutative diagram*

$$\begin{array}{ccccc}
 Y & \hookrightarrow & X & \longrightarrow & \text{Spec}(k) \\
 \downarrow & & \downarrow & & \downarrow \\
 Y_{A/J} & \hookrightarrow & X_{A/J} & \longrightarrow & \text{Spec}(A/J) \\
 \downarrow & & \downarrow & & \downarrow \\
 \boxed{???} & \dashrightarrow & X_A & \longrightarrow & \text{Spec}(A)
 \end{array}$$

where all the horizontal morphisms are flat, and $Y \hookrightarrow X, Y_{A/J} \hookrightarrow X_{A/J}$ are closed immersions.

If a flat extension $Y_A \hookrightarrow X_A \rightarrow \text{Spec}(A)$ of $Y_{A/J}$ exists to fill in the blank spot $\boxed{???}$, then the space of flat extensions of $Y_{A/J}$ is in bijection with $J \otimes H^0(Y, \mathcal{N}_{Y/X})$.

Proof. Fix an index set $\mathcal{A}$. For each $\alpha \in \mathcal{A}$, let $\text{Spec}(R_{A,\alpha}) \subset X_A$ be an affine chart, such that the collection of all of these charts covers X_A . Let $Y_A \cap \text{Spec}(R_{A,\alpha})$ be cut out by the ideal $I_{A,\alpha}$. Then $Y_{A/J} \cap \text{Spec}(R_{A,\alpha} \otimes_A A/J)$ is cut out by the ideal $I_{A/J,\alpha} := I_{A,\alpha} \otimes_A A/J$, and we let $R_\alpha = R_{A,\alpha} \otimes_A k$ and $I_\alpha = I_{A,\alpha} \otimes_A k$. We let $\mathcal{U} = \{\text{Spec}(R_\alpha)\}_{\alpha \in \mathcal{A}}$ be the open covering of X .

Locally, a flat extension $Y'_{A,\alpha}$ of $Y_{A/J} \cap \text{Spec}(R_{A,\alpha} \otimes_A A/J)$ is given by an ideal $I'_{A,\alpha}$ of $R_{A,\alpha}$ that is a flat extension of $I_{A/J,\alpha}$. This is determined by a unique element $\psi_\alpha \in J \otimes \text{Hom}_{R_\alpha/I_\alpha}(I_\alpha/I_\alpha^2, R_\alpha/I_\alpha)$, relative to the given extension $I_{A,\alpha}$.

The collection of possible local extensions then forms a Čech cochain $\psi \in \check{C}^0(\mathcal{U}, J \otimes \mathcal{N}_{Y/X})$. If ψ_α is a cocycle then the collection of local extensions $Y'_{A,\alpha}$ glue to give a global extension Y'_A . Conversely, any global extension Y'_A gives a cochain $\psi \in \check{C}^0(\mathcal{U}, J \otimes \mathcal{N}_{Y/X})$, which is a cocycle because the ψ_α s

agree on intersections. And the Čech 0-cocycles are in bijection with the zeroth cohomology, so there is a bijection

$$\{\text{flat extensions } Y'_A \text{ of } Y_{A/J}\} \longleftrightarrow J \otimes H^0(Y, \mathcal{N}_{Y/X})$$

if a flat extension Y_A exists. $\square$

Remark 3.27. Recall Theorem 3.19, the case when $A = k[\varepsilon]/(\varepsilon^2)$. The form of I_ε gives some insight into why flatness is the right condition to put on a family of schemes. We would hope for a deformation of Y to come from a continuously varying family of deformations of each k -point $y \in Y$, and flatness guarantees this.

If $Y_\varepsilon \rightarrow \text{Spec}(k[\varepsilon]/(\varepsilon^2))$ wasn't flat, then in some affine chart I_ε contains a term εf where $f \notin I$. On the open subscheme Y'_f where f doesn't vanish, f is invertible so the ideal $\mathcal{I}_\varepsilon(Y'_f)$ contains ε , and $Y'_f = Y_f$ as schemes, which means Y_ε is reduced on some open subscheme, but may jump up to being non-reduced on a closed subscheme. This is handwaviely discontinuous (it is continuous in an topological sense, but the jumping behaviour isn't very geometric), and in this context, flatness is exactly the right condition to rule out this kind of jumping behaviour.

Remark 3.28. This is all relative to the given global extension Y_A . When we found the tangent space of the Hilbert scheme, we were interested in the case that $A = k[\varepsilon]/(\varepsilon^2)$ and $J = (\varepsilon)$. This is special because there is a splitting $k[\varepsilon]/(\varepsilon^2) = k \oplus (\varepsilon)$, and so $R[\varepsilon]/(\varepsilon^2) = R \oplus \varepsilon R$. There is then a canonical extension of I , given by the image of I under the ring inclusion $R \rightarrow R \oplus R[\varepsilon]$.

Having this notion of a trivial extension (a) guarantees the existence of flat extensions and (b) gives a canonical identification of the vector space $(\varepsilon) \otimes H^0(Y, \mathcal{N}_{Y/X})$ with the space of extensions, which gives the tangent space of the Hilbert scheme a well-defined structure as a vector space. More generally, there is no canonical extension, so the space of flat extensions has no origin — it is an affine space over $J \otimes H^0(Y, \mathcal{N}_{Y/X})$.

When working over an arbitrary local Artinian k -algebra, we need to also verify local and global existence of flat extensions.

Lemma 3.29. [Kol96, I, Prop. 2.5.1] *If local extensions exist, then the obstruction to the existence of a flat extension $Y_A \hookrightarrow X_A \rightarrow \text{Spec}(A)$, in the sense of Lemma 3.26, lives in $J \otimes H^1(Y, \mathcal{N}_{Y/X})$.*

Proof. We will use some of the notation from the proof of Lemma 3.26. We let $R_{\alpha\beta}$ be the coordinate ring of $\text{Spec}(R_\alpha) \cap \text{Spec}(R_\beta)$, and $I_{\alpha\beta}$ be the ideal that

cuts out $Y \cap \operatorname{Spec}(R_{\alpha\beta})$ inside $\operatorname{Spec}(R_{\alpha\beta})$. Suppose local extensions exist. Then on each $\operatorname{Spec}(R_\alpha)$ we can find a $\psi_\alpha \in \operatorname{Hom}_{R_\alpha}(I_\alpha, R_{A,\alpha})$. We can compare them on intersections, and find that

$$\psi_\alpha - \psi_\beta \in J \otimes \operatorname{Hom}_{R_{\alpha\beta}/I_{\alpha\beta}}(I_{\alpha\beta}/I_{\alpha\beta}^2, R_{\alpha\beta}/I_{\alpha\beta}) = \check{C}^1(\mathcal{U}, J \otimes \mathcal{N}_{Y/X}).$$

We let $\psi_\alpha - \psi_\beta = g_{\alpha\beta}$. We have $(\delta g)_{\alpha\beta\gamma} = (\psi_\alpha - \psi_\beta) - (\psi_\alpha - \psi_\gamma) + (\psi_\beta - \psi_\gamma) = 0$, so g is a cocycle. If $g = \delta f$ for some cochain $f \in \check{C}^0(\mathcal{U}, J \otimes \mathcal{N}_{Y/X})$ then we can modify each ψ_α to a $\psi'_\alpha := \psi_\alpha - f_\alpha$. Then $\psi'_\alpha - \psi'_\beta = g_{\alpha\beta} - (\delta f)_{\alpha\beta} = 0$, so the collection of $\psi'_{\alpha\beta}$ s glue to give a global extension Y_A of $Y_{A/J}$.

The obstruction to gluing is given by the space of 1-cocycles modulo coboundaries, i.e. an element of $J \otimes H^1(Y, \mathcal{N}_{Y/X})$. If $h^1(Y, \mathcal{N}_{Y/X}) = 0$, then any choice of local extensions can always be adjusted to glue, but if $h^1 \neq 0$ then this might not be possible. $\square$

The final piece of the puzzle is to understand when local extensions actually exist. We can always find a lifting function ψ_α that describes the lift of the generators, which generate a candidate $I_{A,\alpha}$. But we need to impose the further condition that $A \rightarrow R(X_{A,\alpha})/I_{A,\alpha}$ is flat, which may not be true. A sufficient condition is that $Y \hookrightarrow X$ is locally principal, cut out by one equation. There are other conditions that are useful for other contexts, such as being a local complete intersection, (see [Kol96, pp. I, 2.11–13]), but this is the simplest and all that we need for our application.

Lemma 3.30 (Existence of local extensions). *Let A be a local k -algebra, R_A an A -algebra and $(f) := I \subset R_A \otimes_A A/\mathfrak{m}_A := R$ a principal ideal with f not a zero divisor. Then any ideal $I_A \subset R_A$ that is a lift of I where $A \rightarrow R_A/I_A$ is flat is principal, generated by a f_A that isn't a zero divisor. And conversely, given any lift f_A of f , then $A \rightarrow R_A/(f_A)$ is flat.*

Proof. Suppose $A \rightarrow R_A/I_A$ is flat. There is always some lift $\tilde{f} \in I_A$ of f . Any other element of I_A is of the form $r_A \tilde{f} + m$ for some $r_A \in R_A$ and $m \in \mathfrak{m}_A R_A$, but because $r_A \tilde{f} \in I_A$ then $m \in I_A \cap \mathfrak{m}_A R_A = \mathfrak{m}_A I_A$. So $I_A = (\tilde{f}) + \mathfrak{m}_A I_A$. But then by Nakayama's lemma, $I_A = (\tilde{f})$. So I_A is principal if $A \rightarrow R_A/I_A$ is flat.

And for the other direction, take any lift $\tilde{f}$ of f . Flatness of $A \rightarrow R_A/(\tilde{f})$ is equivalent to $\mathfrak{m}_A I_A = I_A \cap \mathfrak{m}_A R_A$. Any element of I_A is of the form $r_A \tilde{f}$, so suppose $r_A \tilde{f} \in \mathfrak{m}_A \otimes_A R_A$. Then $\overline{r_A} f \equiv 0$ in $R_A \otimes_A A/\mathfrak{m}_A$. Since f wasn't a zero divisor, then $\overline{r_A} \equiv 0$, i.e. $r_A \tilde{f} \in \mathfrak{m}_A R_A$. So then $r_A \tilde{f} \in \mathfrak{m}_A I_A$, i.e. $I_A \cap \mathfrak{m}_A R_A \subset \mathfrak{m}_A I_A$ and $A \rightarrow R_A/I_A$ is flat. $\square$

Keeping this all in mind, we are now ready to state and prove the key theorem:

Theorem 3.31. *Given any smooth family $X_T \hookrightarrow \mathbb{P}_T^3 \rightarrow T$ of cubic surfaces, then the corresponding family of Hilbert schemes of lines $\text{Hilb}_{m+1}(X_T/T) \rightarrow T$ is étale.*

Proof. Because there are finitely many lines on any smooth cubic surface, then $\text{Hilb}_{m+1}(X_T/T) \rightarrow T$ is locally of finite type (it is finite over T). This then reduces to proving the existence and uniqueness of an extension in the sense of Lemma 3.22, above every line $[L] \in \text{Hilb}_{m+1}(X_T/T)$.

$$\begin{array}{ccccc} \text{Spec}(k) & \longrightarrow & \text{Spec}(A/J) & \longrightarrow & H \\ & & \downarrow & \nearrow \exists! & \downarrow \\ & & \text{Spec}(A) & \longrightarrow & T \end{array}$$

Here $X = X_T \times_T \text{Spec}(k) \subset \mathbb{P}^3$ is a smooth cubic surface, and the morphism $\text{Spec}(k) \rightarrow \text{Hilb}_{m+1}(X_T/T)$ gives a line $L \subset X$. The flat families $X_{A/J}, X_A, L_{A/J}$ come from the morphisms $\text{Spec}(A/J) \rightarrow \text{Hilb}_{m+1}(X_T/T), \text{Spec}(A) \rightarrow T$ as before. We want to show there exists a unique lift $L_A \rightarrow X_A$ that is flat over A .

Because X is smooth, then X has an open affine covering by subschemes X_α that are isomorphic to open affine subschemes of $\mathbb{A}^2$. On these charts L is an irreducible curve on $\mathbb{A}^2$, so is cut out by a principal ideal. Then on each open affine set in this covering local extensions exist, so locally any deformation $L_{A/J,\alpha}$ can be extended to a deformation $L_{A,\alpha}$.

The problem then reduces to verifying that local extensions uniquely glue to form a global extension. We know that $L \cong \mathbb{P}^1$ and $\mathcal{N}_{L/X} \cong \mathcal{O}_{\mathbb{P}^1}(-1)$. So $h^0(L, \mathcal{N}_{L/X}) = h^1(L, \mathcal{N}_{L/X}) = 0$, and by Lemmas 3.26 and 3.29 there is a unique flat extension L_A of L_A . $\square$

Theorem 3.32. [Bea78, Lem. IV.14] *Every cubic surface contains a line.*

Proof. Consider the incidence subscheme $Z \subset \text{Hilb}_{m+1}(\mathbb{P}^3) \times |\mathcal{O}_{\mathbb{P}^3}(3)|$, consisting of points $([L], [S])$ where L is a line in S . A cubic surface contains the line cut out by the ideal (x, y) if and only if the equation defining S has no x^3, x^2y, xy^2, y^3 terms — this cuts out a codimension 4 linear subspace of $|\mathcal{O}_{\mathbb{P}^3}(3)|$. And $\text{Hilb}_{m+1}(\mathbb{P}^3)$ is 4-dimensional, so $\dim(Z) = \dim(|\mathcal{O}_{\mathbb{P}^3}(3)|)$.

Because there is a cubic surface with finitely many lines (e.g. the blowup of $\mathbb{P}^2$ at 6 sufficiently general points), then the projection morphism $p : Z \rightarrow |\mathcal{O}_{\mathbb{P}^3}(3)|$ must be surjective because p is a morphism of projective schemes of the same dimension that has a finite fibre. So the fibre above every point $[S] \in |\mathcal{O}_{\mathbb{P}^3}(3)|$ is non-empty. $\square$

Corollary 3.33. *Any smooth cubic surface has 27 lines. The pairwise intersection graph of the lines on any smooth cubic surface are all isomorphic.*

Proof. There is a smooth cubic surface S_0 with 27 distinct lines: the blowup of $\mathbb{P}^2$ at 6 general points. Let f be the equation that cuts out S_0 , and g be the equation that cuts out any other smooth cubic surface S_1 . The morphism $\mathbb{A}^1 \rightarrow |\mathcal{O}_{\mathbb{P}^3}(3)|$ sending $t \mapsto tg + (1-t)f$ is a one-parameter family $S_{\mathbb{A}^1} \rightarrow \mathbb{A}^1$ of cubic surfaces, and there is an open subscheme $T \subset \mathbb{A}^1$ where the associated cubic surface is smooth. The scheme T is connected so, by Theorem 3.31, $\text{Hilb}_{m+1}(S_T/T)$ is étale and every fibre either has 27 distinct points or is empty. Since S_1 has a line (Theorem 3.32), it has 27 lines.

We can find a global ordering for the 27 lines, by taking the 27 distinct sections $\sigma_i : T \rightarrow \text{Hilb}_{m+1}(S_T/T)$. The corresponding families of lines $L_{i,T}$ are flat families of divisors inside a smooth family of projective surfaces. So the intersection number $\chi(L_{i,t} \cdot L_{j,t})$ is independent of t (Corollary 3.12). The ordering of lines on S_T gives the isomorphism between the intersection graphs (send the line $L_{i,0}$ on S_0 to $L_{i,1}$ on S_1). $\square$

Remark 3.34. Some smooth cubic surfaces have points where 3 lines intersect. These are called *Eckardt points*. But Eckardt points can appear and disappear in flat families (they occur on a codimension 1 subvariety), because the constancy of the intersection number only applies to pairwise intersections.

Remark 3.35 (What if $k \neq \mathbb{C}$?). Almost all of this works over other fields, but we have to loosen our claim to *the number of $\bar{k}$ -lines on any flat family of smooth cubic surfaces is constant*. The classical theory of real cubic surfaces was also studied in the 19th century, Cremona showed there are either 3, 7, 15 or 27 real lines on a real cubic surface [Cre68, Chapitre Onzième] (there are two distinct types of smooth real cubic surfaces with 3 lines).

Example 3.36 (The Hilbert scheme of lines on any singular cubic surface with finitely many lines is NOT always étale). A singular cubic surface with finitely many lines is cut out by $wf_2 - f_3 = 0$ inside $\mathbb{P}^3$, where f_2 and f_3 are quadratic and cubic forms in x, y, z . We consider the trivial extension over the ring $k[\varepsilon]/(\varepsilon^2)$, and the non-trivial extension $S_A = \text{Proj}(k[\varepsilon][x, y, z, w]_{\bullet}/(wf_2 - f_3 - \varepsilon^2 w^3))$ over $A = k[\varepsilon]/(\varepsilon^3)$.

A line passing through the singularity at $[0 : 0 : 0 : 1]$ then has no local extensions from $k[\varepsilon]/(\varepsilon^2)$ to A , we provide a detailed calculation in the appendix (Example C.4) for completeness.

We will see in the next section that $f : \text{Hilb}_{m+1}(S_A/A) \rightarrow \text{Spec}(A)$ is flat whenever f is finite, and so f must actually be ramified — lines collide at a singularity.

Example 3.37. For any smooth surface S and integer $n > 0$, the Hilbert scheme $\text{Hilb}_n(S)$ is a smooth variety.

Proof. Any finite subscheme of a smooth surface is contained in a smooth affine chart, so this reduces to showing that $\text{Hilb}_n(\mathbb{A}^2)$ is smooth. Any length n subscheme is finite, so the positive cohomology of the normal sheaf vanishes. And local extensions exist for zero-dimensional subschemes of $\mathbb{A}^2$. $\square$

3.4 Flat collisions of lines

The Hilbert scheme of lines is still quite nice whenever the associated cubic surface has finitely many lines. While singular cubic surfaces with finitely many lines have less than 27 lines, we can still make sense of what happens to the 27 lines on a smooth surface in a limit to a singular surface.

Theorem 3.38. *Let $S_T \rightarrow T$ be any family of cubic surfaces with finitely many lines. The Hilbert scheme $\text{Hilb}_{m+1}(S_T/T)$ is flat over T .*

Proof. Note that a family of cubic surfaces with finitely many lines is just a family of cubic surfaces $S_T \rightarrow T$ such that $\text{Hilb}_{m+1}(S_T/T) \rightarrow T$ is finite.

Recall the coordinate charts for $\text{Hilb}_{m+1}(\mathbb{P}^3) \cong \text{Gr}(2, 4)$ (Examples 3.15, 3.16). Fix an affine scheme $T = \text{Spec}(R)$ and flat family $S_T \hookrightarrow \mathbb{P}U \times T \rightarrow T$ of cubic surfaces with finitely many lines (i.e. the fibre $S_t \hookrightarrow \mathbb{P}U \times \text{Spec}(\kappa(t))$ is a cubic surface with finitely many lines for every point $t \in T$. It is enough to check this on closed points). The Hilbert scheme $\text{Hilb}_{m+1}(S_T/T)$ is a closed subscheme of $\text{Gr}(2, 4) \times T$.

The family $S_T \hookrightarrow \mathbb{P}U \times T$ is cut out by some homogeneous cubic $f \in R[u_0, u_1, u_2, u_3]_3$. For each point $t \in T$, a line $[L] \in \text{Gr}(2, 4)$ is contained in the surface S_t if and only if f vanishes on L . Concretely, the line L is parametrised by coordinates $[u_0(r, s) : u_1(r, s) : u_2(r, s) : u_3(r, s)]$ (by some $[r : s] \in \mathbb{P}^1$), and the image of f under the quotient map $R[u_0, u_1, u_2, u_3] \rightarrow R[r, s]$ (quotienting out by the ideal cutting out $L \hookrightarrow \mathbb{P}^3$) is a cubic $\bar{f} = \alpha_0 r^3 + \alpha_1 r^2 s + \alpha_2 r s^2 + \alpha_3 s^3 \in R[r, s]_3$. The four coefficients α_i are polynomials in R , and $L \subset S_t$ if and only if the coefficients all vanish at t (each α_i vanishes after base change to $\text{Gr}(2, 4) \times \text{Spec}(\kappa(t))$, i.e. $\bar{f}$ vanishes on $L \times \text{Spec}(\kappa(t))$).

The α_i s then generate an ideal sheaf $\mathcal{I} = (\alpha_0, \alpha_1, \alpha_2, \alpha_3)$, and the closed subscheme $\text{Hilb}_{m+1}(S_T/T) \hookrightarrow \text{Gr}(2, 4) \times T$ is cut out by $\mathcal{I}$. But $\text{Hilb}_{m+1}(S_T/T)$ is finite over T , and $\text{Gr}(2, 4) \times T$ is of relative dimension 4. So $\text{Hilb}_{m+1}(S_T/T)$ is a relative global complete intersection (pass to affine charts $G_{ij} \times T \cong \mathbb{A}_T^4$ of

$\mathrm{Gr}(2, 4)$, then $\mathrm{Hilb}_{m+1}(S_T/T) \cap (G_{ij} \times T) = \mathrm{Spec}(R[a, b, c, d]/(\alpha_0, \alpha_1, \alpha_2, \alpha_3))$ in these charts).

It is known that relative global complete intersections are flat [Stacks, §00SK], so $\mathrm{Hilb}_{m+1}(S_T/T) \rightarrow T$ is flat whenever it is finite. $\square$

As a consequence, a line $L \hookrightarrow S$ can be given a clear notion of multiplicity.

Definition 3.39. The multiplicity of a line $L \subset S$ is the length of local ring of $[L]$ in the Hilbert scheme $\mathrm{Hilb}_{m+1}(S)$. The total multiplicity of lines on S is the length of the scheme $\mathrm{Hilb}_{m+1}(S)$.

Defining multiplicity as the length of the local ring was always possible, but flatness allows us to connect this to our intuitive understanding of limits. We first set up some notation for convenience:

Definition 3.40. Let T be a connected scheme and $X_T \rightarrow T$ a flat family and $t_0 \in T(k)$ a closed point. We call $X_{t_0} = X_T \times_T t_0$ the *flat limit*, or *specialisation*, of the family $X_T \rightarrow T$ at $t = t_0$. We will write $T^* = T \setminus \{t_0\}$, and are interested in understanding the flat limit of families at t_0 , given some known structure on $X_{T^*} \rightarrow T^*$.

When S is a cubic surface, a family of smooth cubic surfaces limiting to S is a family $S_T \rightarrow T$, where $S_{t_0} = S$ and $S_{T^*} \rightarrow T^*$ is smooth.

When $T = \mathbb{A}_{\mathbb{C}}^1$ and $X_T \rightarrow T$ is a family of smooth complex varieties, the flat limit at $t = t_0$ is the limit as $t \rightarrow t_0$ in the Euclidean topology.

Proposition 3.41. *The multiplicity of a line $L \subset S$ is the number of distinct lines that collide at L in any family of smooth cubic surfaces $S_T \rightarrow T$ limiting to S .*

Proof. This follows from flatness and properness of $\mathrm{Hilb}_{m+1}(S_T/T)$. Let $S = S_{t_0}$, the fibre of $S_T \rightarrow T$ above some t_0 . The number of lines colliding at L is the number of sections $L' : T \rightarrow \mathrm{Hilb}_{m+1}(S_T/T)$ that are disjoint away from S , but whose fibre above t_0 is L' . This is the length of the local ring $\mathcal{O}_{[L], \mathrm{Hilb}_{m+1}(S)}$. $\square$

Recall Notation 1.43, where set up the notation of lines on a smooth cubic surface. Given an identified six s , we let the lines in s be labelled M_i , the lines in its dual-six be labelled N_i , and the remaining 15 lines L_{ij} are the lines that intersect both M_i and N_j (or equivalently, N_i and M_j , we have $L_{ij} = L_{ji}$).

Definition 3.42. Let $S_T \rightarrow T$ be a family of smooth cubic surfaces limiting to a normal cubic surface S_{t_0} with finitely many lines. A double-six $\{s, \bar{s}\}$ is said to have *collapsed* on S_{t_0} if the lines $M_i(t_0)$ and $N_i(t_0)$ are identified. This is equivalent to the limiting point configuration Γ_s lying on a conic.

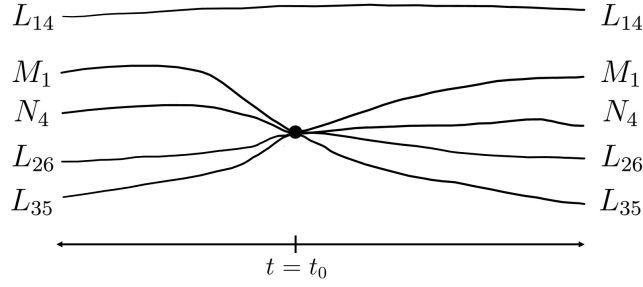


Figure 3.1: Some of the fibres of the Hilbert scheme $\text{Hilb}_{m+1}(S_T/T)$, in Example 3.43. The lines M_1, N_4, L_{26}, L_{35} all collide at $t = t_0$, and the length of the local ring at $[M_1]$ is 4. But the line L_{14} never collides with any other lines — the point $[L_{14}]$ is length one.

Example 3.43. Recall the recurring Example 2.44, of the $4A_1$ cubic surface. We consider a limiting family of smooth cubic surfaces $S_T \rightarrow T$ to $S_{t_0} = S_{4A_1}$. On each of the smooth surfaces S_t , there are 6 lines M_i , 6 lines N_i , and 15 lines L_{ij} . But at $t = t_0$, many lines collide.

The collapsed sixes are $s_{126}, s_{135}, s_{234}, s_{456}$. The lines on S collide as follows:

$$\begin{array}{c|c|c|c|c|c|c|c|c}
 M_1 & M_2 & M_3 & M_4 & M_5 & M_6 & L_{14} & L_{25} & L_{36} \\
 N_4 & N_5 & N_6 & N_1 & N_2 & N_3 & & & \\
 L_{26} & L_{16} & L_{15} & L_{23} & L_{13} & L_{12} & & & \\
 L_{35} & L_{34} & L_{24} & L_{56} & L_{46} & L_{45} & & &
 \end{array} \tag{3.2}$$

The first six lines are all mutually skew lines of multiplicity 4, and each pass through 2 of the singularities on S (the lines of type K (and C and L) in Example 2.30). The last 3 don't pass through any of the singularities, these are the lines of type L only. See Figure 3.1 for a picture of what $\text{Hilb}_{m+1}(S_T/T)$ looks like.

In general, each singularity on a normal cubic surface with finitely many lines corresponds to one double-six collapsing. A line through an A_1 singularity has multiplicity 2, a line through two A_1 singularities has multiplicity 4. A line through an A_2 singularity has multiplicity 3. Once a surface has multiple different singularities, or has A_3 or worse singularities, then different lines through the same singularity will have different multiplicities.

Example 3.44 (Multiplicity of the line through the E_6 singularity). The E_6 cubic surface is given by $wx^2 - xz^2 - y^3 = 0$ (up to a change of coordinates), and contains exactly 1 line: the line cut out by $x = y = 0$. We pass to the chart G_{01} for the Grassmannian, where a line $[L]$ is parametrised by $[ar + bs : cr + ds : r : s]$

in coordinates $(a, b, c, d) \in \mathbb{A}^4 \cong G_{01}$. The coefficients are

$$\alpha_0 = -a - b^3, \quad \alpha_1 = -c + a^2 - 3b^2d, \quad \alpha_2 = 2ac - 3b^2d, \quad \alpha_3 = c^2 - d^3$$

and the dimension of the local ring $k[a, b, c, d]_{(a,b,c,d)}/(\alpha_0, \alpha_1, \alpha_2, \alpha_3)$ is 27 (the computation is too long to include here).

Theorem 3.45. *There are exactly 27 lines on any cubic surface with finitely many lines, counting with multiplicity, and if the surface is smooth then the 27 lines are distinct.*

Proof. We use the same method as the proof of Theorem 3.31, interpolating between two cubic polynomials $tf + (1-t)g$. Let f cut out the E_6 cubic surface S_0 , and g cut out our desired cubic surface with finitely many lines S_1 . On an open subscheme $T \subset \mathbb{A}^1$, the surface S_t has finitely many lines. T is connected. Because $\text{Hilb}_{m+1}(S_T/T) \rightarrow T$ is flat and every fibre is non-empty (every cubic surface contains a line), then the fibres have constant length S_0 and S_1 have the same number of lines (counting multiplicity), which is 27 by Example 3.44. This can also be seen by recalling the tangent space of $\text{Hilb}_{m+1}(S)$ is zero-dimensional whenever S is smooth, so each line on a smooth cubic surface has multiplicity one (and there are 27 of them!). $\square$

Remark 3.46. If we only knew that every smooth cubic surface has 27 distinct lines (by your favourite other method), then this computation actually shows that the Hilbert scheme $\text{Hilb}_{m+1}(S_T/T) \rightarrow T$ has reduced and finite fibres whenever $S_T \rightarrow T$ is a smooth family of cubic surfaces. This means $\text{Hilb}_{m+1}(S_T/T)$ is unramified over T , and because $\text{Hilb}_{m+1}(S_T/T)$ is also T -flat then the Hilbert scheme of lines on a family of smooth cubic surfaces is étale.

Circling back to our main goal in this thesis, we want to understand the determinantal representations of a given cubic surface, up to equivalence. Recall that a determinantal representation $\varphi \in \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ of a smooth cubic surface $S_\varphi \hookrightarrow \mathbb{P}U$ is uniquely determined (up to $\text{PGL}(V) \times \text{PGL}(W)$) by a choice of six skew lines (a six s) and a point on S_φ , the six s is the set of lines of type K on the determinantal representation S_φ . Equivalently, a determinantal representation is uniquely determined (up to $\text{PGL}(U) \times \text{PGL}(W)$) by 6 points $\Gamma_V^\varphi \hookrightarrow \mathbb{P}V$, which is the image of s under the map $\pi_V : S_\varphi \rightarrow \mathbb{P}V$. We can turn this observation into one that also works in families.

Proposition 3.47. *Let $S_{T^*} \hookrightarrow \mathbb{P}U \times T^* \rightarrow T^*$ be a smooth family of cubic surfaces, such that T^* is affine. For each six s on S_{T^*} , there is a morphism*

$\varphi : T^* \rightarrow \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ such that $S_{T^*} \rightarrow T^*$ is cut out by the determinant cubic $\det(\varphi)$, and s is the six lines of type K .

Proof. We label the six as $s = \{M_1, \dots, M_6\}$, recall Notation 1.43. Its dual six is $\bar{s} = \{N_1, \dots, N_6\}$ and the other 15 lines are L_{ij} (these are all families of lines on S_{T^*} , induced by a morphism $L : T^* \rightarrow \text{Hilb}_{m+1}(S_{T^*}/T^*)$). Recall the off-diagonal representation of a smooth cubic surface (Remark 2.47). Consider the family of planes $\varphi_{ij} = \langle M_i, N_j \rangle$ over T^* , given by morphisms $\varphi_{ij} : T^* \rightarrow |\mathcal{O}_{\mathbb{P}U}(1)|$. Take a section $\sigma : T^* \rightarrow S_{T^*}$ that never lies on any of the φ_{ij} s.

For each closed point $t \in T^*(k)$, the equation cutting out $[S_t]$ can be written as $\varphi_{12}\varphi_{23}\varphi_{31}(t) + \lambda(t)\varphi_{21}\varphi_{32}\varphi_{13}(t)$, for some function $\lambda : T^*(k) \rightarrow k$. But we need to make sure that λ can be chosen to vary continuously. To do this, we lift each φ_{ij} to a non-vanishing section $\bar{\varphi}_{ij} : T^* \rightarrow U^\vee$, and σ to a non-vanishing section $\bar{\sigma} : T^* \rightarrow U$. Then the composition $g_{ij} := \bar{\varphi}_{ij} \circ \bar{\sigma}$ is non-vanishing morphism $g_{ij} : T^* \rightarrow \mathbb{A}^1 \setminus \{0\}$, which we view as an element $g_{ij} \in k[T^*]$. Then we let $\lambda(t) = \frac{g_{12}g_{23}g_{31}(t)}{g_{21}g_{32}g_{13}(t)}$. The polynomial $\bar{\varphi}_{12}\bar{\varphi}_{23}\bar{\varphi}_{31}(t) + \lambda(t)\bar{\varphi}_{21}\bar{\varphi}_{32}\bar{\varphi}_{13}(t) \in k[U^\vee]_\bullet$ cuts out S_t for each $t \in T^*(k)$, and the family of tensors

$$\varphi_U = \begin{pmatrix} 0 & \bar{\varphi}_{12} & \lambda\bar{\varphi}_{13} \\ \bar{\varphi}_{21} & 0 & \bar{\varphi}_{23} \\ \bar{\varphi}_{31} & \bar{\varphi}_{32} & 0 \end{pmatrix}$$

gives a morphism $\varphi : T^* \rightarrow \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$, where $\det(\varphi)$ cuts out the family of cubic surfaces $S_{T^*} \rightarrow T^*$. $\square$

Being able to do this construction relies on the six lines $M_1, M_2, M_3, N_1, N_2, N_3$ remaining distinct in the limit to $t = t_0$, and no three becoming coplanar. This is automatic when $S_{T^*} \rightarrow T^*$ is a family of smooth cubic surfaces, but can break down when lines collide.

Because $\mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ and $|\mathcal{O}_{\mathbb{P}U}(3)|$ are projective varieties, then $\varphi : T^* \rightarrow \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ and $f = \det \circ \varphi : T^* \rightarrow |\mathcal{O}_{\mathbb{P}U}(3)|$ both have unique extensions $\hat{\varphi}$ and $\hat{f}$ to T . And because $\det$ is a separated morphism where defined, if $\det(\hat{\varphi}_U(t_0))$ is not the zero polynomial then $\det(\hat{\varphi}_U(t_0)) = \hat{f}(t_0)$. So family of determinantal representations of cubic surfaces extends uniquely to a determinantal representation of the limiting surface, if and only if the determinant of this matrix is non-zero.

Proposition 3.48. *The map sending a tensor φ to the subscheme Γ_V^φ is induced by a rational map $\gamma : \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee) \dashrightarrow \text{Hilb}_6(\mathbb{P}V)$, defined whenever Γ_V^φ is length 6.*

Proof. Consider the ideal sheaf $\mathcal{I}_V \subset \mathcal{O}_{\mathcal{D}} \otimes \mathcal{O}_{\mathbb{P}V}$, generated by the minors of φ viewed as a $4 \times 3 \times 3$ matrix valued in $\mathcal{O}_{\mathcal{D}} \otimes \mathcal{O}_{\mathbb{P}V}$ (recall Remark 2.39). It cuts out a family of length 6 subschemes $\mathcal{P} \hookrightarrow \mathcal{D} \times \mathbb{P}V \rightarrow \mathcal{D}$ that is finite and length 6 over $\mathcal{D}$. This family is flat (constant Hilbert polynomial), so it is induced by a morphism $\mathcal{D} \rightarrow \text{Hilb}_6(\mathbb{P}V)$, which we define to be γ . $\square$

By the same argument, γ is a separated morphism and $\text{Hilb}_6(\mathbb{P}V)$ is proper, so whenever $\hat{\varphi}$ cuts out a length 6 subscheme, then $\gamma(\hat{\varphi}(t_0))$ is the flat limit of the point configuration $\gamma \circ \varphi : T^* \rightarrow \text{Hilb}_6(\mathbb{P}V)$ at $t = t_0$.

Because a general cubic surface is smooth, then any singular cubic surface S_{t_0} is the limit of a family of smooth surfaces, and by applying this construction we can find candidate determinantal representations of a singular cubic surface.

Example 3.49 (A non-determinantal degeneration). Let $S_T \rightarrow T$ be a family of smooth cubic surfaces limiting to an A_1 cubic surface S_{t_0} . Consider the construction from Proposition 3.48, where we pick the six lines $M_1, \dots, M_6$ that pass through the A_1 singularity at $t = t_0$. Then the lines M_i and N_i are identified at $t = t_0$, so the tritangent planes $\varphi_{ij} = \varphi_{ji}$ are equal.

Then $\det(\hat{\varphi}(t_1)) = c\bar{\varphi}_{12}\bar{\varphi}_{23}\bar{\varphi}_{31}(t_1)$ for some $c \in k$, and is reducible. But $[S_{t_0}]$ is irreducible, so $c = 0$ and the tensor $\hat{\varphi}(t_0)$ is not a determinantal representation.

This is a familiar situation: the six lines $M_1, \dots, M_6$ pass through the A_1 singularity if and only if the point configuration Γ_V^φ is contained in a conic, which we already know is not determinantal.

Proposition 3.50. *Let $S_T \hookrightarrow \mathbb{P}U \times T \rightarrow T$ be a family of smooth cubic surfaces limiting to a normal cubic surface $S = S_{t_0}$. Let $s \hookrightarrow S_T \rightarrow T$ be a degenerating six, such that $s(t_0)$ consists of at least 3 distinct lines that don't pass through the same singularity.*

Then the family $S_T \rightarrow T$ is induced by a family of determinantal representations $\hat{\varphi} : T \rightarrow \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$.

Proof. By Proposition 3.47, we can find a morphism $\varphi : T^* \rightarrow \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ where $\det(\varphi)$ cuts out the family of smooth surfaces $S_{T^*} \rightarrow T^*$. This extends uniquely to $\hat{\varphi} : T \rightarrow \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$. We just need to understand what happens in the limit at $t \rightarrow t_0$.

By choosing the section $\sigma : T \rightarrow S_T$ to not pass through any of the hyperplanes φ_{ij} , we ensure that $\lambda(t_0)$ is non-zero and not infinite (both $g_{12}g_{23}g_{31}$ and $g_{21}g_{32}g_{13}$ are non-vanishing on T). So the only way that $\det(\hat{\varphi}(t_0))$ could be undefined was if $\varphi_{12}\varphi_{23}\varphi_{31}(t_0) = -\lambda(t_0)\varphi_{21}\varphi_{32}\varphi_{13}(t_0)$, i.e. if the union of hyperplanes $\varphi_{12}\varphi_{23}\varphi_{31}$ and $\varphi_{21}\varphi_{32}\varphi_{13}$ are identified at $t = t_0$.

The condition of the three lines M_1, M_2, M_3 not passing through the same singularity is equivalent to $M_1, M_2, M_3, N_1, N_2, N_3$ remaining distinct in the limit, and no 3 being coplanar. So $\varphi_{12}\varphi_{23}\varphi_{31}(t_0) \neq \varphi_{21}\varphi_{32}\varphi_{13}(t_0)$, and $\det(\hat{\phi}(t_0)) \neq 0$ is defined and cuts out $[S_{t_0}]$. $\square$

This can be understood from the perspective of the point configuration. The lines M_1, M_2, M_3 remain distinct and don't all pass through the same singularity if and only if the points v_1, v_2, v_3 remain linearly independent in the limit, and the entire subscheme Γ_V doesn't lie on a conic. The reason we need the points v_1, v_2, v_3 to remain linearly independent is that they are the basis vectors in the off-diagonal representation — constructing a degenerating family of determinantal representations where these points collide is possible, but we need to also continuously change basis so that they collide in the limit. This is the main difficulty with lifting families of point configurations and cubic surfaces to tensors, because of the extra freedom of the full $\mathrm{SL}(U) \times \mathrm{SL}(V) \times \mathrm{SL}(W)$ -action.

Corollary 3.51. *If S is a cubic surface with at worst $4A_1$ or $2A_1A_2$ or A_3 or D_4 singularities, then a six s induces a determinantal representation if and only if the double-six $\{s, \bar{s}\}$ has not collapsed.*

Proof. This amounts to using Proposition 3.50 and computing all possible resolutions of singularities $r : \tilde{S} \rightarrow S$ by 6 blowups of $\mathbb{P}V$. When S is a $4A_1$ or $2A_1A_2$ or A_3 or D_4 cubic, then r must blowup at least 3 distinct points (this can be seen by looking at the possible (-2) -curves). These are linearly independent, except for one resolution of the D_4 singularity. But this point configuration is contained in a doubled line, so is not determinantal. $\square$

Remark 3.52. We already know Corollary 3.51, in fact a six s induces a determinantal representation of any normal cubic surface with finitely many lines if and only if $\{s, \bar{s}\}$ has not collapsed, this is just Corollary 2.61. But being able to explicitly reconstruct this fact from degenerating smooth cubic surfaces is nice, the proofs using Gale-duality don't immediately come with an interpretation in terms of degenerations¹.

Example 3.53 (The 50 determinantal representations of a A_1 cubic surface). As an example of Corollary 3.51, the line configuration on an A_1 cubic surface consists of a collapsed double-six $\{s, \bar{s}\}$ and 15 distinct lines L_{ij} . Then all sixes except for $s = \{M_1, \dots, M_6\}$ and $\bar{s} = \{N_1, \dots, N_6\}$ induce determinantal representations, so there are at most 70 inequivalent determinantal representations.

¹The constructions in Chapter 2 can be rigorously extended to work in families, by using the cohomological interpretation of the Gale-transform, and flat base change of cohomology.

However, because each $M_i = N_i$, the $20 = \binom{6}{3}$ degenerate sixes s_{abc} and $\bar{s}_{def}$ are the same (*as sets*). They then induce isomorphic determinantal representations, because the point configurations $\Gamma_{s_{abc}}$ and $\Gamma_{\bar{s}_{def}}$ are isomorphic.

The reason $\Gamma_{s_{abc}} \cong \Gamma_{\bar{s}_{def}}$ is because $\Gamma_{s_{abc}}$ is related to Γ_s by the quadratic transform through the 3 points v_a, v_b, v_c . And $\Gamma_{\bar{s}_{def}}$ is related to $\Gamma_{\bar{s}}$ by blowing up the points w_a, w_b, w_c . But Γ_s is self-associated, so $\Gamma_s = \Gamma_{\bar{s}}$ (up to some projective automorphism) and there is then an isomorphism $\Gamma_{s_{abc}} \cong \Gamma_{\bar{s}_{def}}$. This isomorphism is not Gale-duality — $\Gamma_{s_{abc}}$ is not self-associated because it doesn't lie on a conic.

So there are only 50 different determinantal sixes on S . If two of these 50 sixes induce the same determinantal representation (up to equivalence), then this extends to a non-trivial automorphism of S (recall the proof of Corollary 2.46). But a general A_1 cubic surface has no non-trivial automorphisms, so there are exactly 50 determinantal representations of a general A_1 cubic surface.

Remark 3.54. A $3A_1$ cubic surface has up to 66 inequivalent determinantal representations using this method, more than the upper bound of 50 for a general A_1 cubic. There are $66 = 72 - 2 \cdot 3$ sixes that induce a determinantal representation, and because 3 double-sixes are identified, the symmetry from the A_1 case breaks. Concretely, we cannot do a quadratic transformation blowing up three reduced points of Γ_s , because Γ_s only has two reduced points — the other points have collided!

Dolgachev [Dol12] claims that the number of determinantal representations of a given normal cubic surface with finitely many lines is equal to the number of candidate determinantal representations (72 sixes) minus the number of non-determinantal representations (sixes that collapse), modulo the symmetry by the quadratic transformations. This is only true when S has no non-trivial automorphisms. The $4A_1$ cubic has automorphism group $\mathfrak{S}_4$, and after account for this action, there are only 2 inequivalent determinantal representations, not 64.

Example 3.55 (The 2 determinantal representations of the $4A_1$ cubic surface). We continue Example 3.43. Recall the point configuration in Example 2.2, where the triples $\{v_1, v_2, v_6\}$, $\{v_1, v_3, v_5\}$, $\{v_2, v_3, v_4\}$, $\{v_4, v_5, v_6\}$ are each collinear. We write L_{ijk} for the line in $\mathbb{P}V$ that passes through v_i, v_j, v_k .

The action of $\mathfrak{S}_4$ permutes the four lines $L_{126}, L_{135}, L_{234}, L_{456}$, and acts on the point configuration by extension. Up to the $\mathfrak{S}_4$ -action, there are only 5 non-collapsed sixes $s, \bar{s}, s_{12}, s_{21}, s_{236}, \bar{s}_{345}$. And $s = \bar{s}$ (recall $\Gamma_V^{4A_1} = \Gamma_W^{4A_1}$ as sets), $s_{236} = \bar{s}_{345}$, and $s_{21} \sim s_{234}$ as sets, so there are only 2 possible determinantal representations, induced from the sixes s and s_{21} . The determinantal representation induced by s is the example we used throughout Chapter 2. These are inequivalent: the six s_{21} only has 4 distinct lines on S_{4A_1} .

Chapter 4

The GIT of $4 \times 3 \times 3$ tensors

There are three moduli spaces we are interested in: the moduli space of cubic surfaces, the moduli space of $4 \times 3 \times 3$ tensors, and the moduli space of length 6 subschemes of $\mathbb{P}^2$. Taking the determinant of a general $4 \times 3 \times 3$ tensor φ gives a smooth cubic surface $S_\varphi \hookrightarrow \mathbb{P}U$, and blowing down the lines of type K and C on S_φ give morphisms $\pi_V : S_\varphi \rightarrow \mathbb{P}V$ and $\pi_W : S_\varphi \rightarrow \mathbb{P}W$, which are the blowups of two configurations of 6 points $\Gamma_V \hookrightarrow \mathbb{P}V$ and $\Gamma_W \hookrightarrow \mathbb{P}W$. We have just seen in detail how this construction generalises to certain length 6 subschemes, and these correspondences are all equivariant with respect to the $G := \mathrm{SL}(U) \times \mathrm{SL}(V) \times \mathrm{SL}(W)$ -action on the respective spaces.

A natural question to ask is whether this correspondence is well-behaved with respect to the G -action. We will approach this question from the perspective of Geometric Invariant Theory (GIT). Our main result is a proof that the determinant map

$$\det : \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee) // G \dashrightarrow |\mathcal{O}_{\mathbb{P}U}(3)| // \mathrm{SL}(U) \quad (4.1)$$

is not a morphism, and understanding precisely why this is the case.

We briefly recall GIT, and explain a convex-geometric algorithm that implements the Hilbert-Mumford criterion (Section 4.1). Then we apply this algorithm in Section 4.2 to compute the G -unstable locus of $\mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$. We use our classification to find examples of G -semistable tensors φ whose determinant cubic $\det(\varphi_U)$ is $\mathrm{SL}(U)$ -unstable (the semistable locus maps to the unstable locus), so the map (4.1) is not a morphism. We also provide an alternate geometric proof, using the non-determinantality of 6 points on a conic to show that (4.1) is not proper.

4.1 GIT background

For this section, we are working over $k = \mathbb{C}$. There are several great sources on GIT, everything we discuss here is explained in [Dol03; MFK94], and for a slightly more succinct account there are excellent notes by Victoria Hoskins [Hos16], which is the main reference I used. All proofs are my own. All that is needed for this thesis is understanding how to use the Hilbert-Mumford criterion for semistability, nevertheless we provide some motivation for GIT.

Given a group G acting on a space X (e.g. an algebraic variety, a smooth manifold, etc.), we are often interested in studying the behaviour of this group action, and in particular the orbit space X/G that parametrises elements of X up to G -equivalence.

From topology, we know that whenever G is a topological group acting continuously on a topological space X , we equip X/G with the *quotient topology*, which ensures that the quotient map $\pi : X \rightarrow X/G$ is continuous. Moreover, every G -invariant continuous map $X \rightarrow Y$ factors uniquely through π . Both of these properties are very nice, and one would hope to generalise them to quotients of varieties and schemes. However even very simple quotients can be nasty.

Example 4.1. Consider the action of $G = \mathbb{C}^\times$ on $X = \mathbb{C}^2$, by $t \cdot (x, y) = (tx, y)$, in the Euclidean topology. The quotient space isn't even Hausdorff: the closure of the orbit of $\{(0, 0)\}$ is $\mathbb{C}^2/\mathbb{C}^\times$.

Even with a very simple group action on a very simple space, the quotient $\mathbb{C}^2/\mathbb{C}^\times$ can be much nastier. Instead of trying to preserve the property of being an orbit space, we could instead isolate the other defining property of a topological quotient: any G -invariant continuous map $f : X \rightarrow Y$ factors uniquely through $\pi : X \rightarrow X/G$. A quotient of this form is called the *categorical quotient* (where we replace “continuous map” with “morphism” in our category).

This also runs into issues: the scheme-theoretic analogue of this is the action of $\mathbb{G}_m := \text{Spec}(k[t, t^{-1}])$ on $\mathbb{A}^2 = \text{Spec}(k[x, y])$ arising from the ring homomorphism $k[x, y] \rightarrow k[x, y] \otimes k[t, t^{-1}]$ that sends $(x, y) \mapsto (tx, ty)$. The map on schemes $\mathbb{G}_m \times \mathbb{A}^2 \rightarrow \mathbb{A}^2$ is the multiplication map $(t, (x, y)) \mapsto t \cdot (x, y)$ on closed points. A $\mathbb{G}_m$ -invariant morphism from $\mathbb{A}^2 \rightarrow Y$ for some scheme Y must factor through the inclusion $k[x, y]^{\mathbb{G}_m} \rightarrow k[x, y]$, where $k[x, y]^{\mathbb{G}_m}$ is the subring of $\mathbb{G}_m$ -invariant functions. But $k[x, y]^{\mathbb{G}_m} = k$ (no non-constant function is $\mathbb{G}_m$ -invariant), so the categorical quotient $\mathbb{A}^2/\mathbb{G}_m$ is the point $\text{Spec}(k)$.

What if we threw out the origin, and looked at the quotient $(\mathbb{A}^2 \setminus \{(0, 0)\})/\mathbb{G}_m$? The scheme $\mathbb{A}^2 \setminus \{(0, 0)\}$ isn't affine scheme, so we take the two affine charts $\text{Spec}(k[x, y, y^{-1}])$ and $\text{Spec}(k[x, x^{-1}, y])$. The subrings $k[xy^{-1}]$ and $k[yx^{-1}]$ are

the respective rings of invariants, and any $\mathbb{G}_m$ -invariant morphism factors through the gluing of $\mathrm{Spec}(k[xy^{-1}])$ and $\mathrm{Spec}(k[yx^{-1}])$ — this is $\mathrm{Proj}(k[x, y]) = \mathbb{P}^1$, recovering the well-known fact that projective space is affine space *minus the origin*, quotiented out by scaling.

This example captures the rough philosophy of Geometric Invariant Theory — to produce well-behaved spaces that model a quotient space, we need to throw out some points. By removing a closed subscheme, GIT maintains the universal property of the categorical quotient while also still being an orbit space on some open subscheme.

We deal with projective GIT, where we look at the action of some group G on a projective k -scheme $X = \mathrm{Proj}(R_\bullet)$, and the group action preserves the grading on $R_\bullet$. We consider the graded G -invariant subring $R_\bullet^G = \bigoplus_{i=0}^\infty R_i^G$, and there is the inclusion map $R_\bullet^G \rightarrow R_\bullet$. This induces a rational map

$$X = \mathrm{Proj}(R_\bullet) \dashrightarrow \mathrm{Proj}(R_\bullet^G)$$

which is defined wherever the positive component R_+^G of $R_\bullet^G$ doesn't vanish — the homogeneous ideal of non-constant G -invariant functions on X . These are the points we throw out in the quotient.

Definition 4.2. Let $X = \mathbb{P}^N$ and $G \subset \mathrm{GL}(N+1)$ be a linear algebraic group acting on $\mathbb{P}^N$. The *projective GIT quotient* $X//G$ is defined to be the projective scheme $\mathrm{Proj}(k[x_0, \dots, x_n]_\bullet^G)$.

The *unstable locus* X^{us} of the G -action on X is the closed subvariety of $\mathbb{P}^N$ cut out by the vanishing of the homogeneous ideal $k[x_0, \dots, x_n]_+^G \subset k[x_0, \dots, x_n]$. It is a closed subscheme. When considering different group actions, we also write X^{G-us} for clarity. The *semistable locus* is the complement $X^{ss} := X \setminus X^{us}$.

The GIT quotient is the categorical quotient X^{ss}/G , in the sense that any G -invariant morphism $X^{ss} \rightarrow Y$ for some scheme Y factors uniquely through the quotient $\pi : X^{ss} \rightarrow X//G = X^{ss}/G$. When the G -action is sufficiently nice, there is an open subscheme $X^s \subset X^{ss}$ (known as the *stable locus*), where the quotient X^s/G is an orbit space.

Remark 4.3. The action of a group G needs to be an action of G as a group-variety. This is the analogue of a Lie group, but in the category of varieties. When working with linear algebraic groups, most things that you would hope to be true are true, for space constraints we are forced to omit would have to be a 10-20 page explanation of the key background theory of GIT. In particular, the fact that the ring of invariants $\mathrm{Proj}(R_\bullet^G)$ is even finitely generated is highly

non-trivial (Nagata's Theorem, see [Hos16, Thm. 4.20]), and not true in general, but is true when G is a linear algebraic group.

Example 4.4 (The moduli space of cubic surfaces). [Mum77, 1.14] Consider the GIT quotient of $|\mathcal{O}_{\mathbb{P}^3}(3)| = \mathbb{P}(\text{Sym}^3(U^\vee))$ by the natural action of $\text{SL}(U)$ on $\text{Sym}^3(U^\vee)$. A point $[S] \in |\mathcal{O}_{\mathbb{P}^3}(3)|$ is stable if and only if its associated cubic surface S has at worst A_1 singularities, and semistable if and only if S has at worst A_2 singularities.

The GIT compactification of the moduli space of cubic surfaces is isomorphic to the weighted projective space $\mathbb{P}(1, 2, 3, 4, 5)$ (the ring of invariants is generated by one element in each of the degrees 8, 16, 24, 32 and 40). All strictly semistable cubic surfaces are identified to the same point.

Our goal is to compute the semistable locus of the $\text{SL}(U) \times \text{SL}(V) \times \text{SL}(W)$ -action on $\mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$, and compare it with the semistability of cubic surfaces under the $\text{SL}(U)$ -action. The rest of this section is devoted to understanding how to computationally implement the *Hilbert-Mumford criterion*, which we will use to classify the semistable locus.

In this thesis, we are only concerned with linear actions of a linear subgroup $G \subset \text{GL}(N+1)$ acting on $\mathbb{P}^N$, instead of the more general theory of reductive group actions on projective schemes. Let $n = \dim(G)$ be the dimension of G .

Definition 4.5. A 1-parameter subgroup (1-PS) of an algebraic group G is a non-trivial group homomorphism $\lambda : \mathbb{G}_m \rightarrow G$. Let t be the coordinate on $\mathbb{G}_m = \text{Spec}(\mathbb{C}[t, t^{-1}])$. Any 1-PS $\lambda(\mathbb{G}_m) \subset \text{GL}(N)$ is a diagonal matrix in some basis $\{e_1, \dots, e_N\}$ for $\mathbb{C}^{N+1}$, with entries $t^{r_1}, \dots, t^{r_N}$, where each $r_i \in \mathbb{Z}$. For any closed point $x \in \mathbb{P}^N$, we take a non-zero lift $\tilde{x} = \sum_i x_i e_i$. The action of $\lambda(t)$ on $\tilde{x}$ is

$$\lambda(t) \cdot \tilde{x} = \sum_i t^{r_i} x_i e_i.$$

The *weight vector* $p(\lambda)$ of λ is the vector $(r_1, \dots, r_N) \in \mathbb{Z}^N$. We write $p_i(\lambda) := r_i$ to denote the i^{th} entry. We define $\lambda^n : \mathbb{G}_m \rightarrow G$ to be the 1-PS acting as $\lambda^n(t) \cdot x = \lambda(t^n) \cdot x$ for any $z \in \mathbb{Z}$, and it is immediate that $w(\lambda^z) = z \cdot w(\lambda)$.

A 1-PS is said to *act with only positive weights* on a closed point $x \in \mathbb{P}^N$ if $r_i > 0$ whenever $x_i \neq 0$.

Proposition 4.6 (Hilbert-Mumford Criterion). *A closed point $x \in \mathbb{P}^{N-1}$ is G -unstable if and only if there is a 1-PS λ of G that acts with only positive weights on x .*

Definition 4.7 (Actions of tori on $\mathbb{P}^{N-1}$). A maximal torus of $\mathrm{GL}(N)$ is the subgroup of diagonal matrices in some basis $E = \{e_1, \dots, e_N\}$ of $\mathbb{C}^N$. Any maximal torus of $\mathrm{GL}(N)$ is isomorphic to $\mathbb{G}_m^N$. A n -torus T of $\mathrm{GL}(N)$ is an injective group homomorphism $T : \mathbb{G}_m^n \rightarrow \mathbb{G}_m^N$. It sends a closed point $t \in \mathbb{G}_m^n$ to $T(t) = (\chi_1(t), \dots, \chi_N(t)) \in \mathbb{G}_m^N$, for some collection of group homomorphisms $\chi_i : \mathbb{G}_m^n \rightarrow \mathbb{G}_m$. The group $\mathrm{Hom}_{\mathbb{G}_m}(\mathbb{G}_m^n, \mathbb{G}_m)$ is called the *character group* of the torus $\mathbb{G}_m^n$, and is identified with the $\mathbb{Z}$ -module $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}^n, \mathbb{Z}) = (\mathbb{Z}^n)^\vee$, where the character $\chi_i : \mathbb{G}_m^n \rightarrow \mathbb{G}_m$ sends a point $(t_1, \dots, t_n) \mapsto \prod_{j=1}^n t_j^{\chi_{ij}}$ (viewing χ_i as a vector $(\chi_{i1}, \dots, \chi_{in}) \in (\mathbb{Z}^n)^\vee$). So the data of a n -torus of $\mathrm{GL}(N)$ is the same as N vectors $\chi_i \in (\mathbb{Z}^n)^\vee$.

A 1-PS λ of T is a group homomorphism $\lambda : \mathbb{G}_m \rightarrow \mathbb{G}_m^n$. We call $\mathrm{Hom}_{\mathbb{G}_m}(\mathbb{G}_m, \mathbb{G}_m^n)$ the *cocharacter group* of $\mathbb{G}_m^n$, and the cocharacter group is identified with $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}, \mathbb{Z}^n) = \mathbb{Z}^n$, where $\lambda(t) = (t^{\bar{\lambda}_1}, \dots, t^{\bar{\lambda}_n})$ (viewing λ as $\bar{\lambda} = (\bar{\lambda}_1, \dots, \bar{\lambda}_n) \in \mathbb{Z}^n$, we will use the overline to distinguish between the 1-PS and the vector it is identified with). A 1-PS λ of T acts as $(T \circ \lambda)(\mathbb{G}_m)$ on $\mathbb{C}^N$, where

$$\lambda(t) \cdot \left(\sum_{i=1}^N x_i e_i \right) = \sum_{i=1}^N \chi_i(\lambda(t)) \cdot x_i e_i.$$

We can easily compute this action: $\chi_i(\lambda(t)) = \prod_{j=1}^n \lambda(t)_j^{\chi_{ij}} = \prod_{j=1}^n t^{\chi_{ij} \bar{\lambda}_j}$. So the 1-PS λ acts on e_i with weight $\sum_{j=1}^n \chi_{ij} \bar{\lambda}_j$. The i^{th} weight function is then the function $p_i : \mathbb{Z}^n \rightarrow \mathbb{Z}$, which sends a 1-PS λ to the integer $p_i(\bar{\lambda}) = \sum_{j=1}^n \chi_{ij} \bar{\lambda}_j$. It is easy to see that this is $\mathbb{Z}$ -linear, so the N weight functions p_i can be viewed as elements of $(\mathbb{Z}^n)^\vee$. This recovers the weight vector from Definition 4.5.

Definition 4.8 (Numerical criterion for T -instability). Let $P = \{p_i\}_{i=1}^N \subset (\mathbb{Z}^n)^\vee$ be the set of weight functions on $\mathbb{Z}^n$ as above. For any subset $I \subset \{1, \dots, N\}$ we let $P_I = \{p_i \in P \mid i \in I\}$. A subset $P_I \subset P$ is said to be $\bar{\lambda}$ -unstable if $p_i(\bar{\lambda}) > 0$ for every $i \in I$. A subset $P_I \subset P$ is said to be T -unstable if there exists a $\bar{\lambda} \in \mathbb{Z}^n$ where P_I is $\bar{\lambda}$ -unstable.

Take any closed point $x \in \mathbb{P}^{N-1}$ and a non-zero lift $\tilde{x} = \sum_i x_i e_i \in \mathbb{C}^N$. We define the *support* of x to be the set $\mathrm{Supp}_E(x) = \{p_i \in P \mid x_i \neq 0\}$. The support is independent of the choice of lift.

We say one of the above objects is *semistable* if it is not unstable. Something is T -semistable if it is $\bar{\lambda}$ -semistable for every $\bar{\lambda}$.

Lemma 4.9 (Both definitions of T -instability agree). *A point $x \in \mathbb{P}^{N-1}$ is $T(\mathbb{G}_m^n)$ -unstable if and only if $\mathrm{Supp}_E(x)$ is a T -unstable point set.*

Proof. Choose a lift $\tilde{x} = \sum_{i=1}^N x_i e_i$ of x . A 1-PS λ of T acts with weights

$\{p_i(\bar{\lambda}) \mid x_i \neq 0\}$ on x . And $p_i \in \text{Supp}_E(x)$ if and only if $x_i \neq 0$. So λ acts with positive weights on x if and only if $\text{Supp}_E(x)$ is $\bar{\lambda}$ -unstable, i.e. x is $\lambda(\mathbb{G}_m)$ -unstable if and only if $\text{Supp}_E(x)$ is $\bar{\lambda}$ -unstable.

Then by the Hilbert-Mumford criterion, x is $T(\mathbb{G}_m)$ -unstable if and only if it is T -unstable. $\square$

Lemma 4.10. *Let G be a n -dimensional linear algebraic group acting linearly on $\mathbb{C}^N$. Let $T : \mathbb{G}_m^n \rightarrow G$ be any maximal torus, acting diagonally in a basis for $\mathbb{C}^N$.*

A closed point $x \in \mathbb{P}^{N-1}$ is G -unstable if and only if there is a $g \in G$ where $g \cdot x$ is $T(\mathbb{G}_m^n)$ -unstable.

Proof. Any 1-PS λ of G is contained in some maximal torus T' of G . And any two maximal tori of a linear algebraic group G over $\mathbb{C}$ are conjugate, in the sense that there is a $g \in G$ where $gT'(\mathbb{G}_m^n)g^{-1} = T(\mathbb{G}_m^n)$. So $g\lambda(\mathbb{G}_m)g^{-1} \subset T(\mathbb{G}_m^n)$ is a 1-PS of T .

A closed point $x \in \mathbb{P}^{N-1}$ is $\lambda(\mathbb{G}_m)$ -unstable if and only if $g \cdot x$ is $g\lambda(\mathbb{G}_m)g^{-1}$ -unstable, so by the Hilbert-Mumford criterion x is G -unstable if and only if $g \cdot x$ is T -unstable for some g . $\square$

As a result, finding the unstable locus of the G -action can be reduced to using the numerical criterion for T -instability in a given basis, and then finding the G -orbits of these unstable points. This is still impractical — we would need to check every $\lambda \in \mathbb{Z}^n$, an infinite set. The problem can be reduced further to the following convex geometry problem, where T -instability can be classified by only checking 2^N convex hulls.

Theorem 4.11 (Convex criterion for G -instability). *A closed point $x \in \mathbb{P}^{N-1}$ is G -unstable if and only if there exists a $g \in G$ where $0 \notin \text{conv}(\text{Supp}_E(g \cdot x))$, where we view $\text{Supp}_E(g \cdot x)$ as a subset of $(\mathbb{R}^n)^\vee = \mathbb{R} \otimes_{\mathbb{Z}} (\mathbb{Z}^n)^\vee$ and compute the convex hull there.*

Proof. It suffices to show $\text{Supp}_E(x)$ is T -unstable if and only if $0 \notin \text{conv}(\text{Supp}_E(x))$, by Lemmas 4.9 and 4.10.

Any subset $\text{Supp}_E(x) \subset P$ is a finite set of points with integer coordinates in $(\mathbb{R}^n)^\vee$. So $\text{conv}(\text{Supp}_E(x))$ is a convex polyhedron and has finitely many faces. We let F be the set of faces of $\text{Supp}_E(x)$. Each face is the convex hull of $n - 1$ vertices of $\text{Supp}_E(x)$, and so is contained in the hyperplane cut out by an equation $\bar{\lambda}_f(z) = c_f$ for some $\bar{\lambda} \in \mathbb{R}^n$ and $\bar{\lambda}_0 \in \mathbb{R}$.

Then $\text{conv}(\text{Supp}_E(x))$ is the intersection of finitely many halfspaces $H_f \subset \mathbb{R}^n$ ranging over the finite set $f \in F$, defined by elements $\bar{\lambda}_f \in \mathbb{R}^n$ and $c_f \in \mathbb{R}$ and

$$z \in H \iff \bar{\lambda}_f(z) \geq c_f.$$

The face f is contained in the set $\bar{\lambda}_f(z) = c_f$, so the hyperplanes cut out by $\bar{\lambda}_f(z) = c_f$ bound $\text{conv}(\text{Supp}_E(x))$, and must have rational coefficients because they intersect at least $n + 1$ of the integer points $p_i \in \text{Supp}_E(x)$. By multiplying out by the lowest common positive denominator, we can write the defining inequalities as $\bar{\lambda}_f(z) \geq c_f$ with $\bar{\lambda}_f \in \mathbb{Z}^n$. Notice that $\bar{\lambda}(p_i) = p_i(\bar{\lambda})$.

Now, suppose $0 \notin \text{conv}(\text{Supp}_E(x))$. Then $0 \notin H_f$ for some face f . So we have $0 = \bar{\lambda}_f(0) < c_f$. And $\bar{\lambda}_f(p_i) \geq c_f > 0$ for all $p_i \in \text{Supp}_E(x)$. So $\text{Supp}_E(x)$ is $\bar{\lambda}$ -unstable, hence T -unstable.

Conversely, suppose $\text{Supp}_E(x)$ is T -unstable. Then for some $\bar{\lambda}$, it is $\bar{\lambda}$ -unstable. Then $\bar{\lambda}(p_i) > 0$ for all $p_i \in P_J$. Because the set $\{\bar{\lambda}(p_i) \mid p_i \in \text{Supp}_E(x)\}$ is finite, then it has a non-zero minimum, which we denote as c . Then we have $\bar{\lambda}(p_i) \geq c$ for all $p_i \in \text{Supp}_E(x)$, so the point set $\text{Supp}_E(x)$ is contained in the convex set $H_{(c, \bar{\lambda})} := \{z \in (\mathbb{R}^n)^\vee \mid \bar{\lambda}(z) \geq c\}$. But then $\text{conv}(\text{Supp}_E(x)) \subset H_{(c, \bar{\lambda})}$ and $0 \notin H_{(c, \bar{\lambda})}$. So $0 \notin \text{conv}(\text{Supp}_E(x))$. $\square$

4.2 Applying the convex criterion

We introduce a lot of notation to convert the convex criterion into one that we can easily apply to understand $4 \times 3 \times 3$ semistability, with the goal of writing down an algorithm that implements it to find the unstable locus.

Remark 4.12 (Why $\text{SL}(N)$ instead of $\text{PGL}(N)$?). Throughout this thesis we have been trying to understand $4 \times 3 \times 3$ tensors up to projective equivalence (the $\text{PGL}(U) \times \text{PGL}(V) \times \text{PGL}(W)$ -action) but now we are understanding GIT semistability under the $\text{SL}(U) \times \text{SL}(V) \times \text{SL}(W)$ -action. This is because the action of $\text{PGL}(N)$ on $\mathbb{P}^{N-1}$ does not lift to an action on $\mathbb{C}^N$ (unless $N = 1$), it only lifts to an action on $\text{Sym}^N(\mathbb{C}^N)$, whereas the action of $\text{SL}(N)$ does lift to an action on $\mathbb{C}^N$ (in the language of linearisations, we can only linearise a $\text{PGL}(N)$ -action on the line bundle $\mathcal{O}_{\mathbb{P}^{N-1}}(N)$, whereas we can linearise a $\text{SL}(N)$ -action on the line bundle $\mathcal{O}_{\mathbb{P}^{N-1}}(1)$). We don't choose the $\text{GL}(N)$ action because we need the center $Z(\text{GL}(N)) \cong \mathbb{G}_m$ to act trivially on our linearisation, otherwise everything is unstable by this subgroup, and then this is the same as computing the $\text{PGL}(N)$ -action.

Because $\text{SL}(N)$ is a finite (degree N) cover of $\text{PGL}(N)$, the semistable locus of the two actions are exactly the same, so it doesn't matter which group we take when computing them. We choose the $\text{SL}(N)$ -action because it is easier to work with.

Notation 4.13. Fix positive integers l and m and n . For our purposes, $l = 4$ and

$m = n = 3$, but we explain the general theory as it is not much more difficult.

Let $G = \mathrm{SL}(l) \times \mathrm{SL}(m) \times \mathrm{SL}(n)$. We are computing the semistable locus of $\mathbb{P}(\mathbb{C}^l \otimes \mathbb{C}^m \otimes \mathbb{C}^n)$ under the natural action of G . Let $I = \{1, \dots, l\}$ and $J = \{1, \dots, m\}$ and $K = \{1, \dots, n\}$ be index sets. Fix bases $\{u_i\}_{i \in I}$ for $\mathbb{C}^l$ and $\{v_j\}_{j \in J}$ for $\mathbb{C}^m$ and $\{w_k\}_{k \in K}$ for $\mathbb{C}^n$. Let $e_{(i,j,k)} = u_i \otimes v_j \otimes w_k$. The basis $E = \{e_{(i,j,k)}\}_{(i,j,k) \in I \times J \times K}$ is a basis for $\mathbb{C}^l \otimes \mathbb{C}^m \otimes \mathbb{C}^n$.

Any maximal torus T of $\mathrm{SL}(l) \times \mathrm{SL}(m) \times \mathrm{SL}(n)$ splits as the product of three maximal tori $T = T_l \times T_m \times T_n$, where T_b is a maximal torus $T_b : \mathbb{G}_m^{b-1} \rightarrow \mathbb{G}_m^b$ of $\mathrm{SL}(b)$ (which we choose to be diagonal in the bases we have just given). The torus T_b is determined by b characters $\chi_i^{(b)} \in (\mathbb{Z}^{b-1})^\vee$. We require that $\sum_{i=1}^b \chi_i = 0$, because the determinant of any $g \in \mathrm{SL}(p)$ is 1, and that the χ_i s span $(\mathbb{Z}^{b-1})^\vee$ (that $\sum \chi_i = 0$ is the only linear relation), so the torus is $b - 1$ -dimensional.

A 1-PS $\lambda = (\lambda_l, \lambda_m, \lambda_n)$ of T acts as the product of its components in each maximal torus, i.e. as $T(\lambda(t)) = T_l(\lambda_l(t))T_m(\lambda_m(t))T_n(\lambda_n(t))$. So it acts on a basis vector $e_{(i,j,k)}$ with weight $p_{(i,j,k)}(\bar{\lambda}) = p_i^{(l)}(\bar{\lambda}_l) + p_j^{(m)}(\bar{\lambda}_m) + p_k^{(n)}(\bar{\lambda}_n)$ (the sum of the weights).

Actually implementing this involves choosing coordinate representations for each $\chi_i^{(b)}$. In our computational implementation, we choose coordinates where $\chi_i^{(b)}$ is the i^{th} basis vector of $(\mathbb{Z}^{b-1})^\vee$ if $i \neq b$, and $\chi_b^{(b)} = (-1, \dots, -1) = -\sum_{i=1}^{b-1} \chi_i^{(b)}$.

The next important thing to do is actually understand the outputted results. We phrase this in the form of *block conditions*, which we define as follows:

Definition 4.14 (Block form). To each subset $F \subset I \times J \times K$ (equivalently $P_F \subset P_{I \times J \times K}$), we associate the subset $E_F = \{e_f\}_{f \in F} \subset E$ (note $E = E_{I \times J \times K}$).

A tensor $\varphi \in \mathbb{P}(\mathbb{C}^l \otimes \mathbb{C}^m \otimes \mathbb{C}^n)$ is said to be of *block form* B_F if there exists a $g \in G$ where $g \cdot \varphi \in \mathbb{P}(\mathrm{Span}(E_F))$. If φ is of block form B_F then we write $\varphi \in B_F$. By abuse of notation, we let $B_F \subset \mathbb{P}(\mathbb{C}^l \otimes \mathbb{C}^m \otimes \mathbb{C}^n)$ be the set of tensors of block form B_F . The variety B_F is the orbit of $\mathbb{P}(\mathrm{Span}(E_F))$ under G . A subset $Y \subset \mathbb{P}(\mathbb{C}^l \otimes \mathbb{C}^m \otimes \mathbb{C}^n)$ is said to impose a *block condition* if there exists a block form B_F where $Y = B_F$. Each block condition is a G -invariant closed subset of $\mathbb{P}(\mathbb{C}^l \otimes \mathbb{C}^m \otimes \mathbb{C}^n)$, and the complement of a block condition is a (possibly empty) G -invariant open subset.

A priori, there are 2^{lmn} block conditions on $\mathbb{P}(\mathbb{C}^l \otimes \mathbb{C}^m \otimes \mathbb{C}^n)$ corresponding to each subset $F \subset I \times J \times K$, but many are equivalent under change of basis. We impose the partial ordering on block conditions, where $B \preceq B'$ if every element of B is also of block form B' . Two block forms are said to be equivalent if $B \preceq B'$ and $B' \preceq B$, in which case we write $B \sim B'$.

Definition 4.15 (Visualising a block form). Fix a subset $F \subset I \times J \times K$. The

block tensor $B_F = [B_1 | \cdots | B_l]$ is a collection of $m \times n$ matrices B_i with entries 0 or $*$. The rows of each B_i are indexed by J and columns indexed by K , and the jk^{th} entry of B_i is $*$ if $(i, j, k) \in F$, and 0 otherwise. See the following Example 4.16 for a picture.

The vector space $\text{Span}(E_F)$ consists of elements $\varphi = \sum_{f \in F} \varphi_f e_f$ — writing it out as a block tensor, $\varphi \in \text{Span}(E_F)$ if and only if φ has non-zero entries only where B_F has an $*$.

Whenever $F \subset F'$, we have $B_F \preceq B_{F'}$. But there are many other equivalences.

Example 4.16 (Understanding the poset of block conditions). Consider the case where $\sigma \in \mathfrak{S}_I \times \mathfrak{S}_J \times \mathfrak{S}_K$ is any permutation (where $\mathfrak{S}_X$ is the symmetric group on the set X), then $B_F \sim B_{\sigma(F)}$ by the action of the subgroup $\mathfrak{S}_I \times \mathfrak{S}_J \times \mathfrak{S}_K \subset G$ permuting the basis vectors.

Sometimes if $F \subset F'$ then $B_F \sim B_{F'}$. For example, the condition that the cubic surface S_φ contains a plane of type L is the $4 \times 3 \times 3$ block condition

$$\varphi \in \left[\begin{array}{ccc|ccc|ccc|ccc} * & * & * & * & * & * & * & * & * & * & * & * \\ * & * & * & * & 0 & 0 & * & 0 & 0 & * & 0 & 0 \\ * & * & * & * & 0 & 0 & * & 0 & 0 & * & 0 & 0 \end{array} \right].$$

But by mixing between the basis vectors, we can eliminate some entries to get an equivalent block form

$$\varphi \in \left[\begin{array}{ccc|ccc|ccc|ccc} * & * & * & * & * & * & * & * & * & * & * & * \\ * & * & * & * & 0 & 0 & * & 0 & 0 & 0 & 0 & 0 \\ * & * & * & * & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right].$$

In particular, this shows that any plane of type L contains a rank one point.

Finally, we connect this back to G -instability. Recall that φ is G -unstable if and only if $g \cdot \varphi$ is T -unstable for some $g \in G$.

Definition 4.17. A block condition B_F is *unstable* if $B_F \subset \mathbb{P}(\mathbb{C}^l \otimes \mathbb{C}^m \otimes \mathbb{C}^n)^{G-us}$.

We have arrived at a rough algorithm to compute the unstable locus (and hence its complement, the semistable locus) of $\mathbb{P}(\mathbb{C}^l \otimes \mathbb{C}^m \otimes \mathbb{C}^n)$ under the action of $\text{SL}(l) \times \text{SL}(m) \times \text{SL}(n)$:

- (1) Find the point set $P_{I \times J \times K}$.
- (2) For each subset $P_F \subset P_{I \times J \times K}$, check if $0 \in \text{conv}(P_F)$.

- (3) If $0 \notin \text{conv}(P_F)$, then the block form B_F is an unstable block form.
- (4) Geometrically classify these unstable block forms.

In practice, this needs significant optimisation — there are 2^{36} subsets of $P_{I \times J \times K}$ when $(l, m, n) = (4, 3, 3)$. In particular, we only care about finding the maximal unstable block forms (with respect to the partial ordering $\preceq$), because every unstable tensor will be contained in one of these. We optimise the algorithm by noting that the maximal unstable subsets $F \subset I \times J \times K$ satisfy a partial order condition up to a permutation of coordinates (Example 4.16), where $(i, j, k) \in F$ whenever $(i', j', k') \in F$ and $i \leq i'$ and $j' \leq j$ and $k' \leq k$. This is known as a *flag condition*, and is also known more abstractly in the literature (see [Dol03, §9.5; MFK94, §2.2]), we present it in a more concrete form for our application. It reduces the number of point sets to check from 2^{36} to just $4116 \sim 2^{12}$.

The computational implementation of the naive algorithm (Algorithm 1) and its optimisation (Algorithm 2) is detailed in Appendix B. We use functions from the codebase [Sage] to implement this algorithm. A copy of the code we use is available at the following link, it will compute the $\text{SL}(l) \times \text{SL}(m) \times \text{SL}(n)$ unstable locus when you define `thegroup = [l, m, n]` in the first line, and run all the cells:

<https://github.com/charlotte-sherratt/4x3x3-GIT-quotient>.

4.2.1 3×3 and 3×4 semistability

Let $G = \text{SL}(U) \times \text{SL}(V) \times \text{SL}(W)$. Because any choice of basis is eliminated after taking the GIT quotient, we will often write $\mathbb{P}(4 \times 3 \times 3) = \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ (the projective space of $4 \times 3 \times 3$ tensors), and instead of $\text{SL}(U) \times \text{SL}(V) \times \text{SL}(W)$ we will often write $\text{SL}(4) \times \text{SL}(3) \times \text{SL}(3)$.

Whenever $H \subset G$ is a subgroup, the semistable locus with respect to the H -action always contains the semistable locus with respect to the G -action. We write $\mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)^{H-ss}$ to indicate semistability with respect to a subgroup H . There are three important subgroups of $\text{SL}(4) \times \text{SL}(3) \times \text{SL}(3)$ — the $\text{SL}(4)$ subgroup, the $\text{SL}(3) \times \text{SL}(3)$ subgroup, and the two $\text{SL}(4) \times \text{SL}(3)$ subgroups.

Proposition 4.18. *Let V_1 and V_2 be vector spaces, and let $H \subset \text{GL}(V_2)$ act linearly on V_2 , inducing an action on the projective space $\mathbb{P}(\text{Hom}(V_1, V_2))$. Then a point $x \in \mathbb{P}(\text{Hom}(V_1, V_2))$ is H -unstable if and only if there exists a 1-PS λ of H where $x(v_1) \in \mathbb{P}V_2$ is λ -unstable for every $v_1 \in \mathbb{P}V_1$.*

Proof. The 1-PS acts trivially on every vector of V_1 , so the maximal block forms only depend on V_2 . $\square$

Corollary 4.19. *A $4 \times 3 \times 3$ tensor φ is:*

- (i) *$\mathrm{SL}(4)$ -semistable if and only if $\varphi_U : U \rightarrow V^\vee \otimes W^\vee$ is injective.*
- (ii) *$\mathrm{SL}(3) \times \mathrm{SL}(3)$ -semistable if and only if $\varphi_U(U)$ is not a singular subspace of type C or K or L.*
- (iii) *$\mathrm{SL}(4) \times \mathrm{SL}(3)$ -semistable (for both $\mathrm{SL}(4) \times \mathrm{SL}(3)$ subgroups) if and only if all of the following holds:*
 - φ_U is injective.
 - S_φ contains no rank one line of shared kernel/cokernel.
 - S_φ contains no plane of type C or K.

Proof. Applying the Hilbert-Mumford criterion and converting the problem into a convex geometry one, we find that the unstable block forms for 3×3 matrices under the $\mathrm{SL}(3) \times \mathrm{SL}(4)$ -action are the following:

$$\begin{bmatrix} * & * & * \\ * & * & * \\ 0 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} * & * & 0 \\ * & * & 0 \\ * & * & 0 \end{bmatrix}, \quad \begin{bmatrix} * & * & * \\ * & 0 & 0 \\ * & 0 & 0 \end{bmatrix}.$$

And for 3×4 matrices under the $\mathrm{SL}(3) \times \mathrm{SL}(4)$ -action:

$$\begin{bmatrix} * & * & * & * \\ * & * & * & * \\ * & 0 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} * & * & * & * \\ * & * & 0 & 0 \\ * & * & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} * & * & * & 0 \\ * & * & * & 0 \\ * & * & * & 0 \end{bmatrix}.$$

And the conditions we listed are the complements of these block conditions. This case is simple enough for these block forms to be computed by hand, we use the convex algorithm 2 with $n = 1$. $\square$

Corollary 4.20. *If the image of φ_U is a three-dimensional vector space of skew-symmetric matrices, then φ_U is $\mathrm{SL}(V) \times \mathrm{SL}(W)$ -semistable, but $\mathrm{SL}(U)$ -unstable. This is the only $\mathrm{SL}(V) \times \mathrm{SL}(W)$ -semistable tensor whose determinant cubic is the zero cubic, up to basis change.*

Proof. If the determinant is the zero cubic, then the image $\varphi_U(U) \subset \mathrm{Hom}(V, W^\vee)$ is rank at most two everywhere. Any singular subspace is of type C or K or L (and $\mathrm{SL}(3) \times \mathrm{SL}(3)$ -unstable), *except* for a three-dimensional vector space of skew-symmetric matrices. All surjective maps from U to a three-dimensional vector space of skew-symmetric matrices are G -equivalent. $\square$

Corollary 4.21. *The restriction of the determinant map to the $\mathrm{SL}(V) \times \mathrm{SL}(W)$ -semistable locus $\mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)^{\mathrm{SL}(V) \times \mathrm{SL}(W)-ss}$ is not a morphism.*

If $\det : \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)^{\mathrm{SL}(V) \times \mathrm{SL}(W)-ss} \dashrightarrow |\mathcal{O}_{\mathbb{P}U}(3)|$ was a morphism, that would have been extremely useful. Then after further taking the GIT quotient by $\mathrm{SL}(4)$, the rational map $\det : \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)//G \dashrightarrow |\mathcal{O}_{\mathbb{P}U}(3)|//\mathrm{SL}(U)$ would be a morphism, and the set of semistable determinantal tensors would be exactly the set of determinantal representations of semistable cubic surfaces (by pullback). But because of Corollary 4.21, we cannot use this trick. We will see, after fully classifying the G -semistable locus, that the determinant map is not a morphism between GIT quotients.

4.2.2 $4 \times 3 \times 3$ semistability

Definition 4.22. A cubic surface is said to be *semistable* if it is $\mathrm{SL}(4)$ -semistable.

A tensor $\varphi \in \mathbb{P}(4 \times 3 \times 3)$ is said to be *semistable* if it is G -semistable. A determinantal representation φ of a given cubic surface S is semistable if it is semistable as a tensor.

The above objects are *unstable* if they are not semistable.

For example, we might say a cubic surface is unstable, but has a semistable determinantal representation, if the surface is $\mathrm{SL}(4)$ -unstable but the tensor representing it is G -semistable.

Applying the convex algorithm, we find that:

Theorem 4.23 ($4 \times 3 \times 3$ unstable locus). *A $4 \times 3 \times 3$ tensor $\varphi \in \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ is $\mathrm{SL}(4) \times \mathrm{SL}(3) \times \mathrm{SL}(3)$ -unstable if and only if one of the following is true:*

- (1) φ_U is not injective.
- (2) S_φ contains a plane of type C or a plane of type K .
- (3) S_φ contains a plane of type L .
- (4) φ is of the following block form:

$$\left[\begin{array}{ccc|ccc|ccc|ccc} * & * & * & * & * & * & * & * & * & * & 0 & 0 \\ * & * & * & * & * & * & * & 0 & 0 & 0 & 0 & 0 \\ * & * & * & * & * & * & * & 0 & 0 & 0 & 0 & 0 \end{array} \right].$$

(5) φ is of the following block form:

$$\left[\begin{array}{ccc|ccc|ccc|ccc} * & * & * & * & * & * & * & * & * & * & * & 0 \\ * & * & * & * & * & * & * & 0 & 0 & 0 & 0 & 0 \\ * & * & 0 & * & * & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right].$$

Any unstable determinantal representation of a normal cubic surface with finitely many lines is of block type (4) or (5).

Proof. See Appendix A for the complete classification of the block forms. They reduce to the above four cases, and any other $4 \times 3 \times 3$ -tensor is semistable. The block forms given in Theorem B.1 explicitly give weights where the above forms are unstable, and show that any other unstable tensor is of the above form.

If φ_U is not injective then S_φ is a cone, so contains infinitely many lines. And no normal cubic surface contains a plane, so cannot be of type (2) or (3). $\square$

Corollary 4.24. *If S_φ has finitely many lines and no rank one points, then φ is a semistable tensor.*

Proof. A tensor φ is unstable only if it is either not injective (1), reducible (2,3), or has a rank one point (4,5). Note that any plane of type L has a rank one point, but a general plane of type C or K does not. $\square$

Remark 4.25. Corollary 4.24 is not an if and only if: there are determinantal representations of semistable cubic surfaces that have rank one points. For example, the recurring Example 2.19 is a determinantal representation of the $4A_1$ cubic surface with 4 rank one points. The tensor φ^{4A_1} is $\mathrm{SL}(U) \times \mathrm{SL}(V) \times \mathrm{SL}(W)$ -semistable because the cubic surface S_{4A_1} is $\mathrm{SL}(U)$ -semistable (in fact both φ^{4A_1} and S_{4A_1} are stable). The block conditions (4) and (5) do not include a general tensor with a rank one point.

Corollary 4.26. *There are semistable tensors whose associated cubic surface is unstable, and even reducible. In particular, the A_3 cubic and the union of a plane and a smooth quadric both have semistable determinantal representations.*

- The rational map $\det : \mathbb{P}(4 \times 3 \times 3)^{G-ss} \dashrightarrow |\mathcal{O}_{\mathbb{P}^3}(3)|^{\mathrm{SL}(4)-ss}$ is not a morphism.
- The map $\det : \mathbb{P}(4 \times 3 \times 3)^{G-ss} \dashrightarrow \{\text{normal cubic surfaces}\} \subset |\mathcal{O}_{\mathbb{P}^3}(3)|$ is not a morphism.

Proof. Every A_3 cubic surface has a determinantal representation with no rank one points, found by blowing up a length 6 subscheme that has no length three subscheme contained in a line. This is explicitly computed by [Ng02], it is the $A_3(d)$ representation they provide.

And the $\mathrm{SL}(3) \times \mathrm{SL}(3)$ -semistability of skew-symmetric matrices extends to the G -semistability in some cases where U contains the vector space of skew-symmetric matrices. Consider the tensor

$$\varphi_U = \begin{bmatrix} u_0 & u_1 & u_2 \\ -u_1 & u_0 & u_3 \\ -u_2 & -u_3 & u_0 \end{bmatrix}, \quad \det(\varphi_U) = u_0(u_0^2 - u_1^2 - u_2^2 - u_3^2). \quad (4.2)$$

If S_φ contained a rank one point, then all the 2×2 minors of φ_U would have to vanish. But then $u_0^2 + u_1^2 = u_0^2 + u_2^2 = u_0^2 + u_3^2 = 0$, so $u_1^2 + u_2^2 + u_3^2 = 3u_0^2$. This must lie on the surface S_φ , so $u_0(u_0^2 + 3u_0^2) = 0$. But if $u_0 = 0$ then $u_1 = u_2 = u_3 = 0$, so $\varphi_U(u_0, u_1, u_2, u_3)$ is only zero when $u_0 = u_1 = u_2 = u_3 = 0$, which is not a point in $\mathbb{P}^3$. The map φ_U is injective, and the only plane it contains is a subspace of skew-symmetric matrices, which is not of type C or K or L.

So there exists a semistable tensor φ whose determinant cubic $\det(\varphi_U)$ is reducible and unstable. $\square$

Corollary 4.27. *Taking the determinant of a $4 \times 3 \times 3$ tensor does not induce a morphism of GIT-quotients, the following map is only rational:*

$$\det : \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee) // (\mathrm{SL}(U) \times \mathrm{SL}(V) \times \mathrm{SL}(W)) \dashrightarrow |\mathcal{O}_{\mathbb{P}^U}(3)| // \mathrm{SL}(U).$$

Remark 4.28. The semistability of the determinantal representation (4.2) can be understood from Gale-duality. The subscheme cut out by φ_U is an irreducible conic in $\mathbb{P}^U$ — this is the tensor associated to 6 points on a conic, which is a semistable point configuration in $\mathrm{Hilb}_6(\mathbb{P}^U)$, and associated to a semistable A_1 cubic surface.

Proposition 4.29. *Every A_3 unstable cubic surface has an unstable determinantal representation.*

Proof. The following determinantal representation is a general A_3 cubic surface of block type (4), where $\lambda \neq 0$:

$$\varphi_U = \begin{bmatrix} u_3 & u_2 & -\lambda u_2 \\ u_0 + u_1 + u_2 & u_1 & 0 \\ u_0 + u_1 + u_2 & 0 & u_0 \end{bmatrix}, \quad \det(\varphi_U) = u_3(u_0 u_1) - u_2(u_0 + u_1 + u_2)(u_0 - \lambda u_1) \quad (4.3)$$

which is the normal form seen in [BW79]. To see this is block type (4), look at the non-zero entries of the four matrices $[\varphi_U(u_0)|\varphi_U(u_1)|\varphi_U(u_2)|\varphi_U(u_3)]$. $\square$

Remark 4.30. The failure of the map $\mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee) // G \dashrightarrow |\mathcal{O}_{\mathbb{P}^U}(3)| // \mathrm{SL}(U)$ to be a morphism can also be motivated geometrically, without actually computing the unstable locus.

Recall the constructions of Example 3.49, where we found a one-parameter family of determinantal representations $\hat{\phi} : T \rightarrow \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ where $\det(\phi(t))$ cuts out a smooth cubic surface for every $t \neq t_0$, the determinant $\det(\hat{\phi}(t_0))$ was the zero cubic (undefined), but the limit of the family of cubic surfaces at $t = t_0$ was an A_1 cubic surface S_{t_0} .

Remark 4.31 (Determinantal threefolds). The same algorithm can be applied to classify semistability of $5 \times 3 \times 3$ tensors. An injective $5 \times 3 \times 3$ tensor embeds $\mathbb{P}^4$ as a 4-dimensional linear subspace $\Lambda \hookrightarrow \mathbb{P}(V \otimes W)$. The rank two locus of Λ is a cubic threefold, cut out by the determinant of a $5 \times 3 \times 3$ matrix. It is the dual to φ .

After running the convex algorithm, we find that the $\mathrm{SL}(5) \times \mathrm{SL}(3) \times \mathrm{SL}(3)$ -semistable $5 \times 3 \times 3$ tensors are exactly the duals of the semistable $4 \times 3 \times 3$ tensors. This is not surprising: because $\mathrm{Gr}(m, n) = \mathbb{P}(\mathbb{C}^m \otimes \mathbb{C}^n) // \mathrm{SL}(m)$, then the GIT quotient $\mathbb{P}(U^\vee \times V^\vee \times W^\vee) // (\mathrm{SL}(U) \times \mathrm{SL}(V) \times \mathrm{SL}(W))$ is the quotient of the Grassmannian $\mathrm{Gr}(4, V \otimes W) // \mathrm{SL}(V) \times \mathrm{SL}(W)$, which is isomorphic to the quotient $\mathrm{Gr}(5, V \otimes W) // \mathrm{SL}(V) \times \mathrm{SL}(W)$ — the semistable loci had to be the same by various universal properties of the GIT quotient. Nonetheless, it is a nice sanity check that the algorithm does give the right answer.

4.3 What other moduli spaces might work?

Because there is no morphism from $\mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee) // G$ to $|\mathcal{O}_{\mathbb{P}^U}(3)| // \mathrm{SL}(U)$, the moduli problem of $4 \times 3 \times 3$ tensors does not perfectly model the space of cubic surfaces. There is also not a morphism from $\mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee) // G$ to any moduli space of points in $\mathbb{P}V$, because of the semistability of the tensor (4.2) — the ideal generated by its minors is a conic in $\mathbb{P}V$, not even finite. As such, we might want to refine the objects we are looking at, and consider some other similar moduli spaces.

The two key objects that the $4 \times 3 \times 3$ tensor gives are its two cokernel sheaves, the sheaves $\mathrm{coker}(\tilde{\varphi}_U)$ and $\mathrm{coker}(\tilde{\varphi}_V)$. The latter is the ideal sheaf $\mathcal{I}_V^\varphi(3)$, and instead treating the ideal sheaf $\mathcal{I}_V^\varphi$ as the fundamental object gives us the Hilbert scheme $\mathrm{Hilb}_6(\mathbb{P}^2)$.

The asymptotic GIT semistability of $\text{Hilb}_6(\mathbb{P}^2)$ is known, and we have the following theorem (rephrased in the language of cubic surfaces):

Theorem 4.32. [Dur15, Thm. 4.4.1] (Semistable locus of $\text{Hilb}_6(\mathbb{P}^2)$ when $m \gg 0$)

A length 6 subscheme $\Gamma \in \text{Hilb}_6(\mathbb{P}^2)$ is $\text{SL}(3)$ -semistable when $m \gg 0$ if and only if all of the following conditions hold:

- (1) *No length three subscheme of Γ is contained in a line.*
- (2) *The length of Γ at any point is at most two.*
- (3) *The cubic surface associated to Γ has at worst $3A_2$ singularities.*

As a result, there is a morphism $\text{Hilb}_6(\mathbb{P}^2) // \text{SL}(3) \rightarrow |\mathcal{O}_{\mathbb{P}^3}(3)| // \text{SL}(4)$, for large m . But there are unstable point configurations whose associated cubic surface is semistable (the opposite problem to ours!). As a result, this moduli problem is too fine.

There are two other natural generalisations that arise from the $4 \times 3 \times 3$ tensor. We could take the cokernel sheaf $\text{coker}(\tilde{\varphi}_U)$ as the fundamental object. It is a quotient of $\mathcal{O}_{\mathbb{P}^3}^{\oplus 3}$, and the cubic surface S_φ is recovered by finding its support. A general quotient of $V \otimes \mathcal{O}_{\mathbb{P}^U}$ is a line bundle supported on a cubic surface, and so

Alternatively, one could take the birational map $S_\varphi \dashrightarrow \mathbb{P}V$ to be the fundamental object. This map is given by a line bundle on the blowup of the rank one locus $\text{Bl}_{S_{\varphi, \text{rk } 1}} S_\varphi$, which is a twisted cubic when S is a rank two surface. The moduli problem of the pair (S, C) , where C is a twisted cubic on a cubic surface S , could yield interesting results, and is birational to the moduli space of $4 \times 3 \times 3$ tensors (after quotienting by the appropriate groups).

Remark 4.33. The moduli problem of twisted cubics is promising. Whenever we have a birational map $\pi^{-1} : \mathbb{P}^2 \dashrightarrow S \hookrightarrow \mathbb{P}^3$ given by some linear system of cubics that contracts a conic C to some singularity u , a general $L \subset \mathbb{P}^2$ will intersect C twice. So the resulting twisted cubic $\pi^{-1}(L)$ that defines the inverse map $\pi : S \dashrightarrow \mathbb{P}^2$ has a double point at u . Whereas when a linear system of cubics only contracts lines, a general line L only passes through each of these lines once and so the twisted cubic $\pi^{-1}(L)$ only intersects each singularity with multiplicity one — at least for an A_1 cubic surface, adding non-determinantal line bundles is the same as considering twisted cubics that have multiplicity two at a singularity.

Appendix A

The recurring $4A_1$ example

We provide the complete list of examples for the $4A_1$ cubic surface here as well.

Example 2.2. To start with, consider the following pair of ordered point sets

$$\begin{aligned}\Gamma_V^{4A_1} &= \{[1 : 0 : 0], [0 : 1 : 0], [0 : 0 : 1], [0 : -1 : 1], [1 : 0 : -1], [-1 : 1 : 0]\} \subset \mathbb{P}V \\ \Gamma_W^{4A_1} &= \{[0 : -1 : 1], [1 : 0 : -1], [-1 : 1 : 0], [1 : 0 : 0], [0 : 1 : 0], [0 : 0 : 1]\} \subset \mathbb{P}W.\end{aligned}$$

They are Gale-dual. See Figure A.1 for a picture of the point arrangement.

Example 2.6. In this case, the matrix is

$$A = \begin{pmatrix} 0 & -1 & 1 \\ 1 & 0 & -1 \\ -1 & 1 & 0 \end{pmatrix}.$$

Notice that $A = -A^T$ is skew-symmetric, hence $\Gamma_W^{4A_1}$ is the same as $\Gamma_V^{4A_1}$ as sets. But they are not the same ordered point configuration — the first three points of $\Gamma_V^{4A_1}$ are linearly independent, whereas the first three points of $\Gamma_W^{4A_1}$ are linearly dependent.

Example 2.14. The tensor product $\Gamma_V^{4A_1} \otimes \Gamma_W^{4A_1}$ of the point configurations consists of the 6 points

$$\begin{pmatrix} 0 & 0 & 0 \\ -1 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & -1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & -1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 0 \\ -1 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ -1 & 1 & 0 \end{pmatrix}.$$

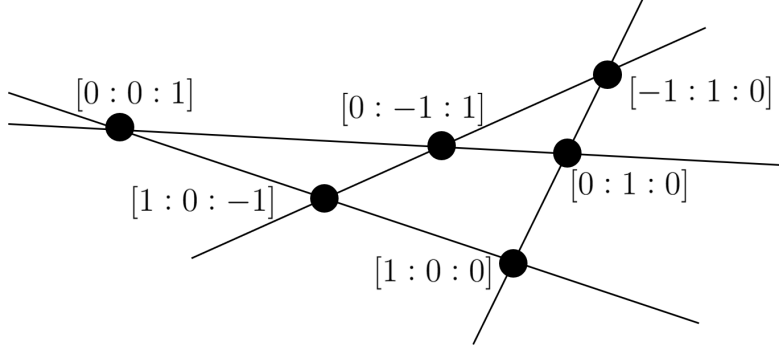


Figure A.1: The point configuration $\Gamma_V^{4A_1}$, where φ^{4A_1} is a determinantal representation of a $4A_1$ cubic surface.

Consider the map $\varphi_U^{4A_1} : U \rightarrow V^\vee \otimes W^\vee$, defined by:

$$\varphi_U = \begin{pmatrix} u_0 & u_1 & u_1 \\ u_1 & u_2 & u_1 \\ u_1 & u_1 & u_1 + u_3 \end{pmatrix}. \quad (\text{A.1})$$

The kernel of the dual map $(\varphi^{4A_1})^\vee : V \otimes W \rightarrow U^\vee$ is the span of $\Gamma_V^{4A_1} \otimes \Gamma_W^{4A_1}$.

Example 2.19. The determinant cubic is

$$\det(\varphi_U^{4A_1}) = u_3(u_0u_2 - u_1^2) - u_1(u_0 - u_1)(u_1 - u_2). \quad (\text{A.2})$$

This is exactly the normal form of the $4A_1$ cubic surface (see [BW79]). We call it *the* $4A_1$ cubic surface, because every cubic surface with $4A_1$ singularities is isomorphic to $S_{4A_1} = S_{\varphi^{4A_1}}$.

There are four rank one points on S_{4A_1} , having coordinates

$$S_{4A_1, \text{rk } 1} = \{[1 : 0 : 0 : 0], [1 : 1 : 1 : 0], [0 : 0 : 1 : 0], [0 : 0 : 0 : 1]\}$$

The rank two loci $\Gamma_V^{\varphi^{4A_1}}$ and $\Gamma_W^{\varphi^{4A_1}}$ are the point configurations $\Gamma_V^{4A_1}$ and $\Gamma_W^{4A_1}$ introduced in Example 2.2.

Example 2.23. The fibre of π_V above $[1 : 0 : 0]$ is the line $[0 : 0 : s : t]$. The fibre above $[1 : 1 : 0]$ is empty. The fibre above $[1 : 2 : 0]$ is the point $[4 : 2 : 1 : 0]$.

Example 2.30. The lines of type C and K (they are of both types at the same time for this specific example, because $\varphi_U^{4A_1}$ is symmetric, and are also of

type L) are

$$s = \{[0 : 0 : s : t], [s : 0 : 0 : t], [s : 0 : t : 0], [s : s : s : t], [s : s : t : 0], [t : s : s : 0]\}.$$

These are ordered to be the fibres above the points in $\Gamma_V^{4A_1}$ in the same order as given in Example 2.2. We use the notation s , because this is a six on S_{4A_1} . Each line $M_i \in s$ passes through exactly two singularities on S_{4A_1} . The lines of type L only are

$$\{[s : t : 0 : t - s], [0 : t : s : t - s], [s + 2t : t : -s : -t]\}.$$

These do not pass through any of the singularities. There are 9 lines on the $4A_1$ cubic surface.

Example 2.41. The 4×3 matrix is

$$\varphi_V^{4A_1} = \begin{pmatrix} v_0 & 0 & 0 \\ v_1 + v_2 & v_0 + v_2 & v_0 + v_1 + v_2 \\ 0 & v_1 & 0 \\ 0 & 0 & v_2 \end{pmatrix}. \quad (\text{A.3})$$

The minors of $\varphi_V^{4A_1}$ are

$$m_0 = v_1 v_2 (v_0 + v_2), \quad m_1 = -v_0 v_1 v_2, \quad m_2 = v_0 v_2 (v_0 + v_2), \quad m_3 = -v_0 v_1 (v_0 + v_1 + v_2)$$

and one can verify that $\det(\varphi_U(m_0, m_1, m_2, m_3)) = 0$ by substituting into (A.2).

Example 2.44. The map $\tilde{m}_V : \text{Bl}_{\Gamma_V^{4A_1}} \mathbb{P}V \rightarrow S_{4A_1}$ contracts the strict transforms of 4 lines: the lines cut out by the equations $v_0 = 0$, $v_0 + v_1 + v_2 = 0$, $v_1 = 0$, $v_2 = 0$. The respective images of these lines are the rank one singularities

$$[1 : 0 : 0 : 0], [1 : 1 : 1 : 0], [0 : 0 : 1 : 0], [0 : 0 : 0 : 1].$$

These singularities are all A_1 singularities, so the surface $\text{Bl}_{\Gamma_V^{4A_1}} \mathbb{P}V$ is smooth — each A_1 singularity is resolved after only one blowup. This is expected: $\text{Bl}_{\Gamma_V^{4A_1}} \mathbb{P}V$ is the blowup of the 6 reduced points $\Gamma_V^{4A_1}$, so had to be smooth.

Example 3.43. We consider a limiting family of smooth cubic surfaces $S_T \rightarrow T$ to $S_{t_0} = S_{4A_1}$. On each of the smooth surfaces S_t , there are 6 lines M_i , 6 lines N_i , and 15 lines L_{ij} . But at $t = t_0$, many lines collide.

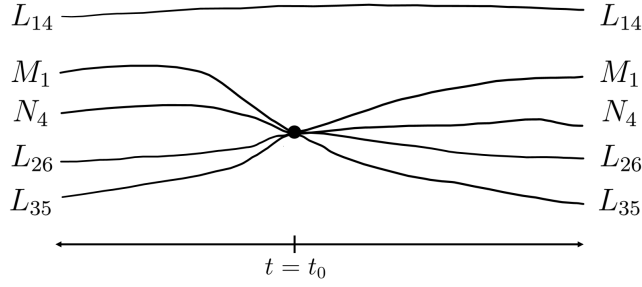


Figure A.2: Some of the fibres of the Hilbert scheme $\text{Hilb}_{m+1}(S_T/T)$, in Example 3.43. The lines M_1, N_4, L_{26}, L_{35} all collide at $t = t_0$, and the length of the local ring at $[M_1]$ is 4. But the line L_{14} never collides with any other lines — the point $[L_{14}]$ is length one.

The collapsed sixes are $s_{126}, s_{135}, s_{234}, s_{456}$. The lines on S collide as follows:

$$\begin{array}{c|c|c|c|c|c|c|c|c}
 M_1 & M_2 & M_3 & M_4 & M_5 & M_6 & L_{14} & L_{25} & L_{36} \\
 N_4 & N_5 & N_6 & N_1 & N_2 & N_3 & & & \\
 L_{26} & L_{16} & L_{15} & L_{23} & L_{13} & L_{12} & & & \\
 L_{35} & L_{34} & L_{24} & L_{56} & L_{46} & L_{45} & & &
 \end{array} \tag{A.4}$$

The first six lines are all mutually skew lines of multiplicity 4, and each pass through 2 of the singularities on S (the lines of type K (and C and L) in Example 2.30). The last 3 don't pass through any of the singularities, these are the lines of type L only. See Figure A.2 for a picture of what $\text{Hilb}_{m+1}(S_T/T)$ looks like.

In general, each singularity on a normal cubic surface with finitely many lines corresponds to one double-six collapsing. A line through an A_1 singularity has multiplicity 2, a line through two A_1 singularities has multiplicity 4. A line through an A_2 singularity has multiplicity 3. Once a surface has multiple different singularities, or has A_3 or worse singularities, then different lines through the same singularity will have different multiplicities.

Example 3.55. Recall the point configuration in Example 2.2, where the triples $\{v_1, v_2, v_6\}, \{v_1, v_3, v_5\}, \{v_2, v_3, v_4\}, \{v_4, v_5, v_6\}$ are each collinear. We write L_{ijk} for the line in $\mathbb{P}V$ that passes through v_i, v_j, v_k .

The action of $\mathfrak{S}_4$ permutes the four lines $L_{126}, L_{135}, L_{234}, L_{456}$, and acts on the point configuration by extension. Up to the $\mathfrak{S}_4$ -action, there are only 5 non-collapsed sixes $s, \bar{s}, s_{12}, s_{21}, s_{236}, \bar{s}_{345}$. And $s = \bar{s}$ (recall $\Gamma_V^{4A_1} = \Gamma_W^{4A_1}$ as sets), $s_{236} = \bar{s}_{345}$, and $s_{21} \sim s_{234}$ as sets, so there are only 2 possible determinantal representations, induced from the sixes s and s_{21} . The determinantal representation induced by s is the example we used throughout Chapter 2. These are

inequivalent: the six s_{21} only has 4 distinct lines on S_{4A_1} .

Remark 4.25. Corollary 4.24 is not an if and only if: there are determinantal representations of semistable cubic surfaces that have rank one points. For example, the recurring Example 2.19 is a determinantal representation of the $4A_1$ cubic surface with 4 rank one points. The tensor φ^{4A_1} is $\mathrm{SL}(U) \times \mathrm{SL}(V) \times \mathrm{SL}(W)$ -semistable because the cubic surface S_{4A_1} is $\mathrm{SL}(U)$ -semistable (in fact both φ^{4A_1} and S_{4A_1} are stable). The block conditions (4) and (5) do not include a general tensor with a rank one point.

Appendix B

Computational classification of the $4 \times 3 \times 3$ unstable locus

B.1 Implementation of the Hilbert-Mumford criterion

Over $\mathbb{C}$, our numerical implementation of the Hilbert-Mumford criterion gives a classification of the unstable locus of the $\mathrm{SL}(l) \times \mathrm{SL}(m) \times \mathrm{SL}(n)$ action on $\mathbb{P}(\mathbb{C}^l \otimes \mathbb{C}^m \otimes \mathbb{C}^n)$. We use the notation of §4.2.

The naive implementation is Algorithm 1. It involves finding $P_{I \times J \times K}$, computing every subset of $P_{I \times J \times K}$, and if the convex hull contains 0 then appending it to a list of unstable loci. We remove any subset of an unstable locus from the set of unstable loci, so that it only contains maximal unstable loci with respect to the partial order of subsets of $P_{I \times J \times K}$. This is a very very slow algorithm for the $4 \times 3 \times 3$ case – it needs to check 2^{36} convex hulls and then check inclusions on a list of $\sim 2^{30}$ elements. The coefficients of h are the 1-PS that unstabilises a given unstable point set, which we print for manual verification if desired.

We implement a significant optimisation to reduce this algorithm to one that runs in just a few minutes. The permutation action of $\mathfrak{S}_I \times \mathfrak{S}_J \times \mathfrak{S}_K$ on the basis vectors of $\mathbb{C}^l$ and $\mathbb{C}^m$ and $\mathbb{C}^n$ induces a permutation of the character $\chi_i^{(a)}$ for each $a \in \{l, m, n\}$. For each 1-PS λ , we choose a permutation of coordinates so that the weights $p_{(i,j,k)}(\bar{\lambda})$ are non-increasing, in the sense that $p_{(i,j,k)}(\bar{\lambda}) \geq p_{(i',j',k')}(\bar{\lambda})$ whenever $i \leq i'$ and $j \leq j'$ and $k \leq k'$. A 1-PS in this form is said to act with *nonincreasing weights*.

Then if the point $p_{(i,j,k)}$ is λ -unstable when λ has nonincreasing weights, the point $p_{(i',j',k')}$ will also be λ -unstable if $i' \leq i$ and $j' \leq j$ and $k' \leq k$. So the data of a maximal unstable locus after permutation of coordinates is only the set of

Algorithm 1 Slow $l \times m \times n$ instability algorithm

Compute $P_{I \times J \times K} \subset (\mathbb{Z}^{l+m+n-3})^\vee$.
set_of_pointsets := {power set of $P_{I \times J \times K}$ }
function IS_UNSTABLE(P_F)
 Compute the convex hull $\text{conv}(P_F)$ of P_F
 if $0 \notin \text{conv}(P_F)$ **then return** true
 else if $0 \in \text{conv}(P_F)$ **then return** false
 end if
end function
unstable_loci = [empty array]
for all $P_F \in \text{set_of_pointsets}$ **do**
 if IS_UNSTABLE(P_F) and $(P_F \not\subset P_{F'} \quad \forall P_{F'} \in \text{unstable_loci})$ **then**
 while $\exists P_{F'} \in \text{unstable_loci}$ where $P_{F'} \subset P_F$ **do**
 Remove $P_{F'}$ from **unstable_loci**
 end while
 Append P_F to **unstable_loci**
 end if
end for
for all $P_F \in \text{unstable_loci}$ **do**
 for all hyperplanes h bounding $\text{conv}(P_F)$ **do**
 if 0 and $\text{conv}(P_F)$ are on the same side of h **then**
 Print the coefficients of $h \quad \triangleright P_{F'}$ is unstable under these weights
 end if
 Print the block matrix form of P_F
 end for
end for
 Manually interpret and classify the not-so-small number of printed block forms.

largest values of (i, j, k) under this partial ordering. As an example, the data of the following maximal unstable $4 \times 3 \times 3$ block form

$$\left[\begin{array}{ccc|ccc|ccc|ccc} * & * & * & * & * & * & * & * & *! & * & *! & 0 \\ * & * & * & * & * & *! & * & *! & 0 & 0 & 0 & 0 \\ * & * & 0 & * & *! & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right] \quad (\text{B.1})$$

is captured by the five entries which we have labelled with an $!$. We call this a *flag condition*, because it can be equivalently rewritten in terms of a sequence of subspaces of $\mathbb{C}^l \otimes \mathbb{C}^m \otimes \mathbb{C}^l$ (which is known as a flag, in the sense of linear algebra). We convert this to a tuple as follows:

- A 2-flag tuple is an element of $\mathbb{Z}^J$. Its entries k_j are elements of $K \cup \{0\}$, and we enforce $k_j \geq k_{j'}$ if $j \leq j'$ due to the lexicographic ordering.
- A 3-flag tuple is an element $F \in (\mathbb{Z}^J)^I$. It consists of l two-flag tuples $(k_{i1}, \dots, k_{im})$ indexed by $i \in I$. We enforce $F_{ij} \geq F_{i'j}$ if $i \leq i'$ due to the lexicographic ordering.
- An element $(i, j, k) \in I \times J \times K$ satisfies the flag condition F if $k \leq F_{ij}$. We let $F \subset I \times J \times K$ be the set of all tuples (i, j, k) satisfying the flag condition F , by abuse of notation.

For example, where we identify the rows of the matrix with J and the columns with K , the block form B.1 is represented by the flag tuple

$$((3, 3, 2), (3, 3, 2), (3, 2, 0), (2, 0, 0)). \quad (\text{B.2})$$

We then have a partial ordering on these tuples, where we set $F \preceq F'$ if and only if $F_{ij} \leq F'_{ij}$ for all i, j . This is equivalent to $F \subset F'$ as subsets of $I \times J \times K$, or equivalently $P_F \subset P_{F'}$.

As a result, it suffices to only check point configurations $P_F \subset P_{I \times J \times K}$ where F is a flag condition, and it is very easy to check $P_F \subset P_{F'}$ by only checking if $F \preceq F'$, instead of computing these as subsets. This cuts down the number of things to check for the $4 \times 3 \times 3$ case from 2^{36} point sets using Algorithm 1 down to 4116 flags using Algorithm 2. This runs in under one minute, and scales well to even the $8 \times 3 \times 3$ unstable locus. Some optimisations can probably be applied to implement this method for larger tensors. The runtime of this algorithm is of the order of $(\dim U)^{\dim V \cdot \dim W}$, significant optimisations are needed to compute the unstable locus for even $4 \times 4 \times 4$ tensors. We are only interested in the $4 \times 3 \times 3$ and $5 \times 3 \times 3$ cases in this thesis because of their relationship to cubic surfaces, so

we do not attempt to implement any further optimisations, and we are somewhat lucky that this case is easily computable with a reasonably simple optimisation.

Algorithm 2 Fast $l \times m \times n$ instability algorithm.

```

Compute  $P_{I \times J \times K} \subset (\mathbb{Z}^{l+m+n-3})^\vee$ 
Compute set_of_3_flags := {set of all 3-flag tuples}
3_flags := [array of 3-flags tuples ordered in a non-increasing order]
function IS_UNSTABLE( $F$ )
    Convert the flag  $F$  to a set of points  $P_F \subset (\mathbb{Z}^{l+m+n-3})^\vee$ 
    Compute the convex hull  $\text{conv}(P_F)$  of  $P_F$ 
    if  $0 \notin \text{conv}(P_F)$  then return true
    else if  $0 \in \text{conv}(P_F)$  then return false
    end if
end function
unstable_loci = [empty array]
for all  $F \in \mathbf{3\_flags}$  do
    if IS_UNSTABLE( $F$ ) and  $(F \not\preceq F' \ \forall F' \in \mathbf{unstable\_loci})$  then
        while  $\exists F' \in \mathbf{unstable\_loci}$  where  $F' \preceq F$  do
            Remove  $F'$  from unstable_loci
        end while
        Append  $F$  to unstable_loci
    end if
end for
for all  $F' \in \mathbf{unstable\_loci}$  do
    Convert the flag  $F'$  to a set of points  $P_{F'}$ 
    for all hyperplanes  $h$  bounding  $\text{conv}(P_{F'})$  do
        if  $0$  and  $\text{conv}(P_{F'})$  are on the same side of  $h$  then
            Print the coefficients of  $h$ 
        end if
    end for
    Print the block matrix form of  $F'$ 
end for
Manually interpret and classify the small number of printed block forms.
```

We provide an implementation of this code using [Sage], available at the following link:

<https://github.com/charlotte-sherratt/4x3x3-GIT-quotient>.

B.2 Block forms for the $4 \times 3 \times 3$ unstable locus

Theorem B.1. *A closed point $\varphi \in \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ is $\mathrm{SL}(U) \times \mathrm{SL}(V) \times \mathrm{SL}(W)$ -unstable if and only if it is of one of the following block forms:*

- (1) φ_U is not injective. $\mathrm{SL}(4)$ -unstable.

Unstable by the weight $(0, 0, 0), (0, 0, 0), (1, 1, 1, -3)$.

$$\left[\begin{array}{ccc|ccc|ccc} * & * & * & * & * & * & * & * & * \\ * & * & * & * & * & * & * & * & * \\ * & * & * & * & * & * & * & * & * \end{array} \right]$$

- (2) S contains a plane of type C or a plane of type K. $\mathrm{SL}(3) \times \mathrm{SL}(4)$ -unstable.

Unstable by the weight $(4, 4, -8), (0, 0, 0), (9, -3, -3, -3)$.

$$\left[\begin{array}{ccc|ccc|ccc} * & * & * & * & * & * & * & * & * \\ * & * & * & * & * & * & * & * & * \\ * & * & * & 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

- (3) S contains a plane of type L. $\mathrm{SL}(3) \times \mathrm{SL}(3) \times \mathrm{SL}(4)$ -unstable.

Unstable by the weight $(8, -4, -4), (8, -4, -4), (9, -3, -3, -3)$.

$$\left[\begin{array}{ccc|ccc|ccc} * & * & * & * & * & * & * & * & * \\ * & * & * & * & 0 & 0 & * & 0 & 0 \\ * & * & * & * & 0 & 0 & * & 0 & 0 \end{array} \right]$$

And two more types with slightly less clear geometric conditions:

- (4) Unstable by the weight $(8, -4, -4), (8, -4, -4), (9, 9, -3, -15)$.

$$\left[\begin{array}{ccc|ccc|ccc} * & * & * & * & * & * & * & 0 & 0 \\ * & * & * & * & * & * & * & 0 & 0 \\ * & * & * & * & * & * & * & 0 & 0 \end{array} \right]$$

- (5) Unstable by the weight $(12, 0, -12), (4, 4, -8), (9, 9, -3, -15)$.

$$\left[\begin{array}{ccc|ccc|ccc} * & * & * & * & * & * & * & * & 0 \\ * & * & * & * & * & * & * & 0 & 0 \\ * & * & 0 & * & * & 0 & 0 & 0 & 0 \end{array} \right]$$

The algorithm finds seven other maximal block forms. These are contained in one of the forms (1-5) in a different basis, but are maximal with respect to their respective bases. We list them here for completeness, if you run the code yourself you will find them:

Contained in type (2):

$$(6) \left[\begin{array}{ccc|ccc|ccc|ccc} * & * & * & * & * & * & * & * & * & * & * & 0 \\ * & * & 0 & * & * & 0 & * & * & 0 & * & 0 & 0 \\ * & 0 & 0 & * & 0 & 0 & * & 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

$$(7) \left[\begin{array}{ccc|ccc|ccc|ccc} * & * & * & * & * & * & * & * & * & * & * & 0 \\ * & * & * & * & * & * & * & * & 0 & * & 0 & 0 \\ * & 0 & 0 & * & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

$$(8) \left[\begin{array}{ccc|ccc|ccc|ccc} * & * & * & * & * & * & * & * & * & * & * & * \\ * & * & * & * & * & * & * & 0 & 0 & * & 0 & 0 \\ * & 0 & 0 & * & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

$$(9) \left[\begin{array}{ccc|ccc|ccc|ccc} * & * & * & * & * & * & * & * & * & * & 0 & 0 \\ * & * & * & * & * & * & * & * & * & * & 0 & 0 \\ * & 0 & 0 & * & 0 & 0 & * & 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

Contained in type (4):

$$(10) \left[\begin{array}{ccc|ccc|ccc|ccc} * & * & * & * & * & * & * & * & * & * & * & * \\ * & * & * & * & * & * & 0 & 0 & 0 & 0 & 0 & 0 \\ * & * & * & * & * & * & 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

$$(11) \left[\begin{array}{ccc|ccc|ccc|ccc} * & * & * & * & * & * & * & * & * & * & 0 & 0 \\ * & * & * & * & * & 0 & * & * & 0 & 0 & 0 & 0 \\ * & * & 0 & * & 0 & 0 & * & 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

Contained in type (5):

$$(12) \left[\begin{array}{ccc|ccc|ccc|ccc} * & * & * & * & * & * & * & * & 0 & * & * & 0 \\ * & * & * & * & * & * & * & * & 0 & * & * & 0 \\ * & * & 0 & * & * & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

Proof. We refer to the code to actually compute these block forms.

Types (6-9) are contained in type (2) because there is a change of basis that converts the bottom row to be of the form

$$\left[* \ 0 \ 0 \mid 0 \ 0 \ 0 \mid 0 \ 0 \ 0 \mid 0 \ 0 \ 0 \right].$$

(10) $\subset$ (4) by changing basis where the last matrix is of the form $e_1 \otimes e_1$. And (11) $\subset$ (4) by taking a linear combination of the middle two matrices that cancels out the middle entry.

(12) $\subset$ (5) comes from the fact that any line of type C and K contains a rank one point.

The remaining five types are inequivalent. $\square$

B.3 The moduli space of determinantal cubic threefolds

The same algorithm can be very easily applied to the case of $n \times 3 \times 3$ tensors for arbitrary integer n . Let $\phi : k^n \rightarrow \text{Hom}(k^3, k^3)$ be a non-zero linear map for some integer $n \geq 5$. If $n > 9$ then ϕ isn't injective, and so is $\text{SL}(n)$ -unstable. Any non-injective tensor is unstable, so we can restrict our attention to injective linear maps $\phi : k^n \rightarrow \text{Hom}(k^3, k^3)$. The cokernel $\text{coker}(\phi) \cong k^{9-n}$, and so up to the $\text{SL}(n)$ -action, an injective $n \times 3 \times 3$ tensor gives a unique $(9 - n) \times 3 \times 3$ tensor. This cuts out a cubic hypersurface in $\mathbb{P}^{9-n}$ that is dual to the surface cut out by $\det(\phi) = 0$. We will briefly discuss this correspondence for cubic surfaces and threefolds. Note that the moduli space of cubic threefolds is $34 - 24 = 10$ -dimensional, so a threefold being determinantal is a codimension 6 condition.

Definition B.2. Let $\varphi \in U^\vee \otimes V^\vee \otimes W^\vee$ be an injective $4 \times 3 \times 3$ tensor as before. The map $\varphi_U : U \hookrightarrow V^\vee \otimes W^\vee$ has cokernel $\text{coker}(\varphi_U) \cong U^\perp \subset V \otimes W$, and $\dim U^\perp = 5$.

We write $\varphi_{U^\perp}^\perp : U^\perp \hookrightarrow V \otimes W$ for the standard inclusion. Let $T_\varphi \hookrightarrow \mathbb{P}U^\perp$ be the three-fold cut out by $\det(\varphi_{U^\perp}^\perp) = 0$. We call this threefold the *dual determinantal threefold* to a determinantal cubic surface S_φ .

We have $U^\perp = \ker(\varphi^\vee)$, so $T_\varphi \hookrightarrow \mathbb{P}(\ker(\varphi^\vee))$ is the rank one and two locus of this linear subspace.

Conversely, any injective linear map $\phi : k^5 \rightarrow V \otimes W$ is of this form, and so every determinantal threefold cut out by an injective tensor has a dual determinantal cubic surface.

Proposition B.3. *Suppose S_φ is normal with finitely many lines. Then T_φ has rank one singularities at all closed points of $\Gamma_{V \otimes W}$.*

Proof. $\Gamma_{V \otimes W} \subset T_\varphi$, these are exactly the rank one points inside $T_\varphi \hookrightarrow \Lambda$. $\square$

In the general case, we have the following isomorphism:

Proposition B.4. [HT10, Props. 10, 19, 22] *There is an isomorphism*

$$\left\{ \begin{array}{l} \text{cubic threefolds with 6} \\ \text{ordered ordinary} \\ \text{double points in} \\ \text{general linear position} \end{array} \right\} / \text{SL}(5) \longrightarrow \left\{ \begin{array}{l} \text{smooth cubic surfaces} \\ \text{with an identified} \\ \text{double-six} \end{array} \right\} / \text{SL}(4)$$

Proof. See [HT10] for a full proof that any threefold with 6 ordinary double points in general linear position is determinantal (proposition 19). Once we find

a determinantal representation, it is uniquely determined by the location of the 6 double points, which are contained in the rank one locus of $\mathbb{P}(V \otimes W)$. These are the Gale dual configuration $\Gamma_V \otimes \Gamma_W$, which produces a unique cubic surface with an identified double-six. $\square$

We now turn to the unstable locus of the $5 \times 3 \times 3$ GIT quotient. Let $\tilde{U}$ be a five-dimensional vector space.

Theorem B.5. *A $5 \times 3 \times 3$ tensor $\varphi^\perp \in \mathbb{P}(\tilde{U} \otimes V \otimes W)$ is $\mathrm{SL}(5) \times \mathrm{SL}(3) \times \mathrm{SL}(3)$ -unstable if and only if it is of one of the following forms:*

- (1) $\varphi^\perp_{\tilde{U}}$ is not injective. This is $\mathrm{SL}(5)$ -unstable.
- (2) T_φ contains a rank one line of shared kernel or cokernel. This also includes the case where T_φ contains a hyperplane of type L.
- (3) T_φ contains a plane of simultaneous type C and K.
- (4) The following block form also includes the case where T_φ contains a hyperplane of type C or K.

$$\left[\begin{array}{ccc|ccc|ccc|ccc|ccc} * & * & * & * & * & * & * & * & * & * & 0 & * & * & 0 \\ * & * & * & * & * & * & * & * & * & * & 0 & * & * & 0 \\ * & * & 0 & * & * & 0 & * & * & 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right].$$

- (5) This is the only asymmetric unstable block type.

$$\left[\begin{array}{ccc|ccc|ccc|ccc|ccc} * & * & * & * & * & * & * & * & * & * & * & 0 & 0 & 0 \\ * & * & * & * & * & * & * & * & * & 0 & 0 & 0 & 0 & 0 \\ * & * & * & * & 0 & 0 & * & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right].$$

Types (2-5) are the annihilators of $4 \times 3 \times 3$ types (2-5), respectively, under an identification $U^\perp = \tilde{U}^\vee$.

Because any four-dimensional rank at most two subspace of $\mathbb{C}^5$ is of type C or K or L, we don't need to separately consider the skew-symmetric case. And because every three-fold contains some planes, the semistability of some determinantal threefolds containing a skew-symmetric plane is expected.

Proof. The code outputs 27 block forms. Classifying them is a very similar process to the $4 \times 3 \times 3$ case. We provide a proof that $5 \times 3 \times 3$ type (4) is the annihilator of $4 \times 3 \times 3$ (4), the ones follow similarly.

Recall the following theorem from linear algebra: let C be a finite-dimensional k -vector space and $A \subset C$ and $B \subset C$ be subspaces. For any subspace $D \subset C$,

let $D^\perp \subset C^\vee$ be the annihilator of a subset D with respect to the dual pairing $C \otimes C^\vee \rightarrow k$. Then $(A \cap B)^\perp = A^\perp + B^\perp$ and

$$\dim(A^\perp \cap B^\perp) \geq \dim C + \dim(A \cap B) - \dim A - \dim B.$$

In this case, fix a basis for V and W and consider the following subspaces (written in block form):

$$A := \begin{bmatrix} * & * & * \\ * & * & * \\ * & * & 0 \end{bmatrix}, \quad B := \begin{bmatrix} * & * & 0 \\ * & * & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad A^\perp = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & * \end{bmatrix}, \quad B^\perp = \begin{bmatrix} 0 & 0 & * \\ 0 & 0 & * \\ * & * & * \end{bmatrix}.$$

Note that $\dim(A) = 8$ and $\dim(B) = 4$.

Fix an injective $5 \times 3 \times 3$ tensor $\varphi^\perp$. We identify $\tilde{U}^\vee$ with a five-dimensional subspace of $V \otimes W$ under the map $\varphi^\perp$. The tensor $\varphi^\perp$ is of block type (4) if and only if $\tilde{U}^\vee \subset A$ and $\dim(\tilde{U}^\vee \cap B) \geq 2$ in some basis for V and W . This is equivalent to $A^\perp \subset (\tilde{U}^\vee)^\perp$ and $\dim((\tilde{U}^\vee)^\perp \cap B^\perp) \geq 2$. And this is equivalent to φ being of $4 \times 3 \times 3$ block type (4). $\square$

Corollary B.6. *An injective $4 \times 3 \times 3$ tensor φ is $\mathrm{SL}(4) \times \mathrm{SL}(3) \times \mathrm{SL}(3)$ -semistable if and only if $\varphi^\perp$ is $\mathrm{SL}(5) \times \mathrm{SL}(3) \times \mathrm{SL}(3)$ -semistable.*

Appendix C

Long computations

C.1 Lines on determinantal cubic surfaces

We want to understand lines in determinantal cubic surfaces (most of this thesis), and under what conditions the determinant cubic is the zero cubic (Chapter 4). These are special cases of rank at most two linear subspaces of $V^\vee \otimes W^\vee$, when $\dim V = \dim W = 3$. The case for rank at most two projective lines is as follows:

Lemma C.1. *The span of two 3×3 pairings A and $B : V \otimes W \rightarrow k$ is rank at most two if and only if (at least) one of the following hold:*

- (i) *They both annihilate either a common non-zero subspace $V' \subset V$, or a common non-zero subspace $W' \subset W$ (either their kernels or cokernels intersect non-trivially, when viewed as matrices in $\text{Hom}(V, W^\vee)$).*
- (ii) *There is a pair of two-dimensional subspaces $V' \subset V, W' \subset W$ where $A|_{V' \otimes W'} = B|_{V' \otimes W'} = 0$.*

Equivalently, this is stating that there exists a basis for V and $W^\vee$ where $A, B \in \text{Hom}(V, W^\vee)$ (and any linear combination) are of at least one of the forms

$$\begin{bmatrix} * & * & * \\ * & * & * \\ 0 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} * & * & 0 \\ * & * & 0 \\ * & * & 0 \end{bmatrix}, \quad \begin{bmatrix} * & 0 & 0 \\ * & 0 & 0 \\ * & * & * \end{bmatrix}$$

corresponding to cases (i), (i) and (ii) respectively, where $*$ denotes a possibly non-zero entry. Because of this presentation, case (ii) is referred to as “type L” in this thesis, and we refer to linear subspaces with shared non-zero subspaces in their kernel (resp. cokernel) as type K (resp. type C).

Proof. It is immediate that if either of the two conditions hold for both A, B then they hold for any linear combination $\alpha A + \beta B$, and so their span is a rank at most two subspace.

For the reverse direction, suppose the span of A, B is rank at most two. We will interchangeably view them as elements of $\text{Hom}(V, W^\vee)$ and $V^\vee \otimes W^\vee$ for notational convenience in this proof. Then there is a non-zero $a \in \ker A, b \in \ker B$, and a, b are linearly independent (otherwise the kernels intersect non-trivially and they satisfy condition (i)). $A + B$ is also rank at most two, so there is a non-zero $c \in \ker(A + B)$, and $Bc = -Ac$. If a, b, c were linearly dependent then $c = \alpha a + \beta b$ for some $\alpha, \beta \in k$, but then $\beta Ab = Ac = -Bc = -\alpha Ba$, meaning Ab, Ba are linearly dependent and any linear combination of A, B map $\text{Span}\{a, b\} \subset V$ to $\text{Span}\{Ab\} \subset W^\vee$. But then we are in case (ii), because the 2×2 -dimensional subspace $\text{Span}\{a, b\} \otimes \text{Span}\{Ab\}^\perp$ is annihilated by both A, B .

So a, b, c are linearly independent and so span V . If Ab, Ba were linearly dependent then A, B have the same image (cokernel) and we are again in case (i), so we have reduced to the case that a, b, c form a basis for V and Ab, Ba are linearly independent. But then $2A + B$ is also rank at most two, and has a non-zero d in its kernel. Then in this basis, $d = \alpha a + \beta b + \gamma c$ for some $\alpha, \beta, \gamma \in k$ not all zero. Applying $(2A + B)$ to both sides, we get

$$(2A + B)d = (2A + B)\gamma c + (2A + B)(\alpha a + \beta b)$$

And expanding, noting that $Ac = (2A + B)c$ because $c \in \ker(A + B)$, we get

$$0 = \gamma Ac + 2\beta Ab + \alpha Ba$$

and the coefficients aren't all zero, so Ac, Ab, Ba are linearly dependent, and hence the images of A, B intersect non-trivially, i.e. we are in case (i) again. Hence all possibilities reduce to cases (i) or (ii), and we are done. $\square$

Remark C.2. While there may be slicker ways to prove this, it is insufficient to consider less than three linear combinations because the rank at most two locus in $\mathbb{P}(V \otimes W) \cong \mathbb{P}^8$ is cut out by the determinant, a cubic in the entries of a matrix (given some basis), and so is degree 3. Any generic line will then intersect it in three distinct points, but by finding a fourth point of intersection we show that the line is actually contained in the rank at most two locus. As an example, consider:

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

in some basis. Then $A, B, A + B$ are all rank at most two, but no other non-

zero linear combination is. Similar results hold for other dimensions (including non-square matrices subject to a maximal rank condition), which can be proved analogously. The above proof also works in all fields apart from $\mathbb{F}_2$, if the field is characteristic two then simply replace 2 with some other element distinct from 0, 1 and the modified proof works.

It would be nice if this result generalised to rank at most two subspaces of higher dimension, but this is false. Lemma 7 of [AL81] proves there is only one further possibility (up to change of basis), which they refer to as $\mathcal{W}_2$.

Lemma C.3. *A line on a determinantal cubic surface with finitely many lines that passes through exactly two singularities is of type L.*

Proof. If one or both of the singularities are rank one then it is immediate — any line of rank at most two containing a rank one point is of type L. So it suffices to show that if both singularities are rank two then they cannot share a kernel (and the same statement for sharing a cokernel holds by dualising).

Suppose the two singular points have a common kernel. Then in some basis, φ is of the form

$$\varphi(s, t) = s \begin{pmatrix} * & * & 0 \\ * & * & 0 \\ 0 & 0 & 0 \end{pmatrix} + t \begin{pmatrix} * & * & 0 \\ * & * & 0 \\ * & * & 0 \end{pmatrix}.$$

when restricted to the line parametrised by coordinates s, t , where $[1 : 0]$ and $[0 : 1]$ are the two singular points. $\varphi(s, t)$ is always rank two, otherwise it would have a third singularity on this line and then the entire line would be singular.

We can change coordinates so that the first matrix is diagonal (there's no need to worry about diagonalisability since we have complete freedom in both $\text{GL}(V), \text{GL}(W)$), and by some row and column operations we can also eliminate the second matrix's entries where the first one is non-zero:

$$\varphi(s, t) = s \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} + t \begin{pmatrix} 0 & a & 0 \\ b & 0 & 0 \\ c & d & 0 \end{pmatrix} = \begin{pmatrix} s & at & 0 \\ bt & s & 0 \\ ct & dt & 0 \end{pmatrix}.$$

Now, we compute the derivative of the determinant cubic, which is $\det(\varphi)' = \text{tr}((\text{Adj } \varphi)\varphi')$, where the prime denotes the derivative taken in an arbitrary direction. Actually computing these matrices, we have

$$\text{Adj } \varphi(s, t) = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ bdt^2 - cst & act^2 - dst & s^2 - abt^2 \end{pmatrix}$$

and so

$$\mathrm{tr}((\mathrm{Adj} \varphi)\varphi') = \varphi'_{13}(bdt^2 - cst) + \varphi'_{23}(act^2 - dst) + \varphi'_{33}(s^2 - abt^2).$$

Because the surface is singular when $s = 0$ and when $t = 0$, then for any directional derivative this expression is a quadratic with roots at $s = 0, t = 0$, and so is a multiple of st , i.e. the coefficients of s^2, t^2 are both always zero. So we always have

$$\begin{aligned} bd\varphi'_{13} + ac\varphi'_{23} &= 0 \\ \varphi'_{33} &= 0 \end{aligned}$$

There are four cases where the first equation is trivial: if $a = b = 0, c = d = 0$, $a = d = 0$ then $\varphi(0, 1)$ is rank one and this is a contradiction, we only need to check the final case where $c = d = 0$. But then the trace is

$$\mathrm{tr}((\mathrm{Adj} \varphi)\varphi') = \varphi'_{33}(s^2 - abt^2) = 0$$

so the entire line is singular. So instead we know that $\varphi'_{33} = 0$ and $\varphi'_{13}, \varphi'_{23}$ are linearly dependent. Because these are homogeneous linear forms, they are completely determined by their derivatives, so $\varphi_{13}, \varphi_{23}, \varphi_{33}$ are all multiples of some homogeneous linear form π . But then $\det(\varphi) = \pi \cdot \text{quadric}$ is reducible, and has infinitely many lines, another contradiction. $\square$

C.2 Miscellaneous calculations

Example C.4 (Calculation for 3.36). Fix any normal singular cubic surface with finitely many lines $S \subset \mathbb{P}^3 = \mathrm{Proj}(k[x, y, z, w]_{\bullet})$. Up to projective equivalence, S is cut out by $wf_2 - f_3 = 0$, where $f_2 \in k[x, y, z]$ is a non-zero homogeneous quadric that vanishes at $(0, 0, 1)$ (if $f_2 = 0$ then S is a cone and has infinitely many lines) and $f_3 \in k[x, y, z]$ is a non-zero homogeneous cubic that vanishes at $(0, 0, 1)$ such that f_2, f_3 have no common components (if they had common components then S would not be irreducible, hence not normal). The equations f_2 and f_3 are projectively equivalent to the normal form given by [BW79], although we won't need this. S has a singularity at $[0 : 0 : 0 : 1]$, and contains the reduced line $L = [0 : 0 : s : t]$ parametrised by $[s : t] \in \mathbb{P}^1$ which passes through the singularity.

Because S is singular at $[0 : 0 : 0 : 1]$, the local ring $\mathcal{O}_{S, [0:0:0:1]}$ is 3-dimensional, but the local ring $\mathcal{O}_{L, [0:0:0:1]}$ is 1-dimensional. So in any affine neighbourhood U

of $[0 : 0 : 0 : 1]$, the ideal $\mathcal{I}|_U$ that cuts out $L \cap U \hookrightarrow S \cap U$ is never a principal ideal. An affine open neighbourhood of $[0 : 0 : 0 : 1]$ is given by $\mathbb{A}^3$ under the coordinates $(x, y, z) = [x : y : z : 1] \in \mathbb{P}^3$. We let $S' = S \cap \mathbb{A}^3$ and $L' = L \cap \mathbb{A}^3$. And let $f = f_2 - f_3$.

Now, let $A = k[\varepsilon]/(\varepsilon^3)$ and $J = (\varepsilon^2) \subset A$. We have $J^2 = 0$, and $\mathfrak{m}_A = (\varepsilon)$ so $\mathfrak{m}_A J = (\varepsilon)(\varepsilon^2) = (\varepsilon^3) = 0$. There is then an isomorphism $A/J \cong k[\varepsilon]/(\varepsilon^2)$. We take the trivial extensions $S_{A/J} = S \times \text{Spec}(A/J)$ and $L_{A/J} = L \times \text{Spec}(A/J)$, and the non-trivial extension $S_A = \text{Proj}(A[x, y, z, w]_{\bullet}/(wf_2 - f_3 + \varepsilon^2 w^3)) \hookrightarrow \mathbb{P}_A^3$. The scheme S_A is a cubic surface inside $\mathbb{P}_A^3$. The ideal that cuts out $L' \hookrightarrow \mathbb{A}^3$ is (x, y) , and the ideal that cuts out $L' \hookrightarrow S'$ is $I := (x, y) \subset k[x, y, z]/(f) = R(S')$. Let $I_{A/J} = (x, y) \subset R(S'_{A/J})$ be the ideal that cuts out $L'_{A/J} = L_{A/J} \cap S'_{A/J}$ inside the affine scheme $S'_{A/J}$.

Because I is not principal, I is generated by x, y and so any flat extension I_A of $I_{A/J}$ is determined by an independent choice of lifts $x - \varepsilon^2 x_\varepsilon \in R(S'_A)$ of x and $y - \varepsilon^2 y_\varepsilon \in R(S'_A)$ of y , where x_ε and y_ε can be chosen to be polynomials in z only. The quotient ring $R(S'_A)/I_A$ can then be computed. Because S is normal, then the surface $f = 0$ is not a cone and so the polynomial f contains a non-zero z term. And because f_2 and f_3 each vanish at $(0, 0, 1)$, then there are no z^2 or z^3 terms.

Then $f(\varepsilon^2 x_\varepsilon, \varepsilon^2 y_\varepsilon, z) = \varepsilon^2 p(z) \in k[z, \varepsilon]/(\varepsilon^3)$ for some non-zero $p(z) \in k[z]$ (since the higher powers of ε vanish), and

$$R(S'_A)/I_A \cong k[z, \varepsilon]/(\varepsilon^2 p(z), \varepsilon^3). \quad (\text{C.1})$$

But this ring is not flat over $k[\varepsilon]/(\varepsilon^3)$. To see this, we tensor (C.1) with the map $(\varepsilon) \rightarrow k[\varepsilon]/(\varepsilon^3)$. This sends $\varepsilon p(z) \otimes \varepsilon \mapsto \varepsilon^2 p(z) = 0 \in R(S'_A)/I_A$. But $\varepsilon p(z)$ is non-zero in $R(S'_A)/I_A$, so $\varepsilon p(z) \otimes \varepsilon \neq 0$, and as a result the sequence

$$0 \longrightarrow R(S'_A)/I_A \otimes (\varepsilon) \longrightarrow R(S'_A)/I_A \quad (\text{not exact!})$$

is not exact.

So for any normal singular cubic surface S with finitely many lines and any line L through a singularity on S , there exist deformations of L that have no flat extensions. The Hilbert scheme of lines $\text{Hilb}_{m+1}(S_A/A) \rightarrow \text{Spec}(A)$ is finite over A (since S has finitely many lines), but there is a map $\text{Spec}(A/J) \rightarrow \text{Hilb}_{m+1}(S_A/A)$ that doesn't extend to a map $\text{Spec}(A) \rightarrow \text{Hilb}_{m+1}(S_A/A)$. So by Lemma 3.22, $\text{Hilb}_{m+1}(S_A/A) \rightarrow \text{Spec}(A)$ is not a smooth morphism.

Appendix D

Important notation

Notation

k	A fixed algebraically closed field of characteristic zero.
U and V and W	k -vector spaces of dimensions 4, 3, and 3, respectively.
φ	An element of $\mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$. A $4 \times 3 \times 3$ tensor. We write φ_U for the adjoint 3×3 matrix in $k[U^\vee]_\bullet$, and φ_V and φ_W as 4×3 matrices in $k[V^\vee]_\bullet$ and $k[W^\vee]_\bullet$, respectively.
$S_\varphi \hookrightarrow \mathbb{P}U$	The cubic surface cut out by homogeneous cubic $\det(\varphi_U) = 0$.
Γ_V^φ and Γ_W^φ	The respective subschemes of $\mathbb{P}V$ and $\mathbb{P}W$ where φ is rank two.
$\mathcal{I}_V^\varphi$ and $\mathcal{I}_W^\varphi$	The ideal sheaves that cut out Γ_V^φ and Γ_W^φ . Generated by the minors of φ_V and φ_W , respectively.

Definition 2.26. A rank at most two linear subspace $\mathbb{P}U' \subset \mathbb{P}(V^\vee \otimes W^\vee)$ is:

- Of *type K* (resp. *type C*) if there is a non-zero vector $\bar{v} \in V$ (resp. $\bar{w} \in W$) annihilated by every element $\bar{u} \in U'$.
- Of *type L* if there is a pair of two-dimensional subspaces $V' \subset V$ and $W' \subset W$ where $\bar{u}(V' \otimes W') = 0$ for every $\bar{u} \in U'$.

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